QUATERNIONIC STRUCTURES ON CONICAL SYMPLECTIC SINGULARITIES

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Abstract. We prove that the coordinate ring of any conical symplectic singularity with a grading induced by the minimizer of the normalized volume admits, for any torus of graded Poisson automorphisms, a quaternionic structure commuting with the maximal compact subgroup of the torus, and hence the weights of the torus on each graded piece, counted with multiplicity, are symmetric under negation. We also prove that the identity component of the group of graded Poisson automorphisms is reductive, that twisting the action on each graded piece by any Chevalley involution yields an isomorphic representation, and that the grading descended to a Hamiltonian reduction by such a torus is again induced by the minimizer of the normalized volume if the reduction, with the descended symplectic form, is a symplectic singularity. We thereby solve Problems 27, 28, and 29 and Parts (1) and (2) of Problem 30 in Xu and Zhuang's list of open problems on K-stability of Fano varieties.

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1. Introduction

We work over the field of complex numbers $\mathbb{C}$.

A *symplectic singularity* is a pair (X, ω) of a normal affine variety $X = \operatorname{Spec} R$ and a closed non-degenerate algebraic 2-form ω on X_{reg} such that for any resolution $h : X' \to X$, the form $h^*\omega$ on $h^{-1}(X_{\mathrm{reg}})$ extends to a regular 2-form on X' [Bea99, Definition 1.1]. Since ω is non-degenerate, $\dim X$ is even, and we write $\dim X = 2n$. A symplectic singularity (X, ω) is *conical* (cf. [XZ26b, Definition 6.1]) if $\dim X > 0$ and there exists a grading $R = \bigoplus_{d \geq 0} R_d$ with $R_0 = \mathbb{C}$ such that ω is homogeneous of some positive degree r, that is, $\lambda_t^* \omega = t^r \omega$ for any $t \in \mathbb{G}_m$, where λ is the $\mathbb{G}_m$-action on X given by $\lambda_t^* f = t^d f$ for $f \in R_d$ (Definitions 2.5 and 2.8). For any such grading, ω induces a Poisson bracket $\{-,-\}$ on R of degree $-r$ (see Lemma 2.11).

Typical examples of conical symplectic singularities are normalizations of nilpotent orbit closures, intersections of Slodowy slices with the nilpotent cone, quotients V/G of a symplectic vector space V by a finite subgroup $G \subset \mathrm{Sp}(V)$, and hyperkähler reductions of symplectic vector spaces when they are symplectic singularities, such as quiver varieties [BS16, Theorem 1.2] (see also [HSS17, Theorems 1.1 and 1.3]). Many of them carry hyperkähler metrics on their regular

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parts: see [Kro90, Theorem 1 and Remarks (i)–(ii)] for nilpotent orbits and their intersections with Slodowy slices, and [HKLR87, Section 3(D) and Theorem 3.2] for hyperkähler quotients.

We now turn to the normalized volume. A symplectic singularity is rational Gorenstein, equivalently canonical of index 1 [Bea99, Proposition 1.3 and footnote 1]. In particular, for a symplectic singularity (X,ω) and a closed point $x \in X$, the normalized volume $\widehat{\mathrm{vol}}_{X\ni x}$ is defined on the space of real valuations of the function field of X centered at x (Definition 2.2). The normalized volume was introduced in [Li15]. A minimizer of $\widehat{\mathrm{vol}}_{X\ni x}$ exists [Blu16, Main Theorem], and it is unique up to rescaling [XZ20, Theorem 1.1]. For a K-semistable log Fano cone (X,ξ) (Definition 2.4) with vertex o, the weight valuation wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{X\ni o}$ [LX17, Theorem 1.3], and a log Fano cone (X,ξ) is K-semistable if and only if wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{X\ni o}$ [LWX18, Theorem 2.5].

For symplectic singularities, a conjecture of Kaledin (see [Kal03, Remark 4.2]) predicts that for any symplectic singularity (X,ω) and any closed point $x \in X$, the formal completion of X at x with the form ω is isomorphic to the formal completion of a conical symplectic singularity (X_0,ω_0) at the vertex of a grading as in Definition 2.8, with the form ω_0. Kaledin's conjecture was proved in [XZ26b, Theorem 1.2] (see [XZ26b, Theorem 6.2] for the compatibility of the formal isomorphism with the symplectic forms). Moreover, by [XZ26b, Theorem 6.2], which strengthens the statement on K-semistability in [XZ26b, Theorem 1.2], for any conical symplectic singularity (X,ω) and any grading of R for which ω is homogeneous of positive degree, there exist a faithful action of an algebraic torus T on X extending the $\mathbb{G}_m$-action defined by the grading, with $t^*\omega \in \mathbb{C}^*\omega$ for any $t \in T$, and a vector ξ in the Reeb cone of T (Definition 2.4) such that (X,ξ) is a K-semistable log Fano cone.

Let (X,ω) be a conical symplectic singularity. In [XZ26a, Problem 25], Xu and Zhuang ask to show that the minimizer of the normalized volume of X at the vertex is a positive multiple of ord_E for a Kollár component E over X [LX16, Definition 1.1] with $a(E,X) = rn$, where $r \in \{1,2\}$ is the degree of ω with respect to the grading defined by ord_E, that E carries an orbifold contact structure, and that (E,Δ_E) is a K-polystable log Fano pair [LWX18, Definition 2.28], where $h : X' \to X$ is the plt blow-up extracting E and Δ_E is defined by $(K_{X'}+E)|_E = K_E + \Delta_E$. The companion paper [Liu26a] proves that the minimizer is a positive multiple of ord_E for a Kollár component E with $a(E,X) \in \{n, 2n\}$ [Liu26a, Theorem 2.1], and that the log Fano cone given by the minimizer and (E,Δ_E) are K-polystable [Liu26a, Theorem 2.2].

Xu and Zhuang explain that an affirmative answer to [XZ26a, Problem 25] would say that any conical symplectic singularity is a "hidden hyperkähler cone" [XZ26a, Section 3.1, second paragraph after Problem 25]. We use hyperkähler cones in the sense of [ES20, Section 2.5]: a *hyperkähler cone* is a normal conical Poisson variety whose Poisson bracket has negative degree, which is symplectic outside a closed subset of codimension at least 2, and whose smooth part carries a hyperkähler structure compatible with the Poisson structure. The examples of conical symplectic singularities given above appear in [ES20, Section 2.5] as examples of hyperkähler cones. Xu and Zhuang proposed the following conjecture.

Conjecture 1.1 ([XZ26b, Conjecture 1.3])**.** *Any conical symplectic singularity is a hyperkähler cone.*

As explained in [XZ26b, paragraph before Conjecture 1.3], in view of the Yau–Tian–Donaldson correspondence for log Fano cones [Li21, Theorem 2.9] and [Hua22, Theorem 1.0.8], Conjecture 1.1 was essentially proposed in [XZ26a, Section 3.1, second paragraph after Problem 25].

Problems 27–31 of Xu–Zhuang's list of open problems [XZ26a] spell out several properties of hyperkähler cones that are expected to hold for all conical symplectic singularities, and the last of them, [XZ26a, Problem 31], concerns short star-products in the sense of [BPR16, Section 3.1, (3.5)] and [ES20, Definition 2.2]. We recall Problems 27–30 of [XZ26a] as Questions 1.2–1.5.

The Poisson automorphisms and the Poisson brackets in Questions 1.3, 1.4, and 1.5 below are taken with respect to $\{-,-\}$. In Questions 1.2 and 1.3 below, a conical $\mathbb{C}^*$-action is the

$\mathbb{G}_m$-action defined by a grading for which ω is homogeneous of positive degree, its weight is the degree of ω, and the action gives a valuation v, or is induced by v, if v is the degree valuation of the grading (Definitions 2.5 and 2.8). In Questions 1.2 and 1.3, an action of a torus is Hamiltonian if it admits a moment map, as in [XZ26b, Lemma 7.8 and the paragraph before it]. For the moment map in this paper, see Definition-Lemma 5.1.

Question 1.2 ([XZ26a, Problem 27])**.** Let $(X = \operatorname{Spec} R, \omega)$ be a conical symplectic singularity with vertex o, and let T be a torus with an algebraic Hamiltonian action on (X, ω) and moment map $\mu : X \to \mathfrak{t}^*$, where $\mathfrak{t} = \operatorname{Lie} T$. Let $Y := \mu^{-1}(0)/\!\!/T$. Assume that the minimizer of $\widehat{\operatorname{vol}}_{X \ni o}$ is given by a conical $\mathbb{C}^*$-action of weight $\ell > 0$ on ω which commutes with T, and that Y, with the symplectic form on Y_{reg} descended from ω, is a symplectic singularity. Show that the minimizer of the normalized volume of Y at the vertex of Y is given by the conical $\mathbb{C}^*$-action on Y descended from the conical $\mathbb{C}^*$-action on X.

Question 1.2 is inspired by the descent of conical hyperkähler metrics under hyperkähler reduction [HKLR87, Section 3(D) and Theorem 3.2], see [XZ26a, paragraph 26, Problem 27, and the two sentences after it]. The assumption on Y holds in many cases, see e.g. [HSS17, Theorems 1.1 and 1.3], [BS16, Theorem 1.2], and [FL26, Theorem 1.1]. The cited results concern symplectic reductions by general reductive groups (in [BS16, Theorem 1.2], the group is a product of general linear groups), and [HSS17, Theorem 1.3] treats the case when the identity component of the group is a torus (for the complex symplectic quotient of [HSS17, Definition 2.5], which is $\mu^{-1}(0)/\!\!/G$ when (V, G) is 1-large [HSS17, Lemma 2.8]). Xu and Zhuang observe that Question 1.2 suggests the following question [XZ26a, Problem 28].

Question 1.3 ([XZ26a, Problem 28])**.** Let $(X = \operatorname{Spec} R, \omega)$ be a conical symplectic singularity, and let $R = \bigoplus_{d \geq 0} R_d$ be the grading defined by the conical $\mathbb{C}^*$-action induced by the normalized volume minimizer ord_E of X at the vertex. Let T be a maximal torus in the group of graded Hamiltonian Poisson automorphisms of R. Show that

$$\dim R_{d,\chi} = \dim R_{d,-\chi}$$

for any $d \geq 0$ and any character χ of T, where $R_{d,\chi} := \{f \in R_d \mid t \cdot f = \chi(t) f \text{ for any } t \in T\}$.

By the convention on Hamiltonian actions before Question 1.3, the torus T in Question 1.3 is maximal among the tori acting on R by graded Poisson automorphisms whose actions are Hamiltonian (see also Section 1.1). To explain Question 1.3, Xu and Zhuang proposed the following question.

Question 1.4 ([XZ26a, Problem 29])**.** In the setting of Question 1.3, let r be the degree of ω and let $T_c \subset T$ be the maximal compact subgroup. Show that there exists a $\mathbb{C}$-antilinear degree-preserving Poisson automorphism $\rho : R \to R$ which commutes with T_c, such that $\rho^2(f) = \exp(2\pi i d/r) f$ for any $d \geq 0$ and $f \in R_d$.

An automorphism ρ as in Question 1.4 is a quaternionic structure in the sense of [ES20, Definitions 3.17(i) and 3.26], after the normalization of the grading in Remark 1.7. Quaternionic structures originate in physics [BPR16, Appendix A]. Since ρ is antilinear and commutes with T_c, ρ maps $R_{d,\chi}$ onto $R_{d,-\chi}$. Thus an affirmative answer to Question 1.4 answers Question 1.3 [XZ26a, Problem 29 and the paragraph after it].

Xu and Zhuang also proposed a stronger form of Question 1.3, motivated by the reductivity of the following group for hyperkähler cones [XZ26a, paragraph before Problem 30]. Let G_R be the identity component of the group of graded automorphisms of R preserving the Poisson bracket (see Definition-Lemma 4.1). Equivalently, G_R is the identity component of the stabilizer of ω in the group of graded automorphisms of R (Lemma 2.12). Since any Poisson derivation of R is Hamiltonian [Los16, Proposition 2.14], the Lie algebra of G_R is isomorphic to R_r (see [XZ26a, paragraph before Problem 30]).

Question 1.5 ([XZ26a, Problem 30])**.** In the setting of Question 1.3, let r be the degree of ω. Then R_r is a Lie algebra under the Poisson bracket, and R_d is an R_r-module for any $d \geq 0$. Let $\mathfrak{h} \subset R_r$ be the subalgebra corresponding to $\operatorname{Lie} T$.

(1) Show that R_r is reductive, so that R_r admits a Chevalley involution $C : R_r \to R_r$ with respect to $\mathfrak{h}$, that is, $C|_{\mathfrak{h}} = -\operatorname{id}_{\mathfrak{h}}$.
(2) Show that for any $d \geq 0$, there exists an isomorphism of R_r-modules $\tau_d : R_d \to R_d^C$, where R_d^C is the R_r-module whose underlying vector space is R_d and on which $a \in R_r$ acts by $f \to \{C(a), f\}$.
(3) Show that there exists a graded Chevalley involution $\tau : R \to R$ which is compatible with C.

Since $C|_{\mathfrak{h}} = -\operatorname{id}_{\mathfrak{h}}$, an isomorphism τ_d as in Question 1.5(2) maps $R_{d,\chi}$ onto $R_{d,-\chi}$. Thus an affirmative answer to Question 1.5(2) answers Question 1.3 [XZ26a, sentence after Problem 30].

We now explain what was known about Questions 1.2–1.5 before this paper. Question 1.2 has an affirmative answer in the following cases.

(1) Hypertoric singularities [XZ26b, Theorem 1.5] (proved as [XZ26b, Theorem 7.10]). A hypertoric singularity is a Hamiltonian reduction of $(\mathbb{C}^{2N}, \sum_i dz_i \wedge dw_i)$ by a subtorus T of the torus $\mathbb{G}_m^N$ acting by $z_i \to t_i z_i$ and $w_i \to t_i^{-1} w_i$ (see [XZ26b, Definition 7.7] for the conditions on T), and [XZ26b, Theorem 1.5] shows that the minimizer of the reduction Y is induced by the standard $\mathbb{G}_m$-action on $\mathbb{C}^{2N}$.
(2) Hyperkähler quotients of a smooth affine complete hyperkähler manifold by a compact group, under the assumptions of [NO25, Setup 1 and Proposition 6.10]. For any point x of such a quotient, the analytic germ at x is isomorphic to the germ at the vertex of the hyperkähler quotient of a holomorphic symplectic representation, which is a conical symplectic singularity, and the minimizer of the normalized volume at x is induced by the $\mathbb{G}_m$-action descended from the scalar multiplication on the representation [NO25, Proposition 6.10(ii)–(iii) and its proof]. Examples include nilpotent orbit closures of type A and their intersections with Slodowy slices, and many quiver varieties and toric hyperkähler varieties [NO25, Example 6.11].

For hypertoric singularities, the method of [XZ26b] is based on the involution $\iota(z_i, w_i) = (w_i, z_i)$ of $\mathbb{C}^{2N}$: ι preserves the zero fiber of the moment map and the line spanned by $\sum_i dz_i \wedge dw_i$, descends to Y, and reverses the weights of the action of $\mathbb{G}_m^N/T$ on Y. The weight reversal is combined with [XZ26b, Lemma 7.1], which states that if (Y, ξ) is a log Fano cone, then any reductive subgroup of the automorphism group of Y which preserves wt_ξ and contains the torus $\langle \xi \rangle$ generated by ξ has $\langle \xi \rangle$ in the center. In [XZ26b], the lemma is applied to the subgroup of the automorphism group of Y generated by ι, the image of $\mathbb{G}_m^N/T$, and the image of the $\mathbb{G}_m$-action on Y descended from the scalar multiplication on $\mathbb{C}^{2N}$. The identity component of the center of the subgroup is the image of the descended $\mathbb{G}_m$-action, so the minimizer is induced by the descended $\mathbb{G}_m$-action.

The method of [NO25] is based on the reduction of the hyperkähler metric of the smooth complete ambient manifold. For general X and Y as in Question 1.2, neither an involution of X with the properties of ι nor a hyperkähler metric on X to be reduced is available. Instead, we only pass the equalities $\dim R_{d,\chi} = \dim R_{d,-\chi}$ of Question 1.3 from X to Y.

For Questions 1.3 and 1.4, the following antilinear maps were constructed.

(1) A hyperkähler cone X, with the grading for which the Poisson bracket has degree -2, carries a quaternionic structure in the sense of [ES20, Definitions 3.17(i) and 3.26]. The quaternionic structure of [ES20, Section 3.8] is induced by the action on X of the unit quaternion $\mathbf{j} \in \mathrm{SU}(2)$, where the action of $\mathrm{SU}(2)$ on X is the one of [BPR16, Appendix A].
(2) The hyperkähler cone over a compact 3-Sasakian manifold [BF20, Proposition 2.1], and a conical symplectic singularity (X, ω) with a hyperkähler cone metric on X_{reg} whose twistor

space is algebraic [Kot26, Proposition 3.33], carry an antiholomorphic automorphism which pulls the symplectic form back to its complex conjugate.

(3) A symplectic vector space V and the quotient V/G by a finite subgroup $G \subset \mathrm{Sp}(V)$ carry quaternionic structures [ES20, Example 3.29], [Kot26, Example 2.8]. For V and V/G, the minimizer is induced by the blow-up of the origin of V [LX16, Example 7.1(1)], so the degrees of the grading used in [ES20] are a positive multiple of the degrees of a minimizing grading (Definition 2.14).

(4) The nilpotent cone carries an antiholomorphic automorphism as in (2), given by the restriction of a Cartan involution [Kot26, Example 2.9]. By [XZ26b, Theorem 1.4], for normalizations of nilpotent orbit closures, in particular for the nilpotent cone, the minimizer is induced by the scaling action, so the degrees of the grading used in [Kot26, Example 2.9] are twice the degrees of a minimizing grading.

For Question 1.5(1), the following results are known.

(1) If X is a hyperkähler cone, then by the twistor construction [HKLR87, Theorem 3.3], a real form of G_R preserves the hyperkähler metric, and hence a Hermitian inner product on each R_d. Thus the real form is compact, and G_R is reductive [ES20, p. 15].

(2) If R is generated by R_1, then (X, ω) is isomorphic to $(\mathbb{C}^{2n}, \omega_{\mathrm{st}})$, where ω_{st} is the standard symplectic form, or to a normal nilpotent orbit closure, and if moreover ω has degree 1, then R_1 is a semisimple Lie algebra [Nam16, Theorem and Proposition 4].

(3) If R is finite over the subalgebra generated by R_r, where r is the degree of ω, then R_r is a semisimple Lie algebra [Nam17, Theorem (i)].

(4) If (X, ξ) is a quasi-regular [XZ26b, Definition 2.12] K-polystable log Fano cone, then the group of automorphisms of X commuting with the $\mathbb{G}_m$-action generated by ξ is reductive [XZ26b, second paragraph after Lemma 7.1], [ABHLX19, Theorem 1.3] (see also Proposition 4.4).

The grading in Question 1.5 is a minimizing grading (Definition 2.14), and the reductivity of R_r depends on the grading. Indeed, let $m \geq 2$ be an integer, and let $X \subset \mathbb{C}^3$ be the surface defined by $xy = z^{m+1}$. Then X is the quotient of $\mathbb{C}^2$ by the cyclic group generated by $(u, v) \to (\zeta u, \zeta^{-1} v)$, where ζ is a primitive $(m+1)$-th root of unity, with $x = u^{m+1}$, $y = v^{m+1}$, and $z = uv$. Let ω be the 2-form on X_{reg} induced by $du \wedge dv$. Then the Poisson bracket on R is the restriction of the Poisson bracket on $\mathbb{C}[u, v]$ with $\{u, v\} = 1$. For the grading of $R = \mathbb{C}[x, y, z]/(xy - z^{m+1})$ with weights $(1, m, 1)$ on (x, y, z), the form ω has degree 1 and $R_1 = \mathbb{C}x \oplus \mathbb{C}z$. A direct calculation shows that

$$\{x, z\} = (m+1)x.$$

Thus R_1 is a two-dimensional non-abelian solvable Lie algebra, and R_1 is not reductive [Nam17, Example]. The grading with weights $(1, m, 1)$ is not a minimizing grading. Indeed, the minimizer is induced by the blow-up of the origin of $\mathbb{C}^2$ [LX16, Example 7.1(1)], so the values on (x, y, z) of the degree valuation of any minimizing grading are proportional to $(m+1, m+1, 2)$, while the degree valuation of the grading with weights $(1, m, 1)$ takes the values $(1, m, 1)$ on (x, y, z). Dividing the weights $(m+1, m+1, 2)$ on (x, y, z) by their greatest common divisor, we obtain a minimizing grading, for which $R_r = \mathbb{C}z$ as $m \geq 2$, and R_r is a one-dimensional, hence reductive, Lie algebra.

For a minimizing grading, the log Fano cone given by the grading is quasi-regular and K-polystable (Theorem 2.17), and the group of automorphisms of X commuting with the $\mathbb{G}_m$-action defined by the grading is the group of graded automorphisms of R. Thus the group of graded automorphisms of R is reductive (see [XZ26b, second paragraph after Lemma 7.1], which uses [ABHLX19, Theorem 1.3]), and Proposition 4.4 proves the reductivity of the identity component of the group. Recall that G_R is the identity component of the stabilizer of ω in the group of graded automorphisms of R, which acts on the space of 2-forms on X_{reg} homogeneous of degree r. The stabilizer of a vector in a representation of a reductive group need not be reductive:

for example, the stabilizer in $\mathrm{SL}_2(\mathbb{C})$ of a non-zero vector of $\mathbb{C}^2$ is unipotent. We prove the reductivity of G_R by a different argument, explained in the sketch of the proof below.

We now explain where our approach to Questions 1.3 and 1.4 starts. In the known cases above, the antilinear maps come from a hyperkähler structure on X_{reg} and the action of $\mathrm{SU}(2)$ rotating the complex structures of the hyperkähler structure. The cones of [BF20, Proposition 2.1] are built on compact 3-Sasakian manifolds, so their vertices are isolated singularities, and [Kot26, Proposition 3.33] takes a hyperkähler cone metric on X_{reg} whose twistor space is algebraic as input. By [NO25, Theorem 6.19], an action of $\mathrm{SU}(2)$ on the regular part of a conical symplectic singularity exists if the regular part carries a hyperkähler cone metric that satisfies [NO25, Conjecture 1.3] and such that the vector fields obtained by rotating the radial vector field [Li21, Definitions 2.1 and 2.2] of the metric by the complex structures of the metric are complete, and by [NO25, Remark 6.18], the completeness of the vector fields is the main purpose of the assumptions (i)–(iii) of [NO25, Theorem 6.16], namely that the singularity is an isolated symplectic singularity (case (i)) or the germ of a smoothable projective irreducible symplectic variety at a point (cases (ii) and (iii)), each with further conditions.

Our starting point is K-stability. Let (X,ω) be a conical symplectic singularity with a minimizing grading (Definition 2.14). The degree valuation of the grading is a minimizer, which implies that the log Fano cone given by the grading is K-semistable [LWX18, Theorem 2.5] (see also [LX17, Theorem 1.1(2)]). The existence theorem [Li21, Theorem 2.9] (see also [LWX18, Remark A.2] and [Hua22, Theorem 1.0.8]) implies that the regular part of any K-polystable log Fano cone, away from the vertex, carries a Ricci-flat Kähler cone metric [Li21, Definitions 2.1 and 2.2]. The K-polystability of the log Fano cone given by the grading, proved in the companion paper [Liu26a] (see Theorem 2.17), makes such a metric available on $X_{\mathrm{reg}}\backslash\{o\}$, where o is the vertex. The metric is only Ricci-flat. Starting from it, we prove the properties of the metric that [NO25, Theorem 6.19] and its proof use to obtain the action of $\mathrm{SU}(2)$: ω is parallel with respect to the metric, which yields a hyperkähler structure from the metric and ω, and the vector fields obtained by rotating the radial vector field are complete.

The goal of this paper is to answer Questions 1.2, 1.3, and 1.4, and Parts (1) and (2) of Question 1.5, for all conical symplectic singularities.

1.1. **Quaternionic structures and weight symmetry.** Let (X,ω) be a conical symplectic singularity, where $X=\operatorname{Spec} R$. By Theorem 2.16(1), (X,ω) admits a minimizing grading (see Definition 2.14).

Fix a minimizing grading $R=\bigoplus_{d\geq 0}R_d$, let r be the degree of ω, and recall that G_R is the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ (see Definition 2.13 and Definition-Lemma 4.1). By Lemma 4.2(2), the Lie algebra of G_R consists of the Hamiltonian derivations $\{a,-\}$ with $a\in R_r$, and we read the group of graded Hamiltonian Poisson automorphisms of R in Questions 1.3–1.5 as G_R. By Remark 1.10(1), the action on R of any algebraic torus in $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ is Hamiltonian in the sense of Questions 1.2 and 1.3, and by Definition-Lemma 4.1(3), any such torus is contained in a maximal torus of G_R. Thus under either reading of Question 1.3, the torus T in Questions 1.3–1.5 is a maximal torus of G_R, and Theorem 1.6, which holds for any algebraic torus in $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$, answers Questions 1.3 and 1.4.

Theorem 1.6. *Let $(X=\operatorname{Spec}R,\omega)$ be a conical symplectic singularity, let $R=\bigoplus_{d\geq 0}R_d$ be a minimizing grading of (X,ω) (see Definition 2.14), and let r be the degree of ω. Let $T\subset\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ be an algebraic torus (see Definition 2.13), and let $T_c\subset T$ be the maximal compact subgroup. Then $r\in\{1,2\}$, and there exists a ring automorphism $\rho:R\to R$ satisfying the following.*

1. *$\rho(zf)=\bar{z}\rho(f)$, $\rho(R_d)=R_d$, and $\rho(\{f,h\})=\{\rho(f),\rho(h)\}$ for any $z\in\mathbb{C}$, $d\geq 0$, and $f,h\in R$.*
2. *$\rho(t\cdot f)=t\cdot\rho(f)$ for any $t\in T_c$ and $f\in R$.*
3. *$\rho^2(f)=\exp(2\pi id/r)f$ for any $d\geq 0$ and $f\in R_d$.*

In particular, $\rho(R_{d,\chi}) = R_{d,-\chi}$ *and* $\dim R_{d,\chi} = \dim R_{d,-\chi}$ *for any* $d \geq 0$ *and any character* χ *of* T.

In Theorem 1.6, $r \in \{1, 2\}$ holds by [Liu26a, Theorem 2.1] (see Theorem 2.16(2)).

Remark 1.7. In Theorem 1.6, for any integer $m \geq 0$, let $R'_m := R_{rm/2}$ if $rm/2$ is an integer, and let $R'_m := 0$ otherwise. Since $r \in \{1, 2\}$, the decomposition $R = \bigoplus_{m \geq 0} R'_m$ is a grading for which the Poisson bracket has degree -2, and $\rho^2(f) = (-1)^m f$ for any $m \geq 0$ and $f \in R'_m$. Thus ρ is a quaternionic structure on R in the sense of [ES20, Definitions 3.17(i) and 3.26].

The automorphism ρ in Theorem 1.6 is constructed from an action of $\mathrm{SU}(2)$ on $X_{\mathrm{reg}} \backslash \{o\}$ (see Remark 3.21), where o is the vertex of the grading, and the choice of ρ depends on T (see the sketch of the proof below).

1.2. **Reductivity and Chevalley involutions.** Parts (1) and (2) of Theorem 1.8 answer Question 1.5(1), and Part (3) answers Question 1.5(2). As explained in Section 1.1, the torus T in Question 1.5 is a maximal torus of G_R.

Theorem 1.8. *Let* $(X = \operatorname{Spec} R, \omega)$ *be a conical symplectic singularity, let* $R = \bigoplus_{d \geq 0} R_d$ *be a minimizing grading of* (X, ω)*, and let* r *be the degree of* ω*. Let* G_R *be the identity component of* $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ *(see Definition-Lemma 4.1), let* $\mathfrak{g}$ *be the Lie algebra* R_r *with the Poisson bracket, let* $T \subset G_R$ *be a maximal torus, and let* $\mathfrak{h} := \{a \in \mathfrak{g} \mid \{a, -\} \in \operatorname{Lie} T\}$*, where* $\operatorname{Lie} T$ *is viewed as a Lie algebra of derivations of* R*. Then the following hold.*

(1) G_R *is a connected reductive group, and the map* $\mathfrak{g} \to \operatorname{Lie} G_R$*,* $a \to \{a, -\}$*, is an isomorphism of Lie algebras. In particular,* $\mathfrak{g}$ *is reductive.*

(2) There exists an algebraic group involution c_0 *of* G_R *with* $c_0(t) = t^{-1}$ *for any* $t \in T$*, and under the isomorphism in (1), the differential of* c_0 *is a Lie algebra involution of* $\mathfrak{g}$ *whose restriction to* $\mathfrak{h}$ *is* $-\mathrm{id}_{\mathfrak{h}}$*.*

(3) For any Lie algebra involution C *of* $\mathfrak{g}$ *with* $C|_{\mathfrak{h}} = -\mathrm{id}_{\mathfrak{h}}$ *and any* $d \geq 0$*, there exists a* $\mathbb{C}$*-linear bijection* $\tau_d : R_d \to R_d$ *such that* $\tau_d(\{a, f\}) = \{C(a), \tau_d(f)\}$ *for any* $a \in \mathfrak{g}$ *and* $f \in R_d$*. Moreover, under the isomorphism in (1),* C *is the differential of a unique algebraic group involution* c *of* G_R*, the involution* c *satisfies* $c(t) = t^{-1}$ *for any* $t \in T$*, and* τ_d *can be taken to satisfy* $\tau_d(g \cdot f) = c(g) \cdot \tau_d(f)$ *for any* $g \in G_R$ *and* $f \in R_d$*.*

Let C be a Lie algebra involution of $\mathfrak{g}$ with $C|_{\mathfrak{h}} = -\mathrm{id}_{\mathfrak{h}}$, let c be the algebraic group involution of G_R with differential C (see Theorem 1.8(3)), and let $d \geq 0$. By Parts (1) and (2) of Theorem 1.8 and Lemma 4.8(2), the representation R_d of G_R is isomorphic to the representation of G_R on the vector space R_d given by $g \cdot_c f := c(g) \cdot f$ if and only if

$$\dim R_{d,\chi} = \dim R_{d,-\chi} \quad \text{for any character } \chi \text{ of } T,$$

which does not involve C and holds by Theorem 1.6. Taking derivatives at the identity of G_R (see Definition-Lemma 4.1(2)), we see that any such isomorphism is an isomorphism $R_d \to R_d^C$ of $\mathfrak{g}$-modules as in Question 1.5(2). Conversely, since $C|_{\mathfrak{h}} = -\mathrm{id}_{\mathfrak{h}}$, any isomorphism $R_d \to R_d^C$ of $\mathfrak{g}$-modules maps $R_{d,\chi}$ onto $R_{d,-\chi}$. Thus the answer to Question 1.5(2) does not depend on the choice of C in Question 1.5(1).

1.3. **Hamiltonian reduction.** Theorem 1.9 answers Question 1.2 (see Remark 1.10).

Theorem 1.9. *Let* $(X = \operatorname{Spec} R, \omega)$ *be a conical symplectic singularity, let* $R = \bigoplus_{d \geq 0} R_d$ *be a grading with* $R_0 = \mathbb{C}$ *such that* ω *is homogeneous of positive degree, and let* $o \in X$ *be the vertex of the grading (see Definitions 2.5 and 2.8). Let* T *be an algebraic torus acting on* R *by graded Poisson automorphisms (see Definition 2.13), let* μ *be the moment map of the* T*-action (see Definition-Lemma 5.1), let* $Y = \operatorname{Spec} S$ *be the Hamiltonian reduction of* X *by* T*, with the grading* $S = \bigoplus_{d \geq 0} S_d$ *and the Poisson bracket induced from* R *(see Definition-Proposition 5.2), and let* $o_Y \in Y$ *be the vertex of the grading of* S*. Assume that the degree valuation of the grading of* R *is a minimizer of* $\widehat{\mathrm{vol}}_{X \ni o}$ *(see Definition 2.2), and that there exists a symplectic form* ω_Y *on* Y_{reg}

such that (Y, ω_Y) is a symplectic singularity and ω_Y induces the Poisson bracket of S. Then the following hold.

(1) If $\dim Y > 0$, then the degree valuation of the grading of S is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$.
(2) If $\dim Y = 0$, then $S = \mathbb{C}$.

Remark 1.10. In Theorem 1.9, let $I := R\mu(\mathfrak{t})$, so that $S = (R/I)^T$, and R/I is the coordinate ring of the scheme-theoretic fiber $\mu^{-1}(0)$ of the morphism $\mu : X \to \mathfrak{t}^*$. We compare Theorem 1.9 with Question 1.2 as follows.

(1) An action of T on R by graded Poisson automorphisms corresponds to an algebraic action of T on X which preserves ω (see Lemma 2.12) and commutes with λ, where λ is the $\mathbb{G}_m$-action defined by the grading. For $a \in \mathfrak{t}$, let a_X be the vector field on X_{reg} with $a_X(f) = a \cdot f$ for any $f \in R$. Following [XZ26b, paragraph before Lemma 7.8], a moment map of such an action as in Question 1.2 is a morphism $\mu' : X \to \mathfrak{t}^*$ with $d\langle \mu', a\rangle = \iota_{a_X}\omega$ on X_{reg} for any $a \in \mathfrak{t}$. By Lemma 2.11 and Definition-Lemma 5.1, we have $a_X = \{\mu(a), -\}$, and hence $\iota_{a_X}\omega = -d\mu(a)$ on X_{reg}. Thus the action is Hamiltonian, and $-\mu$ is a moment map as in Question 1.2 (cf. [XZ26b, Lemma 7.8]).
(2) Let μ' be a moment map of the T-action as in (1). Since X_{reg} is connected, we have $\mu' = -\mu + \beta$ for some $\beta \in \mathfrak{t}^*$. If $\mu'^{-1}(0)$ is non-empty and stable under λ, then $o \in \mu'^{-1}(0)$ since o is contained in the closure of any λ-orbit. Since $\mu(o) = 0$ by Definition-Lemma 5.1(3), we obtain $\beta = \mu'(o) = 0$, and hence $\mu'^{-1}(0) = \mu^{-1}(0)$.
(3) We take the symplectic form on Y_{reg} descended from ω in Question 1.2 to be a symplectic form ω_Y on Y_{reg} inducing the Poisson bracket of S, which is the hypothesis on ω_Y in Theorem 1.9.
(4) Since T is linearly reductive, we have $(R/\sqrt{I})^T = S/(\sqrt{I}/I)^T$, and $(\sqrt{I}/I)^T$ is a nil ideal of S. Since (Y, ω_Y) is a symplectic singularity, S is a normal domain, and in particular reduced, so we have $(R/\sqrt{I})^T = S$. Thus Y is also the quotient by T of the reduced subscheme of $\mu^{-1}(0)$.
(5) If $\dim Y > 0$, then the $\mathbb{G}_m$-action on Y defined by the grading of S is induced by λ, since $S_d = (R_d/(I \cap R_d))^T$ for any $d \geq 0$ (Definition-Proposition 5.2(2)), and ω_Y is homogeneous of degree r with respect to the grading of S by Definition-Proposition 5.2(3) and Lemma 2.12, where $r > 0$ is the degree of ω. Thus this $\mathbb{G}_m$-action is the conical $\mathbb{C}^*$-action on Y descended from the conical $\mathbb{C}^*$-action on X in Question 1.2.

By Theorem 5.6, if $\dim Y > 0$ and

$$\dim R_{d,\chi} = \dim R_{d,-\chi}$$

for any $d \geq 0$ and any character χ of a maximal torus of G_R containing the image of T, then the degree valuation of the grading of S is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$. Thus the equalities of Question 1.3 imply the conclusion of Question 1.2.

Sketch of the proof. Except for Question 1.5(1), all our answers go through ρ (Theorem 1.6), and Question 1.5(1) uses the K-polystability of (E, Δ_E) and no metric.

(1) Theorem 1.6 follows from Theorems 2.16(2), 2.17, and 3.1. Theorem 3.1 constructs ρ from an action of $\mathrm{SU}(2)$ on $M = X_{\mathrm{reg}} \backslash \{o\}$ (Proposition 3.5) integrating the rotations IV, JV, and KV of the radial vector field V of a Ricci-flat Kähler cone metric g (Construction-Propositions 3.3 and 3.7), once ω is parallel and the rotations are complete (Theorems 3.11 and 3.20).
(2) In Theorem 1.8, G_R is reductive by Propositions 4.4 and 4.7, and Part (3) follows from Theorem 1.6 and Lemma 4.8.
(3) Theorem 1.9 follows from Theorem 1.6 for a maximal torus A of G_R containing the image of T and Theorem 5.6, which passes the equalities

$$\dim R_{d,\chi} = \dim R_{d,-\chi} \quad \text{for any } d \geq 0 \text{ and any character } \chi \text{ of } A$$

from X to Y.

The main step is (1) for $n \geq 2$: the link N of g (Set-up 3.14) may be non-compact, and logarithmic cutoff functions on N (Proposition 3.15) show that ω is parallel (Proposition 3.16) and that the flows of bounded Killing fields on N extend to isometries of the metric completion of N (Lemma 3.18), which preserve N by an intrinsic characterization of the smooth points of the completion (Proposition 3.19). Theorem 1.6 only asserts that ρ exists: ρ depends on g, which is constructed for T.

In (a)–(d), we explain where direct or known arguments fail and what replaces them.
(a) Parallelism. Integrating the Bochner identity $\Delta_g|\omega|_g^2 = 2|\nabla\omega|_g^2$ of Lemma 3.9 over the link, as for $n = 1$ (Theorem 3.11), does not work directly for $n \geq 2$: the identity has to be integrated against cutoff functions on N, while $|\omega|_g^2$ is a priori integrable on N but not known to be bounded. The integrability is where we use that ω extends to a resolution of X. Proposition 3.16 tests $\log(1+|\omega|_g^2)$ instead of $|\omega|_g^2$ against the cutoff functions η_j of Proposition 3.15, which satisfy

$$\lim_{j\to+\infty} \int_N \Phi(|d\eta_j|_{g_N}) d\operatorname{vol}_{g_N} = 0 \quad \text{for } \Phi(a) = a^2 \log^2(e+a)$$

and the metric g_N on N induced by g. The factor $\log^2$ in Φ matches the exponential integrability of $\log(1+|\omega|_g^2)$. The vanishing of Δ_E for $n \geq 2$ (Lemma 3.12), the cutoff functions, the argument for parallelism, and the flows of bounded Killing fields from almost every starting point in (b) are adapted from [Liu26a, Proposition 15.3, Theorems 4.4(2) and 5.1, and Lemmas 6.4 and 6.5].
(b) Completeness. In [NO25, Proof of Theorem 6.16], the completeness for isolated singularities follows from the compactness of the link, while N is non-compact if $X_{\mathrm{sing}}\backslash\{o\} \neq \emptyset$. Once ω is parallel, the rotations restrict to bounded Killing fields on N (Theorem 3.20). Lemma 3.18 extends the flow of a bounded Killing field on N to isometries of the metric completion L of N. To compare distances, we follow the integral curves from all points of a path and keep them in the support of a cutoff function by the supremum estimate on rectangles of Lemma 3.17, which needs the factor $\log^2$ in Φ, as finite Dirichlet energy does not bound the supremum in dimension two. By Proposition 3.19(1), the points of N are exactly the points of L with a neighborhood isometric to an open subset of a smooth Riemannian manifold, so isometries of L preserve N, and the Killing field is complete.
(c) Reductivity. By the K-polystability of (E, Δ_E) and [ABHLX19, Theorem 1.3], the identity component H of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive (Proposition 4.4), and G_R is the identity component of the stabilizer of ω in H. Matsushima's criterion (see [Arz05, Theorem 2.1]) implies that the stabilizer of a vector in a representation of a reductive group is reductive if the orbit of the vector is affine, for example closed, but $H \cdot \omega$ is not closed, since $\lambda(\mathbb{G}_m) \subset H$ acts on the closed 2-forms on X_{reg} homogeneous of degree r by non-zero scalars. Instead, Lemma 4.6, which we deduce from [Nam11, Theorem (3.1)], shows that $H \cdot \omega$ is the principal open subset of the space of such forms defined by a polynomial. Thus $H \cdot \omega$ is affine, and G_R is reductive (Proposition 4.7). The maps τ_d in Theorem 1.8(3) are constructed degree by degree, without compatibility with the multiplication of R.
(d) Hamiltonian reduction. In [XZ26b, Proof of Theorem 7.10], an involution of the ambient symplectic vector space descends to Y and reverses the weights of the torus acting on Y, and no such involution of X is available in general. Instead, the components of the moment map μ form a regular sequence (Proposition 5.3), so the Koszul resolution of $R/R\mu(\mathfrak{t})$ passes the equalities $\dim R_{d,\chi} = \dim R_{d,-\chi}$ from A acting on R to the image A_S of A acting on S, and A_S is a maximal torus of the identity component G_S of $\mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ (Proposition 5.5). Thus by Theorem 2.16(1), which uses that Y is a symplectic singularity, a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$ is $\mathrm{wt}_{c\lambda_S+\eta}$, where λ_S is the $\mathbb{G}_m$-action on Y defined by the grading of S, $c \in \mathbb{R}$, and $\eta \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$ (Definition 2.4). The reflection $c\lambda_S + \eta \to c\lambda_S - \eta$ preserves the normalized volume (Theorem 5.6(2)), so $\eta = 0$ by the uniqueness of the minimizer (Theorem 2.3).

Structure of the paper. In Section 2 we recall preliminary material and fix notation. In particular, we introduce minimizing gradings of conical symplectic singularities, and recall from [Liu26a, Theorems 2.1 and 2.2] that ω has degree 1 or 2 with respect to any minimizing grading, and that the log Fano cone defined by a minimizing grading and the log Fano pair on the quotient of $X\backslash\{o\}$ by the $\mathbb{G}_m$-action defined by the grading are K-polystable.

In Section 3 we construct ρ from an action of $\mathrm{SU}(2)$ on $X_{\mathrm{reg}}\backslash\{o\}$, and we obtain the action of $\mathrm{SU}(2)$ from a Ricci-flat Kähler cone metric, treating surfaces and higher dimensions separately.

In Section 4 we show that G_R is a reductive group with Lie algebra isomorphic to R_r, using the K-polystability of the log Fano pair on the quotient of $X\backslash\{o\}$ by the $\mathbb{G}_m$-action defined by the grading, and we construct Chevalley involutions.

In Section 5 we introduce moment maps and Hamiltonian reductions, study the flatness of moment maps and the maximal tori of Hamiltonian reductions, and show that the equalities of Question 1.3 for a maximal torus of G_R containing T imply that the degree valuation of the grading of the Hamiltonian reduction of X by T is a minimizer of the normalized volume, if the reduction, with the descended symplectic form, is a symplectic singularity of positive dimension.

In Section 6 we prove Theorems 1.6, 1.8, and 1.9.

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2. Preliminaries

2.1. Pairs and normalized volumes.

Definition 2.1. A *pair* (X, B) consists of a normal quasi-projective variety X and an $\mathbb{R}$-divisor $B \geq 0$ on X such that $K_X + B$ is $\mathbb{R}$-Cartier. If we do not require that $B \geq 0$, then we say that (X, B) is a *sub-pair*. Let $h : X' \to X$ be any log resolution of (X, B) and E a prime divisor on X', and write $K_{X'} + B' := h^*(K_X + B)$. Then we define

$$a(E, X, B) := 1 - \mathrm{mult}_E B'$$

as the *log discrepancy* of E with respect to (X, B). For any divisorial valuation $v = c\,\mathrm{ord}_E$ for some real number $c > 0$, we denote by $a(v, X, B) := ca(E, X, B)$.

For any (sub-)pair (X, B), we say that (X, B) is *(sub-)lc* if $a(E, X, B) \geq 0$ for any prime divisor E over X, and *(sub-)klt* if $a(E, X, B) > 0$ for any prime divisor E over X. An *lc place* of (X, B) is a prime divisor E over X such that $a(E, X, B) = 0$. An *lc center* of (X, B) is the center of an lc place of (X, B) on X.

Definition 2.2. Let X be a normal variety such that $(X, 0)$ is klt, and let $x \in X$ be a closed point. Denote by $\mathrm{Val}_{X,x}$ the set of real valuations of the function field of X which are trivial on $\mathbb{C}$ and centered at x. For $v \in \mathrm{Val}_{X,x}$, let $\mathrm{vol}_{X\ni x}(v)$ be the volume of v [LLX18, Definition 2.1], and let $a(v, X) \in (0, +\infty]$ be the log discrepancy of v [LLX18, Definition 2.2], which equals $a(v, X, 0)$ when v is divisorial. For a prime divisor E over X, we write $a(E, X) := a(E, X, 0)$. The *normalized volume* of v is

$$\widehat{\mathrm{vol}}_{X\ni x}(v) := a(v, X)^{\dim X}\,\mathrm{vol}_{X\ni x}(v)$$

if $a(v, X) < +\infty$, and $\widehat{\mathrm{vol}}_{X\ni x}(v) := +\infty$ otherwise [LLX18, Definition 2.3]. A *minimizer* of $\widehat{\mathrm{vol}}_{X\ni x}$ is a valuation $v \in \mathrm{Val}_{X,x}$ with $\widehat{\mathrm{vol}}_{X\ni x}(v) \leq \widehat{\mathrm{vol}}_{X\ni x}(w)$ for any $w \in \mathrm{Val}_{X,x}$.

Theorem 2.3 ([Blu16, Main Theorem] and [XZ20, Theorem 1.1])**.** *Let X be a normal variety such that $(X, 0)$ is klt, and let $x \in X$ be a closed point. Then there exists a minimizer v of $\widehat{\mathrm{vol}}_{X\ni x}$, and any minimizer of $\widehat{\mathrm{vol}}_{X\ni x}$ is equal to cv for some real number $c > 0$.*

Definition 2.4. For an algebraic torus T, we denote by $X^*(T)$ and $X_*(T)$ the character and cocharacter lattices of T, and by $\langle -,-\rangle$ the natural pairing between them, extended $\mathbb{R}$-linearly to $X_*(T)\otimes_{\mathbb{Z}}\mathbb{R}$. Let T be an algebraic torus acting faithfully on a normal affine variety $X=\operatorname{Spec} R$, and let $R=\bigoplus_{\alpha\in X^*(T)}R_\alpha$ be the decomposition into character spaces, where $R_\alpha:=\{f\in R\mid t^*f=\alpha(t)f \text{ for any } t\in T\}$. We say that the action is *good* if there exists a unique closed point $o\in X$ which is contained in the closure of any T-orbit [LLX18, Definition 3.1]. In this case we call o the *vertex* of the action. The *Reeb cone* of a good action is

$$\mathfrak{t}^+_{\mathbb{R}}:=\{\xi\in X_*(T)\otimes_{\mathbb{Z}}\mathbb{R}\mid \langle\alpha,\xi\rangle>0 \text{ for any } \alpha\neq 0 \text{ with } R_\alpha\neq 0\},$$

whose elements are called *Reeb vectors*, and for $\xi\in\mathfrak{t}^+_{\mathbb{R}}$, the *weight valuation* $\mathrm{wt}_\xi\in\mathrm{Val}_{X,o}$ is defined by

$$\mathrm{wt}_\xi(f):=\min\{\langle\alpha,\xi\rangle\mid f_\alpha\neq 0\}$$

for any $0\neq f=\sum_\alpha f_\alpha$ with $f_\alpha\in R_\alpha$ [LLX18, Definition 3.2 and Lemma 3.3]. For $\lambda\in X_*(T)\cap\mathfrak{t}^+_{\mathbb{R}}$, the *grading of R defined by λ* is $R=\bigoplus_{d\geq 0}R_d$ with $R_d:=\bigoplus_{\langle\alpha,\lambda\rangle=d}R_\alpha$. If moreover $(X,0)$ is klt, then (X,ξ) is called a *log Fano cone* [LWX18, Definition 2.12]. We refer to [LWX18, Definition 2.23] for the K-semistability of log Fano cones, and to [LWX18, Definition 2.28] for the K-polystability of log Fano pairs. For the K-polystability of log Fano cones, we use the following notion of product. Let T_0 be an algebraic torus acting faithfully on X with $T\subset T_0$. A T_0-equivariant test configuration $(\mathcal{X},\eta)$ of (X,ξ) [LWX18, Definition 2.14] is called a *product* if there exist a T_0-equivariant isomorphism $\kappa:\mathcal{X}\to X\times\mathbb{C}$ commuting with the projections to $\mathbb{C}$ and an action σ of $\mathbb{G}_m$ on X commuting with T_0 such that κ maps η to $\eta_\sigma-t\partial_t$, where η_σ is the vector field on X generating σ and t is the coordinate on $\mathbb{C}$. A product test configuration for the torus T_0 in the sense of [LWX18, Definition 2.14] is a product for which σ factors through T_0. We say that (X,ξ) is *K-polystable for the torus T_0* if (X,ξ) is K-semistable and any T_0-equivariant special test configuration of (X,ξ) with Futaki invariant 0 is a product. We say that (X,ξ) is *K-polystable* if (X,ξ) is K-polystable for the torus T.

Definition 2.5. Let $X=\operatorname{Spec} R$ be a normal affine variety, and let $R=\bigoplus_{d\geq 0}R_d$ be a grading of the $\mathbb{C}$-algebra R with $R_0=\mathbb{C}$. The $\mathbb{G}_m$-action λ *defined by the grading* is given by $\lambda_t^*f=t^df$ for any $t\in\mathbb{G}_m$, $d\geq 0$, and $f\in R_d$. The *vertex* of the grading is the closed point $o\in X$ defined by the ideal $\bigoplus_{d>0}R_d$. The *degree valuation* v_λ of the grading is the valuation of the function field of X with $v_\lambda(f)=\min\{d\mid f_d\neq 0\}$ for any $0\neq f=\sum_d f_d\in R$ with $f_d\in R_d$. We say that the grading is *primitive* if the greatest common divisor of the integers d with $R_d\neq 0$ is 1.

Lemma 2.6. *Let $X=\operatorname{Spec} R$ be a normal affine variety, let $R=\bigoplus_{d\geq 0}R_d$ be a primitive grading with $R_0=\mathbb{C}$, and let λ be the $\mathbb{G}_m$-action defined by the grading. Then λ is a faithful good action on X, and the vertex of λ is the vertex of the grading. Moreover, the Reeb cone of λ contains the identity cocharacter $\mathrm{id}_{\mathbb{G}_m}\in X_*(\mathbb{G}_m)$, and $\mathrm{wt}_{\mathrm{id}_{\mathbb{G}_m}}=v_\lambda$.*

Proof. Let m be the greatest common divisor of the integers d with $R_d\neq 0$. An element $t\in\mathbb{G}_m$ acts trivially on X if and only if $t^d=1$ for any d with $R_d\neq 0$, that is, $t^m=1$. Since the grading is primitive, $m=1$, and thus λ is faithful.

Let o be the vertex of the grading, and let $x\in X$ be a closed point. The orbit map $\mathbb{G}_m\to X$, $t\to\lambda_t(x)$, corresponds to the homomorphism $R\to\mathbb{C}[t,t^{-1}]$ sending $f\in R_d$ to $f(x)t^d$, whose image is contained in $\mathbb{C}[t]$. Since setting $t=0$ sends $\bigoplus_{d>0}R_d$ to 0, the orbit map extends to a morphism $\mathbb{A}^1\to X$ sending 0 to o. Thus o is contained in the closure of the λ-orbit of x. Since $f(\lambda_t(o))=t^df(o)=0$ for any $d>0$ and $f\in R_d$, o is fixed by λ. Hence a closed point contained in the closure of any λ-orbit is contained in the closure of the orbit $\{o\}$, and thus it is equal to o. Therefore λ is good with vertex o.

By the pullback convention in Definition 2.4, R_d is the character space of the character $\chi_d:t\to t^d$ of $\mathbb{G}_m$, and $\langle\chi_d,\mathrm{id}_{\mathbb{G}_m}\rangle=d$. Comparing Definitions 2.4 and 2.5 term by term, we deduce that $\mathrm{id}_{\mathbb{G}_m}$ is contained in the Reeb cone of λ and $\mathrm{wt}_{\mathrm{id}_{\mathbb{G}_m}}=v_\lambda$. □

Lemma 2.7. *Let T be an algebraic torus with a good action on a normal affine variety X such that $(X,0)$ is klt, and let Ω be a nowhere vanishing algebraic $(\dim X)$-form on X_{reg} such that $t^*\Omega = \alpha(t)\Omega$ for some $\alpha \in X^*(T)$ and any $t \in T$. Then $a(\mathrm{wt}_\xi, X) = \langle \alpha, \xi \rangle$ for any $\xi \in \mathfrak{t}^+_{\mathbb{R}}$.*

Proof. By [XZ26b, Lemma 4.2], $\langle \alpha, \xi \rangle$ is the log discrepancy of wt_ξ with respect to Ω in the sense of [XZ26b, Definition 3.6]. Since Ω is nowhere vanishing on X_{reg}, Ω generates the canonical sheaf of X, and thus the log discrepancy of wt_ξ with respect to Ω is equal to $a(\mathrm{wt}_\xi, X)$ by [XZ26b, Definition 3.6] and [KM98, Definition 2.22], where the discrepancy of a prime divisor E over X defined in [KM98, Definition 2.22] is $a(E,X)-1$ in the notation of Definition 2.1. Here $a(\mathrm{wt}_\xi, X)$ is defined since $(X,0)$ is klt [LLX18, Definition 2.2]. □

2.2. **Conical symplectic singularities.**

Definition 2.8. Let $X = \operatorname{Spec} R$ be a normal affine variety, and let ω be a closed non-degenerate algebraic 2-form on X_{reg}. We say that (X,ω) is a *symplectic singularity* if for any resolution $h : X' \to X$, the form $h^*\omega$ on $h^{-1}(X_{\mathrm{reg}})$ extends to a regular 2-form on X' [Bea99, Definition 1.1]. For a grading $R = \bigoplus_{d\geq 0} R_d$ with $R_0 = \mathbb{C}$ and an integer r, we say that ω is *homogeneous of degree r* if $\lambda_t^*\omega = t^r\omega$ for any $t \in \mathbb{G}_m$, where λ is the $\mathbb{G}_m$-action defined by the grading, and we call r the *degree* of ω. We say that a symplectic singularity (X,ω) is *conical* if $\dim X > 0$ and there exists a grading $R = \bigoplus_{d\geq 0} R_d$ with $R_0 = \mathbb{C}$ such that ω is homogeneous of positive degree (cf. [XZ26b, Definition 6.1]).

Lemma 2.9. *Let (X,ω) be a symplectic singularity of dimension $2n$. Then ω^n is a nowhere vanishing algebraic $2n$-form on X_{reg}, the divisor K_X is Cartier, and X has canonical singularities. In particular, $(X,0)$ is klt, and X has rational singularities and is Cohen–Macaulay.*

Proof. Since ω is non-degenerate on X_{reg}, ω^n is a nowhere vanishing algebraic $2n$-form on X_{reg}. By [Bea99, Proposition 1.3], X has rational Gorenstein singularities, so K_X is Cartier. Thus X has canonical singularities by [KM98, Corollary 5.24], $(X,0)$ is klt by [KM98, Definition 2.34], and X is Cohen–Macaulay by [KM98, Theorem 5.10]. □

Set-up 2.10. Let X, R, o, n, λ, ω, and r be as follows.

(1) $R = \bigoplus_{d\geq 0} R_d$ is a finitely generated graded $\mathbb{C}$-algebra which is a normal domain, with $R_0 = \mathbb{C}$.
(2) $X = \operatorname{Spec} R$, λ is the $\mathbb{G}_m$-action defined by the grading, and $o \in X$ is the vertex of the grading.
(3) $\dim X = 2n$ for an integer $n \geq 1$.
(4) ω is a closed non-degenerate algebraic 2-form on X_{reg} which is homogeneous of degree r for a positive integer r.

Set-up 2.10 does not assume that (X,ω) is a symplectic singularity. Indeed, Set-up 2.10 is also applied to the Hamiltonian reduction (S,ω_Y) (Proposition 5.5(1)), and Set-up 5.4 assumes that S is a normal domain and that ω_Y induces the Poisson bracket of S, but not that $(\operatorname{Spec} S, \omega_Y)$ is a symplectic singularity.

Lemma 2.11. *Notation and conditions as in Set-up 2.10. For $f \in R$, let ζ_f be the vector field on X_{reg} with $\iota_{\zeta_f}\omega = -df$. Then the following hold.*

(1) For any $f,h \in R$, the regular function $\zeta_f(h)$ on X_{reg} extends to an element $\{f,h\} \in R$, and $\{-,-\}$ is a Poisson bracket on R.
(2) $\{R_d, R_e\} \subset R_{d+e-r}$ for any $d,e \geq 0$, where $R_m := 0$ for $m < 0$.
(3) R_r is a Lie algebra under $\{-,-\}$, and R_d is an R_r-module via the Poisson bracket for any $d \geq 0$.

Proof. (1) Since ω is non-degenerate, ζ_f is a regular vector field on X_{reg}, and thus $\zeta_f(h)$ is a regular function on X_{reg}. Since X is normal, $X \backslash X_{\mathrm{reg}}$ has codimension at least 2 in X, so any regular function on X_{reg} is contained in the localization of R at any prime ideal of height

one. Thus $H^0(X_{\mathrm{reg}}, \mathcal{O}_X) = R$ by [Har77, Proposition II.6.3A], and $\zeta_f(h)$ extends to an element $\{f,h\} \in R$. On X_{reg}, we have $\{f,h\} = \omega(\zeta_f, \zeta_h)$, so $\{-,-\}$ is antisymmetric. It is a derivation in the second variable since ζ_f is a vector field. Since $d\omega = 0$, [Mic08, Theorem 9.9(2)] implies that $L_{\zeta_f}\omega = d\iota_{\zeta_f}\omega = -d(df) = 0$ and $L_{\zeta_f}dh = d\iota_{\zeta_f}dh = d\{f,h\}$ on X_{reg}, and hence by [Mic08, Theorem 9.9(6)],

$$\iota_{[\zeta_f,\zeta_h]}\omega = L_{\zeta_f}\iota_{\zeta_h}\omega - \iota_{\zeta_h}L_{\zeta_f}\omega = -L_{\zeta_f}dh = -d\{f,h\}.$$

Since ω is non-degenerate, we obtain $[\zeta_f, \zeta_h] = \zeta_{\{f,h\}}$. Applying both sides to $k \in R$, we obtain $\{f,\{h,k\}\} - \{h,\{f,k\}\} = \{\{f,h\},k\}$ on X_{reg}, hence in R, which is the Jacobi identity. Thus $\{-,-\}$ is a Poisson bracket on R.

(2) Let $d, e \geq 0$, $f \in R_d$, and $h \in R_e$. For $t \in \mathbb{G}_m$, pulling back $\iota_{\zeta_f}\omega = -df$ by the automorphism λ_t of X_{reg} and using $\lambda_t^*\omega = t^r\omega$ and $\lambda_t^* f = t^d f$, we obtain $\lambda_t^*\zeta_f = t^{d-r}\zeta_f$. Thus

$$\lambda_t^*\{f,h\} = (\lambda_t^*\zeta_f)(\lambda_t^* h) = t^{d+e-r}\{f,h\} \quad \text{for any } t \in \mathbb{G}_m.$$

Comparing the homogeneous components of both sides, we obtain $\{f,h\} \in R_{d+e-r}$.

(3) By (2), $\{R_r, R_d\} \subset R_d$ for any $d \geq 0$. Thus (3) follows from the Jacobi identity in (1). □

Lemma 2.12. *Let $X = \operatorname{Spec} R$ be a normal affine variety, let ω be a closed non-degenerate algebraic 2-form on X_{reg}, and let $\{-,-\}$ be a Poisson bracket on R such that $\{f,h\} = \zeta_f(h)$ on X_{reg} for any $f, h \in R$, where ζ_f is the vector field on X_{reg} with $\iota_{\zeta_f}\omega = -df$. Let γ be an automorphism of X, and let $c \in \mathbb{C}^*$. Then $\gamma^*\{f,h\} = c\{\gamma^* f, \gamma^* h\}$ for any $f, h \in R$ if and only if $\gamma^*\omega = c^{-1}\omega$ on X_{reg}.*

Proof. For a closed non-degenerate algebraic 2-form ω' on X_{reg} and $f, h \in R$, let ζ'_f be the vector field on X_{reg} with $\iota_{\zeta'_f}\omega' = -df$, and let $\{f,h\}_{\omega'} := \zeta'_f(h)$, which is a regular function on X_{reg}. Pulling back $\iota_{\zeta_f}\omega = -df$ by γ, we obtain $\iota_{\gamma^*\zeta_f}\gamma^*\omega = -d(\gamma^* f)$, and thus

$$\gamma^*\{f,h\} = (\gamma^*\zeta_f)(\gamma^* h) = \{\gamma^* f, \gamma^* h\}_{\gamma^*\omega}$$

for any $f, h \in R$. Since $\{f,h\}_{c^{-1}\omega} = c\{f,h\}$ for any $f, h \in R$ and γ^* is bijective on R, the equality $\gamma^*\{f,h\} = c\{\gamma^* f, \gamma^* h\}$ holds for any $f, h \in R$ if and only if $\{f,h\}_{\gamma^*\omega} = \{f,h\}_{c^{-1}\omega}$ for any $f, h \in R$.

It remains to show that the restriction of $\{-,-\}_{\omega'}$ to R determines ω'. Let $x \in X_{\mathrm{reg}}$ be a closed point, and let ϕ_x be the isomorphism $v \to \iota_v\omega'$ from the tangent space of X at x to the cotangent space of X at x. Then

$$\{f,h\}_{\omega'}(x) = -dh(x)\big(\phi_x^{-1}(df(x))\big)$$

for any $f, h \in R$. Since the differentials at x of the elements of R span the cotangent space of X at x, the restriction of $\{-,-\}_{\omega'}$ to R determines ϕ_x^{-1}, and hence determines ω' at x. □

Definition 2.13. Notation and conditions as in Set-up 2.10. Denote by $\mathrm{Aut}_{\mathrm{gr}}(R)$ the group of $\mathbb{C}$-algebra automorphisms g of R with $g(R_d) = R_d$ for any $d \geq 0$, and by $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ the subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R)$ consisting of the automorphisms g with $g(\{f,h\}) = \{g(f), g(h)\}$ for any $f, h \in R$. For $g \in \mathrm{Aut}_{\mathrm{gr}}(R)$, we denote by γ_g the automorphism of X with $\gamma_g^* f = g(f)$ for any $f \in R$. Since $\gamma_g(X_{\mathrm{reg}}) = X_{\mathrm{reg}}$, we let $\mathrm{Aut}_{\mathrm{gr}}(R)$ act on the algebraic differential forms η on X_{reg} by $g \cdot \eta := \gamma_g^*\eta$. An *algebraic torus acting on R by graded Poisson automorphisms* is an algebraic torus T together with a group homomorphism from T to $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$, written $f \to t \cdot f$ for $t \in T$, such that R_d is a rational representation of T for any $d \geq 0$. We let T act on X so that $t^* f = t \cdot f$ for any $t \in T$ and $f \in R$. We denote by $\mathfrak{t} := \operatorname{Lie} T$, by $T_c \subset T$ the maximal compact subgroup, and by $a \cdot f$ the derivative at the identity of the T-action on $f \in R$ in the direction of $a \in \mathfrak{t}$. If the homomorphism is injective, then we identify T with the image and write $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$. For $d \geq 0$ and $\chi \in X^*(T)$, we write

$$R_{d,\chi} := \{f \in R_d \mid t \cdot f = \chi(t) f \text{ for any } t \in T\}.$$

2.3. **Minimizing gradings.**

Definition 2.14. Let (X,ω) be a symplectic singularity, where $X = \operatorname{Spec} R$. A *minimizing grading* of (X,ω) is a primitive grading $R = \bigoplus_{d\geq 0} R_d$ with $R_0 = \mathbb{C}$ such that ω is homogeneous of positive degree and the degree valuation of the grading is a minimizer of $\widehat{\operatorname{vol}}_{X\ni o}$, where o is the vertex of the grading.

The primitivity in Definition 2.14 ensures that the value group of the degree valuation of the grading is $\mathbb{Z}$.

Definition-Proposition 2.15. Notation and conditions as in Set-up 2.10, and assume that the grading is primitive. Let $E := \operatorname{Proj} R$. For a prime divisor P on E, let m_P be the order of the stabilizer in $\mathbb{G}_m$ of a general point of the preimage of P in $X\backslash\{o\}$, and let

$$\Delta_E := \sum_P \left(1 - \frac{1}{m_P}\right) P.$$

Then the following hold.

(1) The quotient morphism $X\backslash\{o\} \to E$ is a Seifert $\mathbb{G}_m$-bundle [Kol04, Definition 1], and Δ_E is the branch divisor of the Seifert $\mathbb{G}_m$-bundle [Kol04, Definition 39]. Moreover, for any homogeneous generators $f_1,\dots,f_k$ of the $\mathbb{C}$-algebra R and any closed point $y \in X\backslash\{o\}$, the stabilizer of y in $\mathbb{G}_m$ is the group of roots of unity whose order divides the greatest common divisor of the degrees of those f_j with $f_j(y) \neq 0$. In particular, the stabilizer of y in $\mathbb{G}_m$ is finite.

(2) If $(X,0)$ is klt, then E is the exceptional divisor of the birational morphism $h : X' \to X$ obtained by adding the zero section to the Seifert $\mathbb{G}_m$-bundle in (1) [XZ26b, 2.13], $v_\lambda = \operatorname{ord}_E$, $(K_{X'} + E)|_E = K_E + \Delta_E$, and (E,Δ_E) is a log Fano pair.

Proof. (1) For $d > 0$ and $0 \neq f \in R_d$, the preimage of the open subset $D_+(f) = \operatorname{Spec}(R_f)_0$ of E in $X\backslash\{o\}$ is $D(f) = \operatorname{Spec} R_f$, and $(R_f)_0$ is normal, being the ring of $\mathbb{G}_m$-invariants of the normal domain R_f. Thus E is normal, and since the open subsets $D(f)$ cover $X\backslash\{o\}$, the quotient morphism is affine. Let $f_1,\dots,f_k$ be homogeneous generators of the $\mathbb{C}$-algebra R, and let $y \in X\backslash\{o\}$ be a closed point. Since $f_1,\dots,f_k$ define a closed embedding of X into $\mathbb{A}^k$ and $f_j(\lambda_t(y)) = t^{\deg f_j} f_j(y)$ for any j and $t \in \mathbb{G}_m$, we have $\lambda_t(y) = y$ if and only if $t^{\deg f_j} = 1$ for any j with $f_j(y) \neq 0$. Thus the stabilizer of y in $\mathbb{G}_m$ is the group of roots of unity whose order divides the greatest common divisor of the degrees of those f_j with $f_j(y) \neq 0$. Since $y \neq o$ and the ideal $\bigoplus_{d>0} R_d$ is generated by the f_j of positive degree, we have $f_j(y) \neq 0$ for some j with $\deg f_j > 0$, and hence the stabilizer of y in $\mathbb{G}_m$ is finite. Therefore any $\mathbb{G}_m$-orbit in $X\backslash\{o\}$ is closed in $X\backslash\{o\}$, since the boundary of a $\mathbb{G}_m$-orbit consists of fixed points. Since any fiber of $D(f) \to D_+(f)$ contains exactly one closed orbit, the reduced fiber of the quotient morphism over any closed point of E is one orbit. Since the grading is primitive, the greatest common divisor of the degrees of $f_1,\dots,f_k$ is 1, and thus the stabilizer of y in $\mathbb{G}_m$ is trivial if $f_1(y)\cdots f_k(y) \neq 0$. Thus the quotient morphism $X\backslash\{o\} \to E$ satisfies the conditions of [Kol04, Definitions 1 and 3]. Moreover, since the image of $\{f_1\cdots f_k = 0\}\backslash\{o\}$ in E is a proper closed subset, $m_P = 1$ for all but finitely many P. Since m_P is the multiplicity of the fiber over a general point of P in the sense of [Kol04, Definition 3], m_P is the denominator of the coefficient of P in [Kol04, Theorem 7(3)], that is, the ramification index along P of the orbifold in [Kol04, Definition 39]. Thus Δ_E is the branch divisor of the Seifert $\mathbb{G}_m$-bundle.

(2) By Lemma 2.6, λ is a faithful good action on X with vertex o, the Reeb cone of λ contains $\mathrm{id}_{\mathbb{G}_m}$, and $\mathrm{wt}_{\mathrm{id}_{\mathbb{G}_m}} = v_\lambda$. Since $(X,0)$ is klt, $(X,\mathrm{id}_{\mathbb{G}_m})$ is a log Fano cone for the action λ. Thus by [XZ26b, 2.13], adding the zero section to the Seifert $\mathbb{G}_m$-bundle in (1) yields a birational morphism $h : X' \to X$ with exceptional divisor E such that $v_\lambda = \operatorname{ord}_E$, and (E,Δ') is a log Fano pair, where Δ' is the different defined by $(K_{X'} + E)|_E = K_E + \Delta'$. It suffices to show that $\Delta' = \Delta_E$. Let P be a prime divisor on E, let $m := m_P$, and let $\mu_m \subset \mathbb{G}_m$ be the group of

m-th roots of unity. Since X is normal, the singular locus of X is a λ-stable closed subset of codimension at least 2, so $X\backslash\{o\}$ is smooth along the fiber over a general point of P. Thus by [Kol04, 14, 25, and Claim 26], analytically locally near a general point of P, X' is isomorphic to $(\mathbb{A}^2/\mu_m)\times\mathbb{A}^{2n-2}$, where μ_m acts on $\mathbb{A}^2$ with coordinates (u,y) by $\epsilon\cdot(u,y)=(\epsilon u,\epsilon^{-b}y)$ for an integer b coprime to m, and E is the image of $\{u=0\}\times\mathbb{A}^{2n-2}$. Thus the coefficient of P in Δ' is $1-\frac{1}{m}$ by [Kol+92, Proposition 16.6] (case (16.6.3)), and hence $\Delta'=\Delta_E$. □

The grading in Question 1.3 is a minimizing grading of (X,ω): it is primitive since the value group of ord_E is $\mathbb{Z}$, ω is homogeneous of positive degree with respect to the grading, and the degree valuation ord_E of the grading is a minimizer of $\widehat{\mathrm{vol}}_{X\ni o}$, where o is the vertex of the grading. By Theorem 2.16(1), the grading in Questions 1.3–1.5 exists. The proof of Theorem 2.16(1) uses that [XZ26b, Theorem 6.2 and Proposition 6.5] yield an equivariant algebraic isomorphism between X and a K-semistable log Fano cone whose Reeb vector may a priori be irrational, and that the Reeb vector is a positive multiple of a cocharacter since any minimizer of the normalized volume is divisorial [Liu26a, Theorem 2.1].

Theorem 2.16. *Let (X,ω) be a symplectic singularity of dimension $2n\geq 2$, where $X=\operatorname{Spec}R$. Then the following hold.*

*(1) Let T_X be an algebraic torus with a good action on X such that $t^*\omega\in\mathbb{C}^*\omega$ for any $t\in T_X$. Then there exist an algebraic torus T with a good action on X such that $T_X\subset T$ and $t^*\omega\in\mathbb{C}^*\omega$ for any $t\in T$, and a cocharacter $\lambda\in X_*(T)\cap\mathfrak{t}^+_{\mathbb{R}}$, such that the grading of R defined by λ is a minimizing grading of (X,ω). In particular, if (X,ω) is conical, then (X,ω) admits a minimizing grading.**

(2) Let $R=\bigoplus_{d\geq 0}R_d$ be a minimizing grading of (X,ω), let λ be the $\mathbb{G}_m$-action defined by the grading, let r be the degree of ω, and let $E=\operatorname{Proj}R$ be as in Definition-Proposition 2.15. Then $v_\lambda=\mathrm{ord}_E$, $a(E,X)=nr$, and $r\in\{1,2\}$.

Proof. (1) Let o be the vertex of the action of T_X. By [XZ26b, Theorem 6.2] (see [XZ26b, Definition 2.12] for the terminology), there exist an algebraic torus T_1 with a good action on a normal affine variety X_0 with vertex o_0, a vector ξ_0 in the Reeb cone of T_1 such that (X_0,ξ_0) is a K-semistable log Fano cone, and a closed non-degenerate algebraic 2-form ω_0 on $(X_0)_{\mathrm{reg}}$, such that (X,ω) at o and (X_0,ω_0) at o_0 are formally isomorphic, and $t^*\omega_0\in\mathbb{C}^*\omega_0$ for any $t\in T_0$, where $T_0\subset T_1$ is the smallest subtorus with $\xi_0\in X_*(T_0)\otimes_{\mathbb{Z}}\mathbb{R}$ (the torus $\langle\xi_0\rangle$ in [XZ26b, Theorem 6.2]). Since T_1 acts faithfully on X_0, so does T_0. Since the closure of any T_1-orbit contains o_0 and any T_1-invariant element of the coordinate ring $\mathbb{C}[X_0]$ of X_0 is constant on the closure of any T_1-orbit, the ring of T_1-invariants of $\mathbb{C}[X_0]$ is $\mathbb{C}$. Thus there exist generators $f_1,\dots,f_k$ of the $\mathbb{C}$-algebra $\mathbb{C}[X_0]$ and non-zero characters $\alpha_1,\dots,\alpha_k$ of T_1 such that f_i is contained in the character space of α_i. Since $\langle\alpha_i,\xi_0\rangle>0$ for any i and $X_*(T_0)\otimes_{\mathbb{Z}}\mathbb{Q}$ is dense in $X_*(T_0)\otimes_{\mathbb{Z}}\mathbb{R}$, there exists $\eta\in X_*(T_0)$ with $\langle\alpha_i,\eta\rangle>0$ for any i. For any closed point $x\in X_0$ and $s\in\mathbb{G}_m$, we have $f_i(\eta(s)\cdot x)=s^{\langle\alpha_i,\eta\rangle}f_i(x)$, so the morphism $\mathbb{G}_m\to X_0$, $s\to\eta(s)\cdot x$, extends to a morphism $\mathbb{A}^1\to X_0$ sending 0 to a common zero of $f_1,\dots,f_k$. Since $f_1,\dots,f_k$ generate the $\mathbb{C}$-algebra $\mathbb{C}[X_0]$, we have $\mathbb{C}[X_0]=\mathbb{C}+(f_1,\dots,f_k)$, and since $f_1,\dots,f_k$ have a common zero, the ideal $(f_1,\dots,f_k)$ is the maximal ideal of a closed point $p\in X_0$. Thus p is the only common zero of $f_1,\dots,f_k$, so p is contained in the closure of any T_0-orbit, hence in the closure of any T_1-orbit. Since the ideal $(f_1,\dots,f_k)$ is T_1-stable, p is fixed by T_1. Since o_0 is contained in the closure of the T_1-orbit $\{p\}$, we have $o_0=p$. Thus o_0 is contained in the closure of any T_0-orbit, and since o_0 is fixed by T_0, the action of T_0 on X_0 is good with vertex o_0 in the sense of Definition 2.4. Moreover, any non-zero character β of T_0 whose character space in $\mathbb{C}[X_0]$ is non-zero is the restriction of a non-zero character α of T_1 whose character space is non-zero, and $\langle\beta,\xi_0\rangle=\langle\alpha,\xi_0\rangle>0$. Hence ξ_0 is contained in the Reeb cone of T_0.

Since $t^*\omega\in\mathbb{C}^*\omega$ for any $t\in T_X$ and $t^*\omega_0\in\mathbb{C}^*\omega_0$ for any $t\in T_0$, the actions of T_X on X and of T_0 on X_0 are good and dilating with respect to ω and ω_0 respectively, in the sense of [XZ26b, Definitions 2.12 and 6.1].

By [XZ26b, Proposition 6.5], there exist an algebraic torus $T \supset T_X$ with a good action on X which is dilating with respect to ω, an action of T on X_0 extending the action of T_0, and a T-equivariant isomorphism $g : X \to X_0$ with $g(o) = o_0$. In particular, T acts faithfully on X, and $t^*\omega \in \mathbb{C}^*\omega$ for any $t \in T$. We regard T_0 as a subtorus of T, and let $\xi \in X_*(T) \otimes_{\mathbb{Z}} \mathbb{R}$ be the image of ξ_0. Since the action of T_X is good, o is the unique closed point contained in the closure of any T_X-orbit. Since any element of T_X maps the closure of any T_X-orbit onto itself, o is fixed by T_X. Let $t \in T$. Since $T_X \subset T$ and T is commutative, $t \cdot o$ is fixed by T_X, and thus o, which is contained in the closure of the T_X-orbit $\{t \cdot o\}$, is equal to $t \cdot o$. Hence o is fixed by T. Since any T-orbit contains a T_X-orbit, o is contained in the closure of any T-orbit, and any closed point contained in the closure of any T-orbit is contained in the closure of the T-orbit $\{o\}$, that is, equal to o. Therefore the action of T on X is good with vertex o in the sense of Definition 2.4.

We show that $\xi \in \mathfrak{t}^+_{\mathbb{R}}$. Let $\alpha \in X^*(T)$ be a non-zero character with $R_\alpha \neq 0$. If $\alpha|_{T_0}$ is trivial, then R_α is contained in the ring of T_0-invariants of R, which is $\mathbb{C}$ [LLX18, paragraph after Definition 3.1] since the action of T_0 on X_0 is good and g is T-equivariant. This contradicts $R_\alpha \neq 0$, since $R_\alpha \cap \mathbb{C} = 0$ for $\alpha \neq 0$. Thus $\alpha|_{T_0}$ is a non-zero character of T_0 whose character space in R contains $R_\alpha \neq 0$, and $\langle \alpha, \xi \rangle = \langle \alpha|_{T_0}, \xi_0 \rangle > 0$ since ξ_0 is contained in the Reeb cone of T_0.

Since (X_0, ξ_0) is K-semistable, wt_{ξ_0} is a minimizer of $\widehat{\mathrm{vol}}_{X_0 \ni o_0}$ by [LWX18, Theorem 2.5] (see also [LX17, Theorem 1.3]). Since g is T-equivariant and $\langle \alpha, \xi \rangle = \langle \alpha|_{T_0}, \xi_0 \rangle$ for any $\alpha \in X^*(T)$, we have $\mathrm{wt}_\xi(g^*f) = \mathrm{wt}_{\xi_0}(f)$ for any $0 \neq f \in \mathbb{C}[X_0]$. Since g is an isomorphism with $g(o) = o_0$, it preserves log discrepancies and volumes of valuations. Thus wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{X \ni o}$, and any minimizer of $\widehat{\mathrm{vol}}_{X \ni o}$ is a positive multiple of wt_ξ by Theorem 2.3.

By [Liu26a, Theorem 2.1], $\mathrm{wt}_\xi = c\,\mathrm{ord}_F$ for a real number $c > 0$ and a prime divisor F over X with $a(F, X) \in \{n, 2n\}$. For any $\alpha \in X^*(T)$ and $0 \neq f \in R_\alpha$, we have

$$\langle \alpha, \xi \rangle = \mathrm{wt}_\xi(f) = c\,\mathrm{ord}_F(f) \in c\mathbb{Z}.$$

The characters α with $R_\alpha \neq 0$ generate $X^*(T)$: otherwise, they generate a proper sublattice M' of $X^*(T)$, and $\bigcap_{\beta \in M'} \mathrm{Ker}\,\beta$ is a non-trivial diagonalizable subgroup of T [Mil17, Theorem 12.9] which acts trivially on R, contradicting the faithfulness of the action of T. Thus $\lambda := c^{-1}\xi$ is contained in $\mathrm{Hom}(X^*(T), \mathbb{Z}) = X_*(T)$, and $\lambda \in \mathfrak{t}^+_{\mathbb{R}}$ since $c > 0$.

Let $R = \bigoplus_{d \geq 0} R_d$ be the grading of R defined by λ. Since $\langle \alpha, \lambda \rangle > 0$ for any $\alpha \neq 0$ with $R_\alpha \neq 0$, R_0 is the ring of T-invariants of R, which is $\mathbb{C}$ since the action of T is good [LLX18, paragraph after Definition 3.1]. By the pullback convention in Definition 2.4, the $\mathbb{G}_m$-action defined by the grading is the composition of λ with the action of T on X. Since o is fixed by T, any element of R_α with $\alpha \neq 0$ vanishes at o. Thus the maximal ideal $\bigoplus_{d>0} R_d$ defines o, that is, o is the vertex of the grading. By Definitions 2.4 and 2.5, the degree valuation of the grading is

$$v_\lambda = \mathrm{wt}_\lambda = c^{-1}\,\mathrm{wt}_\xi = \mathrm{ord}_F\,.$$

Since the value group of v_λ is generated by the integers d with $R_d \neq 0$ and the value group of ord_F is $\mathbb{Z}$, the grading is primitive. Since $\widehat{\mathrm{vol}}_{X \ni o}(c^{-1}v) = \widehat{\mathrm{vol}}_{X \ni o}(v)$ for any $v \in \mathrm{Val}_{X,o}$ [Li15, Section 2.1, paragraph after Remark 2.2], v_λ is a minimizer of $\widehat{\mathrm{vol}}_{X \ni o}$.

Since $t^*\omega \in \mathbb{C}^*\omega$ for any $t \in T$, there exists $\chi \in X^*(T)$ with $t^*\omega = \chi(t)\omega$ for any $t \in T$, and thus ω is homogeneous of degree $r := \langle \chi, \lambda \rangle$ with respect to the grading. Lemma 2.7 applied to $\Omega = \omega^n$ and $\alpha = n\chi$ implies

$$a(F, X) = a(v_\lambda, X) = \langle n\chi, \lambda \rangle = nr.$$

Since $(X, 0)$ is klt (Lemma 2.9), $a(F, X) > 0$, and thus $r > 0$. Therefore the grading of R defined by λ is a minimizing grading of (X, ω).

Finally, assume that (X, ω) is conical, let $R = \bigoplus_{d \geq 0} R_d$ be a grading as in Definition 2.8, and let m be the greatest common divisor of the integers d with $R_d \neq 0$. The group of m-th roots of unity acts trivially on X through the $\mathbb{G}_m$-action defined by the grading, hence on ω, so m divides the degree of ω. Dividing the degrees by m, we obtain a primitive grading with degree-0

part $\mathbb{C}$ for which ω is homogeneous of positive degree. By Lemma 2.6, the $\mathbb{G}_m$-action defined by the primitive grading is faithful and good, and the image T_X of the action satisfies $t^*\omega \in \mathbb{C}^*\omega$ for any $t \in T_X$. Applying the first part to T_X, we obtain a minimizing grading of (X, ω).

(2) Since X is a normal affine variety of dimension $2n$, the grading is primitive with $R_0 = \mathbb{C}$, and ω is homogeneous of degree $r > 0$, the conditions of Set-up 2.10 hold. Since $(X, 0)$ is klt by Lemma 2.9, Definition-Proposition 2.15(2) implies $v_\lambda = \mathrm{ord}_E$, so the value group of v_λ is $\mathbb{Z}$. Since v_λ is a minimizer of $\widehat{\mathrm{vol}}_{X\ni o}$, [Liu26a, Theorem 2.1] implies that $v_\lambda = c\,\mathrm{ord}_F$ for a real number $c > 0$ and a prime divisor F over X with $a(F, X) \in \{n, 2n\}$. Comparing the value groups of v_λ and $c\,\mathrm{ord}_F$, we obtain $c = 1$, so $\mathrm{ord}_E = \mathrm{ord}_F$ and $a(E, X) = a(F, X) \in \{n, 2n\}$. By Lemma 2.6, λ is a good action on X whose Reeb cone contains $\mathrm{id}_{\mathbb{G}_m}$, and $\mathrm{wt}_{\mathrm{id}_{\mathbb{G}_m}} = v_\lambda$. Since $\lambda_t^*\omega^n = t^{nr}\omega^n$ for any $t \in \mathbb{G}_m$, Lemma 2.7 applied to $\Omega = \omega^n$ (Lemma 2.9) implies

$$a(E, X) = a(v_\lambda, X) = a(\mathrm{wt}_{\mathrm{id}_{\mathbb{G}_m}}, X) = nr.$$

Thus $nr \in \{n, 2n\}$, that is, $r \in \{1, 2\}$. □

For a minimizing grading of a conical symplectic singularity (X, ω), since the degree valuation v_λ is a minimizer of $\widehat{\mathrm{vol}}_{X\ni o}$ and normalized volumes are invariant under rescaling [Li15, Section 2.1, paragraph after Remark 2.2], the log Fano cone $(X, \frac{2}{r}\lambda)$ is K-semistable by [LWX18, Theorem 2.5] (see also [LX17, Theorem 1.1(2)]), where λ is the $\mathbb{G}_m$-action defined by the grading, regarded as a cocharacter of the torus $\lambda(\mathbb{G}_m)$ (Lemma 2.6), and r is the degree of ω. We take the Reeb vector of the cone to be $\frac{2}{r}\lambda$ rather than λ, since by Theorem 2.16(2) and Lemma 2.7, the log discrepancy of $\mathrm{wt}_{(2/r)\lambda}$ is $\frac{2}{r} \cdot nr = 2n = \dim X$, which is the normalization of the Reeb vector of a Ricci-flat Kähler cone metric in [Li21, (75) and the sentences before and after it] used in Construction-Proposition 3.7. The following theorem shows that both the cone and (E, Δ_E) are K-polystable.

Theorem 2.17 ([Liu26a, Theorem 2.2])**.** *Let (X, ω) be a conical symplectic singularity, where $X = \operatorname{Spec} R$, let $R = \bigoplus_{d\ge 0} R_d$ be a minimizing grading of (X, ω), let λ be the $\mathbb{G}_m$-action defined by the grading, let r be the degree of ω, and let $E = \operatorname{Proj} R$ and Δ_E be as in Definition-Proposition 2.15. Then the log Fano cone $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$, and (E, Δ_E) is a K-polystable log Fano pair.*

3. Quaternionic structures

In this section, we prove Theorem 3.1.

Theorem 3.1. *Notation and conditions as in Set-up 2.10, and let $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ be an algebraic torus. Assume that the grading $R = \bigoplus_{d\ge 0} R_d$ is primitive, that (X, ω) is a symplectic singularity, and that the log Fano cone $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$. Then there exists a ring automorphism $\rho : R \to R$ satisfying the following.*

(1) $\rho(zf) = \bar{z}\rho(f)$, $\rho(R_d) = R_d$, and $\rho(\{f, h\}) = \{\rho(f), \rho(h)\}$ for any $z \in \mathbb{C}$, $d \ge 0$, and $f, h \in R$.
(2) $\rho(t \cdot f) = t \cdot \rho(f)$ for any $t \in T_c$ and $f \in R$.
(3) $\rho^2(f) = \exp(2\pi i d/r) f$ for any $d \ge 0$ and $f \in R_d$.

In particular, $\rho(R_{d,\chi}) = R_{d,-\chi}$ for any $d \ge 0$ and $\chi \in X^*(T)$.

The automorphism ρ in Theorem 1.6 is constructed by Theorem 3.1, and Proposition 3.5 constructs ρ from an action of $\mathrm{SU}(2)$ on $X_{\mathrm{reg}}\backslash\{o\}$ which rotates the complex structure of $X_{\mathrm{reg}}\backslash\{o\}$. The action of $\mathrm{SU}(2)$ is constructed from a Ricci-flat Kähler cone metric on $X_{\mathrm{reg}}\backslash\{o\}$, which exists by the K-polystability of the cone and [Li21, Theorem 2.9] (Construction-Proposition 3.7). We construct the action of $\mathrm{SU}(2)$ in Subsection 3.2 when X is a surface and in Subsections 3.3, 3.4, and 3.5 when X is of dimension at least 4, and we prove Theorem 3.1 at the end of this section.

3.1. **Quaternionic structures from actions of** SU(2)**.** In this subsection, we construct ρ from an action of SU(2) on $X_{\mathrm{reg}}\backslash\{o\}$ (Proposition 3.5).

Set-up 3.2. Notation and conditions as in Set-up 2.10. Let T, M, and I be as follows.

(1) The grading $R = \bigoplus_{d\geq 0} R_d$ is primitive.
(2) $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ is an algebraic torus.
(3) $M := X_{\mathrm{reg}}\backslash\{o\}$, and I is the complex structure of M.

The metric g used later is given by [Li21, Theorem 2.9] (Construction-Proposition 3.7) and is not constructed from ω. The following construction obtains a second complex structure from g and $\mathrm{Re}\,\omega$ by a polar decomposition, whose positive part is in general not a constant multiple of the identity (see Example 3.8), and it keeps ω itself, so that ρ preserves the Poisson bracket $\{-,-\}$ defined by ω.

Construction-Proposition 3.3. Let (M, g, I) be a Kähler manifold, and let ω be a holomorphic symplectic form on M. Let D be the endomorphism of the real tangent bundle TM with $g(Dv, w) = \mathrm{Re}\,\omega(v, w)$ for any tangent vectors v, w, let $P := (-D^2)^{1/2}$ be the positive square root, and let $J := P^{-1}D$ and $K := IJ$. Then the following hold.

(1) $I^2 = J^2 = K^2 = -\mathrm{id}_{TM}$ and $IJ = -JI = K$.
(2) J and K are orthogonal with respect to g, and P commutes with I, J, and K.
(3) $\omega(v, w) = g(PJv, w) + ig(PKv, w)$ for any tangent vectors v, w.
(4) If $\nabla\omega = 0$, where ∇ is the Levi-Civita connection of g, then D, P, J, and K are parallel.

Proof. Statements (1)–(3) are pointwise linear algebra. Let $\alpha := \mathrm{Re}\,\omega$ and $\beta := \mathrm{Im}\,\omega$. Since ω is of type $(2, 0)$, we have $\omega(Iv, w) = i\omega(v, w) = \omega(v, Iw)$, hence

$$\alpha(Iv, w) = -\beta(v, w) = \alpha(v, Iw)$$

for any tangent vectors v, w. Since I is skew-adjoint with respect to g, we get

$$g(DIv, w) = g(Dv, Iw) = -g(IDv, w),$$

that is, $DI = -ID$. Since α is alternating, D is skew-adjoint. If $Dv = 0$, then $\alpha(v, w) = 0$ and $\beta(v, w) = -\alpha(v, Iw) = 0$ for any w, so $v = 0$ because ω is non-degenerate. Thus $-D^2$ is self-adjoint and positive definite. Hence P is well defined, smooth, self-adjoint, and positive definite, and P commutes with any endomorphism commuting with $-D^2$, in particular with D and I.

(1) We have $J^2 = P^{-2}D^2 = -\mathrm{id}_{TM}$ and $IJ = P^{-1}ID = -P^{-1}DI = -JI$. Thus

$$K^2 = IJIJ = -I^2J^2 = -\mathrm{id}_{TM}\,.$$

(2) Since P is self-adjoint and commutes with D, and D is skew-adjoint, the adjoint of J is $-DP^{-1}$, and

$$(-DP^{-1})J = -D^2P^{-2} = \mathrm{id}_{TM}\,.$$

Thus J is orthogonal, and so is $K = IJ$. Since P commutes with I and D, it commutes with $J = P^{-1}D$ and $K = IJ$. In particular, $D = PJ$ is the polar decomposition of D.

(3) We have $\alpha(v, w) = g(Dv, w) = g(PJv, w)$ and

$$\beta(v, w) = -\alpha(Iv, w) = -g(DIv, w) = g(IDv, w) = g(IPJv, w) = g(PKv, w).$$

(4) Assume that $\nabla\omega = 0$. Since $\nabla g = 0$ and $\nabla\alpha = 0$, we have $\nabla D = 0$. Parallel transport along a path is a linear isometry which intertwines the values of $-D^2$ at the start and at the end of the path, hence intertwines the values of P by the uniqueness of the positive square root. Thus P is parallel. Since $\nabla I = 0$, the endomorphisms $J = P^{-1}D$ and $K = IJ$ are parallel. □

We use the notions of a Kähler cone metric, the radius function, and the radial vector field as in [Li21, Definition 2.1], and the notion of a Ricci-flat Kähler cone metric as in [Li21, Definition 2.2], where $Y\backslash\{o\}$ is replaced by M. Indeed, [Li21, Section 2.1, first paragraph] assumes that the singularity is isolated, while the weak Ricci-flat Kähler cone potentials in the paragraph before

[Li21, Theorem 2.9] are considered on the regular part of a singularity which is not necessarily isolated.

Lemma 3.4. *Notation and conditions as in Set-up 3.2. Let g be a Kähler cone metric on M with complex structure I, radius function s, and radial vector field $V = s\partial_s$, and let ∇ be the Levi-Civita connection of g. Then $L_V g = 2g$, $\nabla V = \mathrm{id}_{TM}$, $|V|_g = s$, and the gradient of s with respect to g is V/s.*

Proof. By [Li21, Definition 2.1], $M = (0, +\infty) \times \{s = 1\}$ and $g = ds^2 + s^2 g_1$, where g_1 is the Riemannian metric on $\{s = 1\}$ induced by g, and the flow of V at time t is $(s, x) \to (e^t s, x)$. Thus $L_V g = 2g$ and $|V|_g = s$. Moreover, $g(V/s, \cdot) = ds$, that is, the gradient of s is V/s, so $g(V, \cdot) = d(s^2/2)$ is closed. Since ∇ is torsion-free and $\nabla g = 0$, for vector fields ξ, ξ' on M we have

$$2g(\nabla_\xi V, \xi') = (L_V g)(\xi, \xi') + d\big(g(V, \cdot)\big)(\xi, \xi') = 2g(\xi, \xi').$$

Hence $\nabla V = \mathrm{id}_{TM}$. □

The formula for ρ^2 in Proposition 3.5(3) below follows from the identity $\exp(\pi\mathbf{j}) = -1 = \exp(\pi\mathbf{i})$ in SU(2), so Proposition 3.5 assumes an action of SU(2).

Proposition 3.5. *Notation and conditions as in Set-up 3.2. Let g be a Kähler cone metric on M with complex structure I, radius function $s > 0$, and radial vector field $V = s\partial_s$, such that the flow of V at any time $t \in \mathbb{R}$ is $\lambda_{\exp(2t/r)}$, the form ω is parallel with respect to g, and T_c acts on M by isometries of g. Let D, P, J, and K be as in Construction-Proposition 3.3. Assume that there exists a smooth action of* SU(2) *on M such that for some $\epsilon \in \{1, -1\}$, the one-parameter subgroups* $\exp(t\mathbf{i})$, $\exp(t\mathbf{j})$, *and* $\exp(t\mathbf{k})$ *of* SU(2) *act by the flows of ϵIV, ϵJV, and ϵKV, respectively, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit quaternions. Let $b : M \to M$ be the flow of JV at time $\pi/2$. Then the following hold.*

(1) b is antiholomorphic, and $b^*\omega = \bar{\omega}$.
(2) There exists a unique map $\rho : R \to R$ with $\rho(f)|_M = \overline{f \circ b}$ for any $f \in R$, and ρ is a ring automorphism with $\rho(zf) = \bar{z}\rho(f)$, $\rho(R_d) = R_d$, and $\rho(\{f, h\}) = \{\rho(f), \rho(h)\}$ for any $z \in \mathbb{C}$, $d \geq 0$, and $f, h \in R$.
(3) $\rho(t \cdot f) = t \cdot \rho(f)$ for any $t \in T_c$ and $f \in R$, and $\rho^2(f) = \exp(2\pi i d/r) f$ for any $d \geq 0$ and $f \in R_d$.
(4) $\rho(R_{d,\chi}) = R_{d,-\chi}$ and $\dim R_{d,\chi} = \dim R_{d,-\chi}$ for any $d \geq 0$ and $\chi \in X^*(T)$.

Proof. The proof is long, so we divide it into several steps. Let ∇ be the Levi-Civita connection of g, and let Fl_t^{JV} be the flow of JV at time $t \in \mathbb{R}$. By assumption, Fl_t^{JV} is the action of $\exp(\epsilon t\mathbf{j})$, so Fl_t^{JV} is defined on M for any t, and $b = \mathrm{Fl}_{\pi/2}^{JV}$.

Step 1. In this step we prove (1), and we show that $b \circ \lambda_a = \lambda_a \circ b$ for any real number $a > 0$.

By Lemma 3.4, $\nabla V = \mathrm{id}_{TM}$ and $L_V g = 2g$. By Construction-Proposition 3.3(4), D, P, J, and K are parallel, and $\nabla I = 0$ since g is Kähler. Thus $\nabla_Y(JV) = JY$ for any vector field Y. Since J is skew-adjoint, JV is a Killing vector field, so Fl_t^{JV} is an isometry of g for any t.

Let Q be a parallel endomorphism of TM. For any vector field W on M, since ∇ is torsion-free, we have

$$\begin{aligned}(L_{JV}Q)W &= [JV, QW] - Q[JV, W] \\ &= \nabla_{JV}(QW) - JQW - Q\nabla_{JV}W + QJW = (QJ - JQ)W.\end{aligned}$$

Thus $L_{JV}Q = QJ - JQ$. Applying the last identity to $Q \in \{P, J, I, K\}$ and using (1) and (2) of Construction-Proposition 3.3, we get

$$L_{JV}P = 0, \quad L_{JV}J = 0, \quad L_{JV}I = 2K, \quad \text{and} \quad L_{JV}K = -2I.$$

Since

$$\frac{d}{dt}(\mathrm{Fl}_t^{JV})^* Q' = (\mathrm{Fl}_t^{JV})^*(L_{JV}Q')$$

for any endomorphism Q' of TM, at any point of M the pair $((\mathrm{Fl}_t^{JV})^*I, (\mathrm{Fl}_t^{JV})^*K)$ solves a linear ordinary differential equation with constant coefficients whose value at $t=0$ is (I,K). Hence

$$(\mathrm{Fl}_t^{JV})^*I = \cos(2t)I + \sin(2t)K, \quad (\mathrm{Fl}_t^{JV})^*K = \cos(2t)K - \sin(2t)I,$$
$$(\mathrm{Fl}_t^{JV})^*J = J, \quad (\mathrm{Fl}_t^{JV})^*P = P,$$

and $(\mathrm{Fl}_t^{JV})^*g = g$. For $t=\pi/2$, we get $b^*I=-I$, $b^*K=-K$, $b^*J=J$, $b^*P=P$, and $b^*g=g$. Thus b is antiholomorphic. Moreover, by Construction-Proposition 3.3(3), for any tangent vectors v, w,

$$(b^*\omega)(v,w) = g(PJdb(v), db(w)) + ig(PKdb(v), db(w)) = g(PJv,w) - ig(PKv,w) = \overline{\omega(v,w)}.$$

This proves (1).

Since $\nabla_V(JV) = JV$ and $\nabla_{JV}V = JV$, we have $[V, JV] = 0$, so the flows of V and JV commute. Since the flow of V at time t is $\lambda_{\exp(2t/r)}$, we get $b \circ \lambda_a = \lambda_a \circ b$ for any real number $a > 0$.

Step 2. In this step we construct $\rho(f) \in R_d$ for $f \in R_d$, and we show that ρ is an antilinear ring homomorphism with $\rho(f)|_M = \overline{f \circ b}$ for any $f \in R$.

Fix $d \geq 0$ and $f \in R_d$, and let $h := \overline{f \circ b}$ on M. Since b is antiholomorphic, h is holomorphic. By **Step 1**, for any real number $a > 0$ and $x \in M$,

$$h(\lambda_a(x)) = \overline{f(\lambda_a(b(x)))} = \overline{a^d f(b(x))} = a^d h(x).$$

Since X is normal, the complex analytic space X^{an} associated with X is normal [Gro03, Exposé XII, Proposition 2.1]. The set $X_{\mathrm{sing}} \cup \{o\}$ is an analytic subset of X^{an} of codimension at least 2, because X is normal and $\dim X \geq 2$. Thus h extends to a holomorphic function H on X^{an} [Dem12, Chapter II, Corollary (7.8)], which is unique since M is dense in X^{an}. By continuity, $H(\lambda_a(x)) = a^d H(x)$ for any real number $a > 0$ and $x \in X$. For a fixed $x \in X$, the function $z \to H(\lambda_z(x)) - z^d H(x)$ is holomorphic on $\mathbb{C}^*$ and vanishes on the positive real numbers, so it vanishes on $\mathbb{C}^*$ by the identity theorem. Hence

$$H(\lambda_z(x)) = z^d H(x) \quad \text{for any } z \in \mathbb{C}^* \text{ and } x \in X.$$

Let $x_1, \ldots, x_k$ be homogeneous generators of the $\mathbb{C}$-algebra R of degrees $w_1, \ldots, w_k \geq 1$. They define a closed embedding of X into $\mathbb{C}^k$ which maps o to 0, and λ_z is the restriction of $(Z_1, \ldots, Z_k) \to (z^{w_1}Z_1, \ldots, z^{w_k}Z_k)$. The germ of H at o is the restriction of a convergent power series $F(Z) = \sum_\nu c_\nu Z^\nu$ [Dem12, Chapter II, (2.6), (4.28), and Definition (5.2)]. Let Δ be a polydisc centered at 0 on which F converges and such that $H = F$ on $X \cap \Delta$. For $Z \in \Delta$, let

$$F_d(Z) := \frac{1}{2\pi}\int_0^{2\pi} e^{-id\theta} F(e^{iw_1\theta}Z_1, \ldots, e^{iw_k\theta}Z_k)d\theta = \sum_{w_1\nu_1 + \cdots + w_k\nu_k = d} c_\nu Z^\nu,$$

where the second equality follows from termwise integration. Since $w_j \geq 1$ for any j, the last sum is finite, so F_d is a polynomial. For $x \in X \cap \Delta$, the point $\lambda_{e^{i\theta}}(x)$ is contained in $X \cap \Delta$, and $F(\lambda_{e^{i\theta}}(x)) = H(\lambda_{e^{i\theta}}(x)) = e^{id\theta}H(x)$, so $F_d(x) = H(x)$. Let $p \in R_d$ be the restriction of F_d to X. Then $H = p$ on $X \cap \Delta$. For any $x \in X$, the point $\lambda_a(x)$ tends to o as $a \to 0^+$, because $w_j \geq 1$ for any j. Choosing $a > 0$ with $\lambda_a(x) \in \Delta$, we get

$$a^d H(x) = H(\lambda_a(x)) = p(\lambda_a(x)) = a^d p(x),$$

so $H(x) = p(x)$. Thus $H = p \in R_d$.

We let $\rho(f) := H$ for $f \in R_d$, and extend ρ additively to R. Then $\rho(f)|_M = \overline{f \circ b}$ for any $f \in R$, and $\rho(R_d) \subset R_d$ for any $d \geq 0$. Since M is a non-empty Zariski open subset of the integral scheme X, two elements of R which coincide on M are equal. Thus ρ is the unique map with $\rho(f)|_M = \overline{f \circ b}$ for any $f \in R$, and the identities $\rho(zf + f') = \bar{z}\rho(f) + \rho(f')$, $\rho(ff') = \rho(f)\rho(f')$, and $\rho(1) = 1$ for $z \in \mathbb{C}$ and $f, f' \in R$ follow from the corresponding identities on M.

Step 3. In this step we show that $\rho(\{f,h\}) = \{\rho(f), \rho(h)\}$ for any $f, h \in R$.

For $f \in R$, under the isomorphism $TM \to T^{1,0}M$, $v \to \frac{1}{2}(v - \sqrt{-1}Iv)$, we also write ζ_f for the real vector field on M corresponding to the vector field ζ_f of Lemma 2.11. Since ω is non-degenerate and of type $(2,0)$, ζ_f is the unique real vector field on M with $\iota_{\zeta_f}\omega = -df$, where ω is regarded as a complex valued 2-form on TM. Since $dh(Iv) = \sqrt{-1}dh(v)$ for any holomorphic function h and any real tangent vector v, we have $v(h) = \frac{1}{2}(v - \sqrt{-1}Iv)(h)$, and thus $\zeta_f(h) = \{f, h\}$ on M for any $h \in R$ by Lemma 2.11(1).

Fix $f \in R$, and let $Y := b^*\zeta_f$, that is, $Y(x) = (d_x b)^{-1}(\zeta_f(b(x)))$ for $x \in M$. Then $\iota_Y(b^*\omega) = b^*(\iota_{\zeta_f}\omega) = -d(f \circ b)$. Since Y is real and $b^*\omega = \bar{\omega}$ by (1), we get

$$\iota_Y\omega = \overline{\iota_Y\bar{\omega}} = -d\overline{f \circ b} = -d\rho(f) \quad \text{on } M.$$

Thus $Y = \zeta_{\rho(f)}$. Hence for any $h \in R$, on M we have

$$\{\rho(f), \rho(h)\} = \zeta_{\rho(f)}\big(\overline{h \circ b}\big) = \overline{Y(h \circ b)} = \overline{(\zeta_f h) \circ b} = \overline{\{f, h\} \circ b} = \rho(\{f, h\}).$$

Since $\{\rho(f), \rho(h)\}$ and $\rho(\{f, h\})$ are elements of R which coincide on M, they are equal.

Step 4. In this step we prove the formula for ρ^2, and we conclude the proof of (2).

By assumption, $b^2 = \mathrm{Fl}_\pi^{JV}$ is the action of $\exp(\epsilon\pi\mathbf{j}) = -1 = \exp(\epsilon\pi\mathbf{i})$, which is the flow of IV at time π. Since λ is a holomorphic action, the vector field generating $t \to \lambda_{e^{it}}$ is I applied to the vector field generating $t \to \lambda_{e^t}$. Since V generates $t \to \lambda_{\exp(2t/r)}$, the flow of IV at time t is $\lambda_{\exp(2it/r)}$. Thus $b^2 = \lambda_{\exp(2\pi i/r)}$. For $d \geq 0$, $f \in R_d$, and $x \in M$, we get

$$\rho^2(f)(x) = \overline{\rho(f)(b(x))} = f(b^2(x)) = f(\lambda_{\exp(2\pi i/r)}(x)) = \exp(2\pi id/r)f(x).$$

Thus $\rho^2(f) = \exp(2\pi id/r)f$, since both sides are elements of R which coincide on M.

Hence ρ^2 is bijective on R_d for any $d \geq 0$. Since $\rho(R_d) \subset R_d$, we get $R_d = \rho^2(R_d) \subset \rho(R_d) \subset R_d$, so $\rho(R_d) = R_d$, and ρ is injective on R_d. Thus ρ is bijective, and (2) follows from **Steps 2 and 3**.

Step 5. In this step we prove the rest of (3) and (4).

Let $t \in T$. Since t preserves $\{-,-\}$, Lemma 2.12 with $c = 1$ implies that $t^*\omega = \omega$ on X_{reg}. Now assume that $t \in T_c$. Then t is holomorphic on M, t preserves V because t commutes with λ, and t is an isometry of g by assumption. Thus t preserves D, hence preserves P by the uniqueness of the positive square root, and preserves $J = P^{-1}D$. Hence t preserves JV, so $t \circ b = b \circ t$. For any $f \in R$ and $x \in M$, we get

$$\rho(t \cdot f)(x) = \overline{f(t(b(x)))} = \overline{f(b(t(x)))} = (t \cdot \rho(f))(x).$$

Thus $\rho(t \cdot f) = t \cdot \rho(f)$. This proves (3).

Let $d \geq 0$, $\chi \in X^*(T)$, and $f \in R_{d,\chi}$. Since T_c is compact, $|\chi(t)| = 1$ for any $t \in T_c$. Thus

$$t \cdot \rho(f) = \rho(t \cdot f) = \rho(\chi(t)f) = \overline{\chi(t)}\rho(f) = \chi(t)^{-1}\rho(f) \quad \text{for any } t \in T_c.$$

Since R_d is a rational representation of T, we can write $\rho(f) = \sum_\psi f_\psi$ with $f_\psi \in R_{d,\psi}$, where ψ runs through $X^*(T)$. Then $(\psi(t) - \chi(t)^{-1})f_\psi = 0$ for any $t \in T_c$ and ψ. Since a character of $T \cong \mathbb{G}_m^{\dim T}$ is a Laurent monomial, distinct characters of T have distinct restrictions to $T_c \cong (S^1)^{\dim T}$. Thus $f_\psi = 0$ for $\psi \neq -\chi$, that is, $\rho(f) \in R_{d,-\chi}$. Hence $\rho(R_{d,\chi}) \subset R_{d,-\chi}$ for any d and χ. For $h \in R_{d,-\chi}$, the element $f' := \rho(\exp(-2\pi id/r)h)$ is contained in $R_{d,\chi}$, and $\rho(f') = \rho^2(\exp(-2\pi id/r)h) = h$ by (3). Thus $\rho(R_{d,\chi}) = R_{d,-\chi}$. Since ρ is an injective antilinear map, $\dim R_{d,\chi} = \dim R_{d,-\chi}$. This proves (4). □

3.2. **Ricci-flat cone metrics and the surface case.** In this subsection, we construct a Ricci-flat Kähler cone metric on M (Construction-Proposition 3.7), and we show that the conditions of Proposition 3.5 hold when $n = 1$ (Theorem 3.11).

We construct the metric to be invariant under the maximal compact subgroup of the torus generated by $\lambda(\mathbb{G}_m)$ and T, since Proposition 3.5 uses that T_c acts by isometries to show that ρ commutes with T_c.

Set-up 3.6. Notation and conditions as in Set-up 3.2, and let $dd^c := \frac{\sqrt{-1}}{2\pi}\partial\bar{\partial}$. Assume the following.

(1) (X, ω) is a symplectic singularity.
(2) The log Fano cone $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$ (Definition 2.4).

Later we use (1)–(4) of Construction-Proposition 3.7 only as follows: (1) for the flows of V and IV, (2) for the integrability of $|\omega|_g^2$ on $s^{-1}(1)$ (Proposition 3.16) and in Proposition 3.19(1), and (4) for the cutoff functions of Proposition 3.15. By s_0 in (3), $s^{-1}(1)$ is compact if X is a surface (Theorem 3.11(1)), and otherwise, by the bounds in (3), any subset of M where s and $1/s$ are bounded has compact closure in X (Propositions 3.16 and 3.19).

Construction-Proposition 3.7. Notation and conditions as in Set-up 3.6. Let T' be the image of the action of $\mathbb{G}_m \times T$ on R given by $(t, t_1) \cdot f = t^d(t_1 \cdot f)$ for $t \in \mathbb{G}_m$, $t_1 \in T$, $d \geq 0$, and $f \in R_d$, let $T'_c \subset T'$ be the maximal compact subgroup, and let $E = \operatorname{Proj} R$ and Δ_E be as in Definition-Proposition 2.15. Then T' is a torus, T'_c contains the images of T_c and $\lambda(S^1)$, where $S^1 \subset \mathbb{C}^* = \mathbb{G}_m(\mathbb{C})$ is the unit circle, and there exists a T'_c-invariant Ricci-flat Kähler cone metric g on M with complex structure I, radius function s, and radial vector field $V = s\partial_s$, satisfying the following.

(1) The flow of V at any time $t \in \mathbb{R}$ is $\lambda_{\exp(2t/r)}$, and the flow of IV at any time $t \in \mathbb{R}$ is $\lambda_{\exp(2it/r)}$.
(2) The function s^2 extends to a locally bounded plurisubharmonic function on X with value 0 at o, where a function on X is called *plurisubharmonic* if it is locally the restriction of a plurisubharmonic function [Dem12, Chapter I, Definition (5.1)] under a local embedding of X into some $\mathbb{C}^k$.
(3) There exist a proper continuous function $s_0 : X \to [0, +\infty)$ with $s_0^{-1}(0) = \{o\}$ and $s_0(\lambda_{\exp(2t/r)}(x)) = e^t s_0(x)$ for any $t \in \mathbb{R}$ and $x \in X$, and a real number $c \geq 1$, such that $c^{-1}s_0 \leq s \leq cs_0$ on M.
(4) The form $dd^c \log s^2$ on M is the pullback of a form Θ on the open subset $M/\mathbb{C}^*$ of E, and there exist a Kähler form ω_E on E, that is, locally the restriction of a Kähler form under a local embedding of E into some $\mathbb{C}^k$, and a bounded function u on E such that $\Theta = \omega_E + dd^c u$ on $M/\mathbb{C}^*$ and $\omega_E + dd^c u$ is a weak Kähler–Einstein metric on the log Fano pair (E, Δ_E), that is, a positive constant multiple of $\omega_E + dd^c u$ is a bounded weak solution of [HL20, (3.1)] with constant weight 1.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that T' is a torus whose maximal compact subgroup T'_c contains the images of T_c and $\lambda(S^1)$.

Since R is finitely generated, there exists an integer $N_0 \geq 1$ such that R is generated by $R_1, \ldots, R_{N_0}$. Since $\mathbb{G}_m \times T$ acts on R_d by a rational representation for any d, restricting the action to $R_1, \ldots, R_{N_0}$ defines a homomorphism of algebraic groups

$$\mathbb{G}_m \times T \to \prod_{1 \leq d \leq N_0} \mathrm{GL}(R_d),$$

and T' is identified with the image of the homomorphism, since an automorphism of R is determined by the images of $R_1, \ldots, R_{N_0}$. The image is closed [Mil17, Proposition 1.68], and it is a quotient of a torus, hence a torus [Mil17, Definition 12.14 and the paragraph after it]. The subgroup T'_c consists of the elements of T' at which any character of T' takes a value of modulus 1. The image of a compact subgroup of T' under a character is a compact subgroup of $\mathbb{C}^*$, hence is contained in S^1. Thus any compact subgroup of T' is contained in T'_c. In particular, T'_c contains the images of T_c and $\lambda(S^1)$.

Step 2. In this step we obtain a T'_c-invariant weak Ricci-flat Kähler cone potential from [Li21, Theorem 2.9].

By Lemma 2.9, $(X,0)$ is klt and ω^n is a nowhere vanishing algebraic $2n$-form on X_{reg}. By Lemma 2.6, λ is a faithful good action with vertex o, and the identity cocharacter is contained in the Reeb cone of λ. Since $\lambda_t^*\omega^n = t^{rn}\omega^n$ for $t \in \mathbb{G}_m$, Lemma 2.7 applied to the torus $\lambda(\mathbb{G}_m)$ and $\Omega = \omega^n$ implies

$$a(\mathrm{wt}_{(2/r)\lambda}, X) = \frac{2}{r} \cdot rn = 2n = \dim X.$$

Thus the Reeb vector $\frac{2}{r}\lambda$ is normalized in the sense of [Li21, (75) and the sentence after it].

By Set-up 3.6(2), $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$ (Definition 2.4), where only special test configurations are tested. A special test configuration is weakly special [LWX18, the sentence after Definition 2.16], and on a weakly special test configuration the Ding invariant equals the Futaki invariant [LWX18, Remark A.2]. Thus $(X, \frac{2}{r}\lambda)$ satisfies the condition of Ding-polystability [LWX18, Definition A.1] for $\lambda(\mathbb{G}_m)$-equivariant special test configurations, with products in the sense of Definition 2.4 for $T_0 = \lambda(\mathbb{G}_m)$, and by the last sentence of [Li21, Theorem 2.9], it suffices to test special test configurations there. The test configurations in [Li21, Theorem 2.9] are equivariant with respect to a fixed torus [Li21, Section 2.5, second paragraph], and we take the fixed torus to be T'. Since $\lambda(\mathbb{G}_m) \subset T'$, a T'-equivariant special test configuration is $\lambda(\mathbb{G}_m)$-equivariant. Thus by Set-up 3.6(2), the Futaki invariant of a T'-equivariant special test configuration is non-negative, and if the Futaki invariant is 0, then the test configuration is a product for $T_0 = \lambda(\mathbb{G}_m)$ in the sense of Definition 2.4. We show that such a test configuration is also a product for $T_0 = T'$.

Let $(\mathcal{X}, \eta)$ be a T'-equivariant special test configuration of $(X, \frac{2}{r}\lambda)$ whose Futaki invariant is 0. We identify $\mathcal{X} \times_{\mathbb{C}} \mathbb{C}^*$ with $X \times \mathbb{C}^*$ through the isomorphism in [LWX18, Definition 2.14(2)], which is induced by the action of $\mathbb{C}^*$ generated by η. Thus the action of $z \in \mathbb{C}^*$ generated by η is $(x,t) \to (x, z^{-1}t)$ on $X \times \mathbb{C}^*$, and T' acts on $X \times \mathbb{C}^*$ through the first factor [LWX18, Definition 2.14(3)]. Since $\mathcal{X}$ is flat over $\mathbb{C}$, the coordinate ring $\mathbb{C}[\mathcal{X}]$ is a $\mathbb{C}[t]$-subalgebra of $R[t,t^{-1}]$ stable under the actions of $\mathbb{C}^*$ and T'. For $p \in \mathbb{Z}$, let

$$\mathcal{F}^p := \{f \in R \mid t^{-p} f \in \mathbb{C}[\mathcal{X}]\}.$$

Since $\mathbb{C}[\mathcal{X}]$ is stable under the action of $\mathbb{C}^*$, we have $\mathbb{C}[\mathcal{X}] = \bigoplus_{p\in\mathbb{Z}} \mathcal{F}^p t^{-p}$, and $\mathcal{F}^p$ is T'-stable for any $p \in \mathbb{Z}$.

Let $\kappa : \mathcal{X} \to X \times \mathbb{C}$ and σ be as in Definition 2.4 for $T_0 = \lambda(\mathbb{G}_m)$, and denote by σ_z the action of $z \in \mathbb{C}^*$ on X. Since κ maps η to $\eta_\sigma - t\partial_t$, κ maps the action of $z \in \mathbb{C}^*$ generated by η to the automorphism $(x,t) \to (\sigma_z(x), z^{-1}t)$ of $X \times \mathbb{C}$. Let β be the automorphism of X with $\kappa(x,1) = (\beta(x),1)$ for any $x \in X$. Since (x,t) is the image of $(x,1)$ under the action of t^{-1} generated by η, we have

$$\kappa(x,t) = (\sigma_{t^{-1}}(\beta(x)), t) \quad \text{for any } x \in X \text{ and } t \in \mathbb{C}^*.$$

Since κ is $\lambda(\mathbb{G}_m)$-equivariant, β commutes with λ_a for any $a \in \mathbb{C}^*$. As σ_z also commutes with λ_a, there exists a cocharacter σ' of $G := \mathrm{Aut}_{\mathrm{gr}}(R)$ with $\gamma_{\sigma'(z)} = \beta^{-1} \circ \sigma_z \circ \beta$ for any $z \in \mathbb{C}^*$ (Definition 2.13). For $w \in \mathbb{Z}$ and a cocharacter μ of G, let

$$R_w(\sigma) := \{f \in R \mid f \circ \sigma_z = z^w f \text{ for any } z \in \mathbb{C}^*\}$$

and

$$R_w(\mu) := \{f \in R \mid \mu(z)(f) = z^w f \text{ for any } z \in \mathbb{C}^*\}.$$

Then $R = \bigoplus_{w\in\mathbb{Z}} R_w(\sigma)$, the map $f \to f \circ \beta$ is a bijection from $R_w(\sigma)$ onto $R_w(\sigma')$, and $\kappa^* f = t^{-w}(f \circ \beta)$ for any $f \in R_w(\sigma)$, where f is regarded as a function on $X \times \mathbb{C}$. Since $\kappa^* : R[t] \to \mathbb{C}[\mathcal{X}]$ is an isomorphism, we deduce that

$$\mathbb{C}[\mathcal{X}] = \bigoplus_{w\in\mathbb{Z}} R_w(\sigma')t^{-w}\mathbb{C}[t], \quad \text{that is,} \quad \mathcal{F}^p = \bigoplus_{w\geq p} R_w(\sigma') \text{ for any } p \in \mathbb{Z}.$$

By Definition-Lemma 4.1(1), G is a closed subgroup of $\prod_{1\leq d\leq N_0} \mathrm{GL}(R_d)$, and R_d is a rational representation of G for any $d \geq 0$. Moreover, G is smooth [Mil17, Theorem 3.23]. Let $P_G(\sigma')$

be the subgroup of G consisting of the elements h such that $\lim_{z\to 0}\sigma'(z)h\sigma'(z)^{-1}$ exists in G, let $U_G(\sigma')$ be the subgroup of $P_G(\sigma')$ consisting of the elements h such that the limit is id_R, and let $Z_G(\sigma')$ be the centralizer of $\sigma'(\mathbb{G}_m)$ in G [Mil17, Propositions 13.28 and 13.29]. For $h\in G$, $d\geq 0$, and $w,w'\in\mathbb{Z}$, the component of $\sigma'(z)h\sigma'(z)^{-1}$ from $R_d\cap R_w(\sigma')$ to $R_d\cap R_{w'}(\sigma')$ is $z^{w'-w}$ times the component of h. Since the diagonal blocks of an invertible block triangular matrix are invertible, we deduce that $h\in P_G(\sigma')$ if and only if $h(\mathcal{F}^p)\subset\mathcal{F}^p$ for any $p\in\mathbb{Z}$. Thus $T'\subset P_G(\sigma')$, as $\mathcal{F}^p$ is T'-stable for any p.

By [Mil17, Theorem 13.33(a)(b) and Example 14.13], $U_G(\sigma')$ is a normal unipotent subgroup of $P_G(\sigma')$, and $P_G(\sigma')=U_G(\sigma')\rtimes Z_G(\sigma')$. Let $q: P_G(\sigma')\to Z_G(\sigma')$ be the projection with kernel $U_G(\sigma')$. Since $T'\cap U_G(\sigma')$ is trivial [Mil17, Corollary 14.17], q restricts to an isomorphism from T' onto $q(T')$ [Mil17, Corollary 3.35]. Let $H:=q^{-1}(q(T'))$. Then $q|_H: H\to q(T')$ is surjective with kernel $U_G(\sigma')$, and $q(T')$ is a torus. Thus H is trigonalizable, and $U_G(\sigma')$ is the normal unipotent subgroup H_u of H with H/H_u of multiplicative type [Mil17, Proposition 16.2, Theorem 16.6, and Notation 16.7]. The inclusion $q(T')\to H$ and the inverse of the isomorphism $T'\to q(T')$ are sections of $q|_H$. Hence by [Mil17, Theorem 16.27(b)], there exists $v\in U_G(\sigma')$ with $vT'v^{-1}=q(T')\subset Z_G(\sigma')$.

Let $\sigma'':=v^{-1}\sigma' v$. For any $\tau\in T'$, $v\tau v^{-1}\in Z_G(\sigma')$ commutes with $\sigma'(z)$, so τ commutes with $\sigma''(z)$ for any $z\in\mathbb{C}^*$. Moreover, $R_w(\sigma'')=v^{-1}(R_w(\sigma'))$ for any $w\in\mathbb{Z}$, and $v^{-1}(\mathcal{F}^p)=\mathcal{F}^p$ for any $p\in\mathbb{Z}$ as $v\in P_G(\sigma')$. Thus

$$\mathcal{F}^p=\bigoplus_{w\geq p}R_w(\sigma'')\quad\text{for any } p\in\mathbb{Z},$$

so the homomorphism of $\mathbb{C}[t]$-algebras $R[t]\to\mathbb{C}[\mathcal{X}]$ mapping $f\in R_w(\sigma'')$ to $t^{-w}f$ for any $w\in\mathbb{Z}$ is an isomorphism. Since T' commutes with σ'', T' preserves $R_w(\sigma'')$ for any w, so the isomorphism $\mathcal{X}\to X\times\mathbb{C}$ induced by the homomorphism is T'-equivariant. A direct calculation shows that the isomorphism maps the action of $z\in\mathbb{C}^*$ generated by η to the automorphism $(x,t)\to(\gamma_{\sigma''(z)}(x),z^{-1}t)$ of $X\times\mathbb{C}$. As the action $z\to\gamma_{\sigma''(z)}$ of $\mathbb{G}_m$ on X commutes with T', $(\mathcal{X},\eta)$ is a product for $T_0=T'$ in the sense of Definition 2.4.

Thus $(X,\frac{2}{r}\lambda)$ satisfies the condition of Ding-polystability for T'-equivariant special test configurations, with products in the sense of Definition 2.4 for $T_0=T'$. Hence by [Li21, Theorem 2.9], $(X,\frac{2}{r}\lambda)$ admits a T'_c-invariant weak Ricci-flat Kähler cone potential in the sense of the paragraph before [Li21, Theorem 2.9], that is, a locally bounded function on X which is a smooth radius function on M and satisfies the Ricci-flat Kähler cone equation of [Li21, Definition 2.2] in the sense of pluripotential theory, that is, the square of the potential is a locally bounded plurisubharmonic function on X whose Monge–Ampère measure satisfies the equation.

Step 3. In this step we define g, s, and V, and we prove (1).

Let s be the restriction of the potential in **Step 2** to M, and let g be the Kähler cone metric on M with radius function s and radial vector field $V=s\partial_s$ [Li21, Definitions 2.1 and 2.2]. Then g is Ricci-flat, and g is T'_c-invariant because the radius function in [Li21, Definition 2.1] is invariant under the compact torus there, which is T'_c. Since the Reeb vector of g is $\frac{2}{r}\lambda$, that is, $IV=I(s\partial_s)$ is the real holomorphic vector field associated with $\frac{2}{r}\lambda$ [Li21, Section 2.1, first paragraph, and Definition 2.1(2)], the flow of IV at time t is $\lambda_{\exp(2it/r)}$. Since λ is a holomorphic action, the vector field generating $t\to\lambda_{e^{it}}$ is I applied to the vector field generating $t\to\lambda_{e^t}$. Since $V=-I(IV)$, the flow of V at time t is $\lambda_{\exp(2t/r)}$. This proves (1).

Step 4. In this step we prove (3).

Let $x_1,\dots,x_k$ be homogeneous generators of the $\mathbb{C}$-algebra R of degrees $w_1,\dots,w_k\geq 1$, and let

$$s_0:=\Big(\sum_{j=1}^k|x_j|^{2/w_j}\Big)^{r/4}.$$

Then s_0 is continuous, and $s_0^{-1}(0) = \{o\}$ since o is the only common zero of $x_1, \dots, x_k$. Since $x_1, \dots, x_k$ define a closed embedding of X into $\mathbb{C}^k$ and $\{s_0 \le C\}$ is bounded in $\mathbb{C}^k$ for any $C > 0$, the function s_0 is proper. Since $x_j(\lambda_a(x)) = a^{w_j} x_j(x)$, we have $s_0(\lambda_a(x)) = |a|^{r/2} s_0(x)$ for any $a \in \mathbb{C}^*$ and $x \in X$, in particular $s_0(\lambda_{\exp(2t/r)}(x)) = e^t s_0(x)$. Since $V = s\partial_s$, (1) implies $s(\lambda_{\exp(2t/r)}(x)) = e^t s(x)$ for any $t \in \mathbb{R}$ and $x \in M$. Thus s/s_0 is invariant under λ_a for any real number $a > 0$, and $\lambda_a(x) \in \{s_0 = 1\}$ for $x \in X\backslash\{o\}$ and $a = s_0(x)^{-2/r}$. Hence it suffices to bound s from above and below by positive constants on $\{s_0 = 1\} \cap M$, where $\{s_0 = 1\}$ is compact. Since s^2 is locally bounded on X by **Step 2**, s is bounded from above on $\{s_0 = 1\} \cap M$. The weak potential in [Li21, Theorem 2.9] is induced by a weak Kähler–Einstein potential on (E, Δ_E), which is bounded by [HL20, Corollary 3.6] (see [Li21, p. 35]). Namely, $s = s_1 e^{\psi/2}$ on M, where s_1 is a smooth radius function on $\mathbb{C}^k\backslash\{0\}$ with Reeb vector $\frac{2}{r}\lambda$ and ψ is the pullback to M of the bounded potential on (E, Δ_E) [Li21, p. 35]. Since s_1 is continuous and positive on the compact set $\{s_0 = 1\}$ and ψ is bounded, s is bounded from below by a positive constant on $\{s_0 = 1\} \cap M$. This proves (3).

Step 5. In this step we prove (2).

By **Step 2**, s^2 extends to a locally bounded plurisubharmonic function φ on X. It remains to show that $\varphi(o) = 0$. Fix $x \in M$, and let $\gamma : \mathbb{C} \to X$ be the holomorphic map with $\gamma(z) = \lambda_z(x)$ for $z \neq 0$ and $\gamma(0) = o$. In the coordinates $x_1, \dots, x_k$ of **Step 4**, we have $\gamma(z) = (z^{w_1} x_1(x), \dots, z^{w_k} x_k(x))$. By [Dem12, Chapter I, Theorem (5.11)] applied to a local plurisubharmonic extension of φ near o, the function $\varphi \circ \gamma$ is subharmonic near 0. Since $\gamma(z) \in M$ for $z \neq 0$, $\varphi = s^2 \ge 0$ on M, and φ is upper semicontinuous, the sub-mean value inequality and (3) imply

$$0 \le \varphi(o) \le \frac{1}{2\pi} \int_0^{2\pi} s(\lambda_{\epsilon e^{i\theta}}(x))^2 d\theta \le c^2 s_0(\lambda_\epsilon(x))^2 = c^2 \epsilon^r s_0(x)^2$$

for any real number $0 < \epsilon \ll 1$. Letting $\epsilon \to 0$, we get $\varphi(o) = 0$. This proves (2).

Step 6. In this step we prove (4).

Fix a positive integer ℓ such that the $\mathbb{C}$-algebra $\bigoplus_{j \ge 0} R_{j\ell}$ is generated by R_ℓ [Bou98, Chapter III, §1, Proposition 3], and let $f_1, \dots, f_m$ be a basis of R_ℓ consisting of eigenvectors of T', which exists since R_ℓ is a rational representation of T'. If $x \in X$ and $f_i(x) = 0$ for any i, then for any homogeneous element $h \in R$ of positive degree, h^ℓ is a polynomial in $f_1, \dots, f_m$ without constant term, so $h(x) = 0$, and hence $x = o$. Let

$$\tilde{s} := \Big(\sum_{i=1}^m |f_i|^2\Big)^{r/(4\ell)} \quad \text{on } X\backslash\{o\}.$$

Then $\tilde{s}$ is continuous and positive on $X\backslash\{o\}$, smooth on M, and T'_c-invariant, and $\tilde{s}(\lambda_{\exp(2t/r)}(x)) = e^t \tilde{s}(x)$ since $f_i(\lambda_a(x)) = a^\ell f_i(x)$. Since $(R[f^{-1}])_0 = ((\bigoplus_{j \ge 0} R_{j\ell})[f^{-1}])_0$ for any $0 \neq f \in R_\ell$, we have $E = \operatorname{Proj} \bigoplus_{j \ge 0} R_{j\ell}$, and the surjection $\mathbb{C}[Z_1, \dots, Z_m] \to \bigoplus_{j \ge 0} R_{j\ell}$, $Z_i \to f_i$, induces a closed embedding $E \to \mathbb{P}^{m-1}$. The morphism $X\backslash\{o\} \to \mathbb{P}^{m-1}$ given by $x \to [f_1(x) : \cdots : f_m(x)]$ is the composite of the quotient morphism $X\backslash\{o\} \to E$ and the closed embedding $E \to \mathbb{P}^{m-1}$. Thus $dd^c \log \tilde{s}^2 = \frac{r}{2\ell} dd^c \log \sum_i |f_i|^2$ is the pullback of the Kähler form $\omega_E := \frac{r}{2\ell} \omega_{FS}|_E$ on E, where ω_{FS} is the Fubini–Study form on $\mathbb{P}^{m-1}$ normalized so that $dd^c \log \sum_i |Z_i|^2$ is the pullback of ω_{FS}.

Let $u_M := \log(s^2/\tilde{s}^2)$ on M. Under the flow of V at time t, both s and $\tilde{s}$ are multiplied by e^t, and both are invariant under the flow of IV, which is contained in $\lambda(S^1) \subset T'_c$ by (1) and **Step 1**. Thus u_M is invariant under $\lambda(\mathbb{C}^*)$, so u_M is the pullback of a function u on $M/\mathbb{C}^*$. The function $\tilde{s}/s_0$ is continuous and positive on the compact set $\{s_0 = 1\} \subset X\backslash\{o\}$ and invariant under λ_a for any real number $a > 0$, so $\tilde{s}/s_0$ is bounded from above and below by positive constants on $X\backslash\{o\}$. By (3), u is bounded. Moreover, $dd^c \log s^2 = dd^c \log \tilde{s}^2 + dd^c u_M$ is the pullback of $\Theta := \omega_E + dd^c u$ on $M/\mathbb{C}^*$.

The quotient $M \to M/\mathbb{C}^*$ is induced by the Reeb vector field IV of g, so the θ-term in the equation (105) of [Li21] vanishes, and [Li21, Proposition 2.7] implies that a positive constant multiple of Θ satisfies the Kähler–Einstein equation of (E, Δ_E) on $M/\mathbb{C}^*$.

We extend u to E. Since E is normal, the complex space E^{an} is locally irreducible. Let $E^\star \subset M/\mathbb{C}^*$ be the image of the open subset of M on which $\lambda(\mathbb{G}_m)$ acts freely. Then $E\backslash E^\star$ is an analytic subset of E with empty interior, and the quotient morphism has local holomorphic sections over $E^\star$. Let U be the intersection of E with the open subset $\{Z_{i_0} \neq 0\}$ of $\mathbb{P}^{m-1}$ for some i_0, and let

$$u_U := u + \frac{r}{2\ell} \log \frac{\sum_i |Z_i|^2}{|Z_{i_0}|^2} \quad \text{on } U \cap (M/\mathbb{C}^*),$$

so that $dd^c u_U = \Theta$ there. The pullback of u_U to M is $\log s^2 - \frac{r}{2\ell} \log |f_{i_0}|^2$, whose dd^c is $dd^c \log s^2 \geq 0$. Thus, by composing with local holomorphic sections, u_U is plurisubharmonic on $U \cap E^\star$, and u_U is bounded near any point of U since u is bounded. By [Dem85, Théorème 1.7], u_U restricted to $U \cap E^\star$ extends uniquely to a plurisubharmonic function on U, which is equal to u_U on $U \cap (M/\mathbb{C}^*)$ since u is continuous on $M/\mathbb{C}^*$. Thus u extends uniquely to a bounded ω_E-plurisubharmonic function on E, which we still denote by u.

On a resolution of E, the Monge–Ampère measure of the bounded potential u charges no analytic subset of E, by [Dem12, Chapter III, Proposition (3.11)] applied to the plurisubharmonic function $\log \sum_{j=1}^p |y_j|^2$ for local equations $y_1, \dots, y_p$ of the analytic subset, and the right-hand side of the Kähler–Einstein equation has an integrable density since (E, Δ_E) is klt (Definition-Proposition 2.15(2)) [HL20, (1.8), (1.9), and Definition 1.3]. Thus the Kähler–Einstein equation holds on E, and a positive constant multiple of $\omega_E + dd^c u$ is a bounded weak solution of [HL20, (3.1)] with constant weight 1. This proves (4). □

Example 3.8. Let $X = \mathbb{C}^4$ with coordinates z_1, w_1, z_2, w_2 of degree 1, and let

$$\omega = dz_1 \wedge dw_1 + dz_2 \wedge dw_2,$$

so that $r = 2$. Let $T \subset \mathrm{Sp}_4$ be the diagonal maximal torus, acting by $(z_1, w_1, z_2, w_2) \to (t_1 z_1, t_1^{-1} w_1, t_2 z_2, t_2^{-1} w_2)$. For real numbers $a_1, b_1, a_2, b_2 > 0$, the flat metric

$$g = a_1 |dz_1|^2 + b_1 |dw_1|^2 + a_2 |dz_2|^2 + b_2 |dw_2|^2$$

is a T_c'-invariant Ricci-flat Kähler cone metric on $M = \mathbb{C}^4 \backslash \{0\}$ with

$$s^2 = a_1 |z_1|^2 + b_1 |w_1|^2 + a_2 |z_2|^2 + b_2 |w_2|^2,$$

and g satisfies (1)–(4) of Construction-Proposition 3.7: for (3) take

$$s_0 = (|z_1|^2 + |w_1|^2 + |z_2|^2 + |w_2|^2)^{1/2},$$

and for (4), since λ_z is the multiplication by z on $\mathbb{C}^4$, we have $E = \mathbb{P}^3$ and $\Delta_E = 0$, and Θ is the Fubini–Study form of the Hermitian form s^2. Moreover, $\nabla \omega = 0$ for the Levi-Civita connection ∇ of g. Let D and P be as in Construction-Proposition 3.3 for (M, g, I) and ω, where I is the complex structure of M. A direct calculation shows that

$$D(v) = \Big(-\frac{\overline{dw_1(v)}}{a_1}, \frac{\overline{dz_1(v)}}{b_1}, -\frac{\overline{dw_2(v)}}{a_2}, \frac{\overline{dz_2(v)}}{b_2} \Big)$$

in the coordinates (dz_1, dw_1, dz_2, dw_2), so $-D^2$ is the multiplication by $(a_1 b_1)^{-1}$ on the tangent vectors v with $dz_2(v) = dw_2(v) = 0$, and by $(a_2 b_2)^{-1}$ on those with $dz_1(v) = dw_1(v) = 0$. Thus P is $(a_1 b_1)^{-1/2}$ on the first block and $(a_2 b_2)^{-1/2}$ on the second block, and P is not a constant multiple of the identity if $a_1 b_1 \neq a_2 b_2$.

Lemma 3.9. *Let (M, g, I) be a Kähler manifold with vanishing Ricci curvature, let ∇ be the Levi-Civita connection of g, let $\Delta_g := \mathrm{tr}_g \nabla^2$ be the Laplace–Beltrami operator of g, and let ω be a holomorphic 2-form on M. Let $q := |\omega|_g^2$, where $|\cdot|_g$ is the Hermitian norm induced by g. Then*

$$\Delta_g q = 2 |\nabla \omega|_g^2 \quad \textit{and} \quad |dq|_g^2 \leq 2q |\nabla \omega|_g^2.$$

Proof. Let $m := \dim_{\mathbb{C}} M$. In local holomorphic coordinates $z_1, \dots, z_m$, write $\partial_j := \partial/\partial z_j$, $\nabla_j := \nabla_{\partial_j}$, $\nabla_{\bar{k}} := \nabla_{\partial_{\bar{k}}}$, and $g_{j\bar{k}} := g(\partial_j, \partial_{\bar{k}})$, and let $\langle -, - \rangle$ be the Hermitian inner product on 2-forms induced by g, which is conjugate linear in the second variable. Since ω is holomorphic and g is Kähler, $\nabla_{\bar{k}}\omega = 0$. Hence $\partial_{\bar{k}} q = \langle \omega, \nabla_k \omega \rangle$, and

$$\partial_j \partial_{\bar{k}} q = \langle \nabla_j \omega, \nabla_k \omega \rangle + \langle \omega, \nabla_{\bar{j}} \nabla_k \omega \rangle.$$

Since $\nabla_{\bar{j}}\omega = 0$, we have $\nabla_{\bar{j}} \nabla_k \omega = [\nabla_{\bar{j}}, \nabla_k]\omega$, which is the curvature of g acting on ω. After contracting with $g^{j\bar{k}}$, the second term becomes the action of the Ricci curvature of g on ω paired with ω, which is zero. On functions $\Delta_g = 2g^{j\bar{k}}\partial_j \partial_{\bar{k}}$, and $g^{j\bar{k}}\langle \nabla_j \omega, \nabla_k \omega \rangle = |\nabla \omega|_g^2$ since $\nabla_{\bar{k}}\omega = 0$. Thus $\Delta_g q = 2|\nabla \omega|_g^2$.

For a real tangent vector e, we have $\nabla_{Ie}\omega = i\nabla_e \omega$ since $\nabla_{\bar{k}}\omega = 0$. Hence $dq(e) = 2\operatorname{Re}\langle \nabla_e \omega, \omega \rangle$ and $dq(Ie) = -2\operatorname{Im}\langle \nabla_e \omega, \omega \rangle$, so by the Cauchy–Schwarz inequality,

$$dq(e)^2 + dq(Ie)^2 = 4|\langle \nabla_e \omega, \omega \rangle|^2 \leq 4q|\nabla_e \omega|_g^2.$$

Let $e_1, Ie_1, \dots, e_m, Ie_m$ be an orthonormal basis of a tangent space of M. Summing the last inequality over $e = e_1, \dots, e_m$ and using $|\nabla_{Ie_a}\omega|_g = |\nabla_{e_a}\omega|_g$, we get

$$|dq|_g^2 \leq 4q \sum_{a=1}^{m} |\nabla_{e_a}\omega|_g^2 = 2q \sum_{a=1}^{m} \left(|\nabla_{e_a}\omega|_g^2 + |\nabla_{Ie_a}\omega|_g^2 \right) = 2q|\nabla \omega|_g^2.$$

□

The following lemma integrates IV, JV, and KV to an action of SU(2) when ω is parallel and any real linear combination of IV, JV, and KV is complete. We use Lemma 3.10 in Theorem 3.11 when $n = 1$ and in Theorem 3.20 when $n \geq 2$.

Lemma 3.10. *Notation and conditions as in Set-up 3.6. Let g, s, and V be as in Construction-Proposition 3.7, let ∇ be the Levi-Civita connection of g, and let J and K be as in Construction-Proposition 3.3 for (M, g, I) and ω. Assume that $\nabla\omega = 0$, and that any real linear combination of IV, JV, and KV is a complete vector field on M. Then IV, JV, and KV integrate to a smooth action of* SU(2) *on M such that* $\exp(t\mathbf{i})$, $\exp(t\mathbf{j})$, *and* $\exp(t\mathbf{k})$ *act by the flows of IV, JV, and KV, respectively, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit quaternions. Moreover, g and the action of* SU(2) *satisfy the conditions of Proposition 3.5.*

Proof. By Lemma 3.4, $\nabla V = \mathrm{id}_{TM}$. Since $\nabla\omega = 0$, Construction-Proposition 3.3(4) implies that J and K are parallel, and $\nabla I = 0$ since g is Kähler. Thus for $Q, Q' \in \{I, J, K\}$, we have $\nabla(QV) = Q$ and $\nabla_{QV}(Q'V) = Q'QV$. Since ∇ is torsion-free, $[QV, Q'V] = Q'QV - QQ'V$. Thus by Construction-Proposition 3.3(1),

$$[IV, JV] = -2KV, \quad [JV, KV] = -2IV, \quad \text{and} \quad [KV, IV] = -2JV.$$

In the quaternions, $[\mathbf{i}, \mathbf{j}] = 2\mathbf{k}$, $[\mathbf{j}, \mathbf{k}] = 2\mathbf{i}$, and $[\mathbf{k}, \mathbf{i}] = 2\mathbf{j}$. Thus the linear map ϕ from the Lie algebra of SU(2), which is the space of imaginary quaternions, to the Lie algebra of vector fields on M, given by $\phi(a\mathbf{i} + b\mathbf{j} + c\mathbf{k}) := -(aI + bJ + cK)V$ for $a, b, c \in \mathbb{R}$, is a homomorphism of Lie algebras. By assumption, $\phi(\eta)$ is complete for any η in the Lie algebra of SU(2). Since SU(2) is simply connected, Palais' theorem [Mic08, Theorem 6.5] implies that there exists a smooth left action of SU(2) on M whose fundamental vector field mapping is $-\phi$. Thus the fundamental vector fields of $\mathbf{i}$, $\mathbf{j}$, and $\mathbf{k}$ are IV, JV, and KV, respectively. For η in the Lie algebra of SU(2) and $x \in M$, the curve $t \to \exp(t\eta) \cdot x$ is an integral curve of the fundamental vector field of η, because $\exp((t + t')\eta) \cdot x = \exp(t'\eta) \cdot (\exp(t\eta) \cdot x)$ for any $t, t' \in \mathbb{R}$. Thus $\exp(t\mathbf{i})$, $\exp(t\mathbf{j})$, and $\exp(t\mathbf{k})$ act by the flows of IV, JV, and KV, respectively.

Finally, by Construction-Proposition 3.7, g is a Kähler cone metric on M with complex structure I, radius function s, and radial vector field V, and by Construction-Proposition 3.7(1), the flow of V at any time $t \in \mathbb{R}$ is $\lambda_{\exp(2t/r)}$. By assumption, ω is parallel with respect to g. By Construction-Proposition 3.7, T'_c contains the image of T_c. Since g is T'_c-invariant, T_c acts on M

by isometries of g. Thus g and the action of $\mathrm{SU}(2)$ satisfy the conditions of Proposition 3.5 with $\epsilon = 1$. □

When $n = 1$, the set $X\backslash\{o\}$ is smooth and the set $\{s = 1\}$ is compact, so the parallelism of ω and the completeness of the relevant vector fields follow from classical arguments on compact manifolds.

Theorem 3.11. *Notation and conditions as in Set-up 3.6. Let g, s, and V be as in Construction-Proposition 3.7, let ∇ be the Levi-Civita connection of g, and let J and K be as in Construction-Proposition 3.3 for (M, g, I) and ω. Assume that $n = 1$. Then the following hold.*

(1) $M = X\backslash\{o\}$, and the set $\{s = 1\}$ is compact.
(2) $\nabla\omega = 0$.
(3) The vector fields IV, JV, and KV integrate to a smooth action of $\mathrm{SU}(2)$ on M such that $\exp(t\mathbf{i})$, $\exp(t\mathbf{j})$, and $\exp(t\mathbf{k})$ act by the flows of IV, JV, and KV, respectively, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit quaternions.

In particular, g and the action of $\mathrm{SU}(2)$ in (3) satisfy the conditions of Proposition 3.5.

Proof. For $t \in \mathbb{R}$, let Fl_t^V be the flow of V at time t, which is $\lambda_{\exp(2t/r)}$ by Construction-Proposition 3.7(1). Since $V = s\partial_s$, we have $s(\mathrm{Fl}_t^V(x)) = e^t s(x)$ for any $x \in M$. By Lemma 3.4, $L_V g = 2g$, $|V|_g = s$, and V/s is the gradient of s.

(1) Since X is normal, X is regular in codimension one [Har77, Theorem II.8.22A], and X_{sing} is a proper closed subset of X [Har77, Theorem I.5.3]. Since $\dim X = 2$, the set X_{sing} is finite. Moreover, X_{sing} is invariant under λ. Let $x \in X\backslash\{o\}$. By Definition-Proposition 2.15(1), the stabilizer of x in $\mathbb{G}_m$ is finite, so the λ-orbit of x is one-dimensional. Thus the finite λ-invariant set X_{sing} contains no point of $X\backslash\{o\}$, that is, $X_{\mathrm{sing}} \subset \{o\}$, and $M = X\backslash\{o\}$.

Let s_0 be as in Construction-Proposition 3.7(3). Since s_0 is proper and $s_0^{-1}(0) = \{o\}$, the set $\{s_0 = 1\}$ is compact and contained in $X\backslash\{o\} = M$. Since $s_0(\mathrm{Fl}_t^V(x)) = e^t s_0(x)$ and $s(\mathrm{Fl}_t^V(x)) = e^t s(x)$ for any $t \in \mathbb{R}$ and $x \in M$, the continuous maps

$$\{s_0 = 1\} \to \{s = 1\},\ x \to \mathrm{Fl}^V_{-\log s(x)}(x), \quad \text{and} \quad \{s = 1\} \to \{s_0 = 1\},\ y \to \mathrm{Fl}^V_{-\log s_0(y)}(y)$$

are inverse to each other. Thus $\{s = 1\}$ is compact.

(2) Let $q := |\omega|_g^2$. Since $\lambda_a^*\omega = a^r\omega$ for $a \in \mathbb{C}^*$, we have $(\mathrm{Fl}_t^V)^*\omega = e^{2t}\omega$, that is, $L_V\omega = 2\omega$. Since $L_V g = 2g$, we have $(\mathrm{Fl}_t^V)^* g = e^{2t} g$. Hence q is invariant under the flow of V, and $dq(V) = 0$. The set $A := \{1 \le s \le 2\}$ is compact, since A is the image of $[0, \log 2] \times \{s = 1\}$ under $(t, y) \to \mathrm{Fl}_t^V(y)$. Since g is Ricci-flat, Lemma 3.9 implies $\Delta_g q = 2|\nabla\omega|_g^2$. The boundary of A is $\{s = 1\} \cup \{s = 2\}$. Since V/s is the gradient of s and $|V/s|_g = 1$, the outward unit normal vector of A is V/s along $\{s = 2\}$ and $-V/s$ along $\{s = 1\}$. By the divergence theorem,

$$2\int_A |\nabla\omega|_g^2 d\,\mathrm{vol}_g = \int_A \Delta_g q\, d\,\mathrm{vol}_g = \int_{\{s=2\}} \frac{dq(V)}{s} d\sigma - \int_{\{s=1\}} \frac{dq(V)}{s} d\sigma = 0,$$

where $d\sigma$ is the measure induced by g on the boundary. Thus $\nabla\omega = 0$ on A. Since $(\mathrm{Fl}_t^V)^* g = e^{2t} g$ and the Levi-Civita connections of g and $e^{2t}g$ coincide, Fl_t^V preserves ∇. Hence $(\mathrm{Fl}_t^V)^*(\nabla\omega) = \nabla((\mathrm{Fl}_t^V)^*\omega) = e^{2t}\nabla\omega$. Since any point of M is the image of a point of A under Fl_t^V for some t, we get $\nabla\omega = 0$ on M.

(3) Let $a, b, c \in \mathbb{R}$. Since $aI + bJ + cK$ is skew-adjoint, $g((aI + bJ + cK)V, V) = 0$. Since V/s is the gradient of s, the vector field $(aI + bJ + cK)V$ is tangent to $\{s = e^u\}$ for any $u \in \mathbb{R}$. The set $\{s = e^u\}$ is the image of $\{s = 1\}$ under Fl_u^V, so $\{s = e^u\}$ is compact by (1). Thus the restriction of $(aI + bJ + cK)V$ to $\{s = e^u\}$ is complete [Mic08, 3.8] for any u, and $(aI + bJ + cK)V$ is complete on M. By (2) and Lemma 3.10, (3) holds, and g and the action of $\mathrm{SU}(2)$ in (3) satisfy the conditions of Proposition 3.5. □

3.3. **Logarithmic cutoffs and parallelism.** In this subsection, we assume that $n \geq 2$, construct cutoff functions on the link N (Set-up 3.14 and Proposition 3.15), and use them to prove that ω is parallel with respect to the Ricci-flat Kähler cone metric g (Proposition 3.16). When $n \geq 2$, the set $X_{\mathrm{sing}}\backslash\{o\}$ may be non-empty, so N need not be compact, and neither the integration of the Bochner identity (Lemma 3.9) nor the completeness of Killing fields can be handled as on a compact manifold. The vanishing of Δ_E for $n \geq 2$ (Lemma 3.12), the cutoff functions, the argument for parallelism, and the flows of bounded Killing fields from almost every starting point in Subsection 3.4 are adapted from [Liu26a, Proposition 15.3, Theorems 4.4(2) and 5.1, and Lemmas 6.4 and 6.5], and we write them out completely here.

The cutoff functions in Proposition 3.15 are constructed through a log resolution of E, after removing from E the singular locus, the image of $X_{\mathrm{sing}}\backslash\{o\}$, and the image of the $\mathbb{G}_m$-orbits with non-trivial stabilizers, and the removed set must have codimension at least 2 in E. If $\Delta_E \neq 0$, then a non-exceptional divisor has to be removed as well, but the following lemma shows that $\Delta_E = 0$ when $n \geq 2$. When $n = 1$, Δ_E may be non-zero (Example 3.13), and this case is treated in Theorem 3.11 without cutoff functions.

Lemma 3.12. *Notation and conditions as in Set-up 3.2, and let $E = \operatorname{Proj} R$ and Δ_E be as in Definition-Proposition 2.15. Assume that $n \geq 2$. Then $\Delta_E = 0$.*

Proof. Suppose that $\Delta_E \neq 0$. Then there exists a prime divisor Q on E with $m_Q \geq 2$. Let Q' be the preimage of Q in $X\backslash\{o\}$, let $t \in \mathbb{C}^*$ be a primitive m_Q-th root of unity, and let $\gamma := \lambda_t$. By the definition of m_Q, γ fixes a general point of Q'. Let p be a general point of Q'. Since X is normal and $\dim Q' = 2n-1$, we have $p \in M$. As p is general, Q' is smooth at p, and γ fixes any point of Q' near p. Thus the differential $d\gamma_p$ of γ at p is the identity on the hyperplane $T_pQ' \subset T_pX$. Since γ^{m_Q} is the identity, $d\gamma_p$ is diagonalizable. For $a \in \mathbb{C}$, let $W_a \subset T_pX$ be the eigenspace of $d\gamma_p$ with eigenvalue a, and let ζ be the eigenvalue of $d\gamma_p$ on T_pX/T_pQ'.

We show that $\zeta \neq 1$. Suppose that $\zeta = 1$. Then $d\gamma_p$ is the identity. The finite group generated by γ is commutative and of multiplicative type, hence linearly reductive [Mil17, Proposition 12.54], so the fixed locus M^γ of γ in M is a smooth closed subvariety of M [Mil17, Theorems 7.1, 13.1, and 13.7]. Since $d\gamma_p$ is the identity, $f - f \circ \gamma \in \mathfrak{m}_p^2$ for any f in the maximal ideal $\mathfrak{m}_p$ of $\mathcal{O}_{M,p}$, so the cotangent spaces of M^γ and M at p coincide [Mil17, Lemma 13.4], that is, $T_pM^\gamma = W_1 = T_pX$. Thus $M^\gamma = M$, as M is irreducible of dimension $2n$. Hence γ acts trivially on X, which contradicts the faithfulness of λ (Lemma 2.6), as $t \neq 1$.

Thus $\zeta \neq 1$, $W_1 = T_pQ'$, $\dim W_\zeta = 1$, and $T_pX = W_1 \oplus W_\zeta$. Since ω is homogeneous of degree r, we have $\gamma^*\omega = c\omega$ for $c := t^r$. As $\gamma(p) = p$, it follows that $\omega_p(d\gamma_p(v), d\gamma_p(v')) = c\omega_p(v, v')$ for any $v, v' \in T_pX$. Thus

$$\omega_p(v, v') = 0 \quad \text{for any } v \in W_a \text{ and } v' \in W_b \text{ with } ab \neq c.$$

If $c = 1$, then $\omega_p(W_\zeta, W_1) = 0$, and $\omega_p(W_\zeta, W_\zeta) = 0$ as $\dim W_\zeta = 1$, so W_ζ is contained in the kernel of ω_p. If $c = \zeta$, then $\omega_p(W_1, W_1) = 0$, so the kernel of ω_p contains the kernel of the linear map $W_1 \to \operatorname{Hom}(W_\zeta, \mathbb{C})$ given by $v \to \omega_p(v, \cdot)|_{W_\zeta}$, whose dimension is at least $2n - 2 > 0$ as $n \geq 2$. If $c \notin \{1, \zeta\}$, then $\omega_p(W_1, T_pX) = 0$, so W_1 is contained in the kernel of ω_p. In any case, ω_p is degenerate, which contradicts Set-up 2.10(4). □

Example 3.13. Let $\zeta := e^{2\pi i/3}$, and let the group μ_3 of cube roots of unity act on $\mathbb{C}^2$ by $(z_1, z_2) \to (\zeta z_1, \zeta^{-1} z_2)$. Then

$$R := \mathbb{C}[z_1, z_2]^{\mu_3} = \mathbb{C}[x, y, z]/(xy - z^3), \quad \text{where } x = z_1^3,\ y = z_2^3,\ \text{and } z = z_1 z_2.$$

With the grading induced by $\deg z_1 = \deg z_2 = 1$, the degrees of x, y, and z are 3, 3, and 2, so the grading is primitive. Since μ_3 acts freely on $\mathbb{C}^2\backslash\{0\}$, we have $X_{\mathrm{reg}} = X\backslash\{o\}$, and $dz_1 \wedge dz_2$ descends to a closed non-degenerate algebraic 2-form ω on X_{reg}, which is homogeneous of degree $r = 2$. Thus X, ω, and the trivial torus T satisfy Set-up 3.2 with $n = 1$. The quotient $E \cong \mathbb{P}^1/\mu_3 \cong \mathbb{P}^1$ is the curve $(xy = z^3) \subset \mathbb{P}(3, 3, 2)$. The $\mathbb{G}_m$-orbits over $[1 : 0 : 0]$ and $[0 : 1 : 0]$

have stabilizer μ_3, and the other orbits in $X\backslash\{o\}$ have trivial stabilizers. Thus

$$\Delta_E = \frac{2}{3}\big([1:0:0]+[0:1:0]\big) \neq 0.$$

Set-up 3.14. Notation and conditions as in Set-up 3.6. Let g, s, V, T', T'_c, N, g_N, d_N, and Φ be as follows.

(1) $n \geq 2$.
(2) g, s, V, T', and T'_c are as in Construction-Proposition 3.7.
(3) $N := \{s = 1\} \subset M$, g_N is the Riemannian metric on N induced by g, and d_N is the distance on N defined by g_N.
(4) $\Phi : [0,+\infty) \to [0,+\infty)$ is the function $\Phi(a) := a^2\log^2(e+a)$.

Since M is a non-empty Zariski open subset of the irreducible variety X, [Gro03, Exposé XII, Proposition 2.4] implies that M is connected as a complex manifold. Since M is the product of $(0,+\infty)$ and N [Li21, Definition 2.1], the manifold N is connected, and d_N is a distance on N.

The factor $\log^2(e+a)$ in $\Phi(a)$ is used in Proposition 3.16 and Lemma 3.17.

Proposition 3.15. *Notation and conditions as in Set-up 3.14. Then* $\mathrm{vol}(N, g_N) < +\infty$*, and there exist smooth functions* $\eta_j : N \to [0,1]$ *with compact support,* $j = 1, 2, \ldots$*, such that any compact subset of* N *is contained in* $\{\eta_j = 1\}$ *for any* $j \gg 0$*, and*

$$\lim_{j\to+\infty}\int_N \Phi(|d\eta_j|_{g_N})\, d\,\mathrm{vol}_{g_N} = 0.$$

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we remove from E a closed subset of codimension at least 2, take a log resolution of E, and show that $\mathrm{vol}(N, g_N) < +\infty$.

By [Li21, Definition 2.1], we have

$$M = (0,+\infty)\times N, \quad g = ds^2 + s^2 g_N, \quad \text{and} \quad g(I\cdot,\cdot) = \frac{\sqrt{-1}}{2}\partial\bar\partial s^2.$$

By Lemma 3.4, $|V|_g = s$, and since g is I-invariant, $|IV|_g = |V|_g$. By Construction-Proposition 3.7(1), the flows of V and IV at time $t \in \mathbb{R}$ are $\lambda_{\exp(2t/r)}$ and $\lambda_{\exp(2it/r)}$, which are given by elements of T' and of the image of $\lambda(S^1)$ in T'_c (Construction-Proposition 3.7), respectively. Since g is T'_c-invariant and T' is commutative, the function $s = |V|_g$ is T'_c-invariant, and T'_c acts on N by isometries of g_N.

Let $E = \operatorname{Proj} R$ and Δ_E be as in Definition-Proposition 2.15, and let Θ, ω_E, and u be as in Construction-Proposition 3.7(4). By Set-up 3.6(1) and Lemma 2.9, $(X, 0)$ is klt, so (E, Δ_E) is a log Fano pair by Definition-Proposition 2.15(2). Moreover, $\Delta_E = 0$ by Lemma 3.12. Thus E is klt, and $-K_E$ is an ample $\mathbb{Q}$-Cartier divisor.

Fix finitely many homogeneous generators of R. By Definition-Proposition 2.15(1), for any $y \in X\backslash\{o\}$, the stabilizer of y in $\mathbb{G}_m$ is the group of roots of unity whose order divides the greatest common divisor of the degrees of the generators not vanishing at y. Thus the set of points of $X\backslash\{o\}$ with non-trivial stabilizers is a finite union of fixed loci of finite subgroups of $\mathbb{G}_m$, and hence closed. Let $E^\circ \subset E$ be the complement of the union of E_{sing}, the image of $X_{\mathrm{sing}}\backslash\{o\}$, and the image of the set of points of $X\backslash\{o\}$ with non-trivial stabilizers. Then E° is an open subset of $M/\mathbb{C}^*$, and $\mathbb{G}_m$ acts freely on the preimage of E° in M.

We show that $\mathrm{codim}_E(E\backslash E^\circ) \geq 2$. Since R is normal, E is normal, so $\mathrm{codim}_E E_{\mathrm{sing}} \geq 2$. Since X is normal, $\dim X_{\mathrm{sing}} \leq 2n-2$, so the image of $X_{\mathrm{sing}}\backslash\{o\}$ in E is of dimension at most $2n-3$, as the fibers of the quotient morphism $X\backslash\{o\} \to E$ are $\mathbb{G}_m$-orbits (Definition-Proposition 2.15(1)). If the image of the set of points with non-trivial stabilizers contained a prime divisor Q on E, then any point of the preimage of Q would have a non-trivial stabilizer, so $m_Q \geq 2$, which contradicts $\Delta_E = 0$.

Let $h : E' \to E$ be a log resolution of $(E, E\backslash E^\circ)$ which is a composition of blow-ups along smooth centers whose images in E are contained in $E\backslash E^\circ$. Such h exists: we may take for h the

composition of the resolution $E'' \to E$ in [BM97, Theorem 1.6], applied to a closed embedding of E into a projective space, and the principalization in [Kol05, Theorem 35] of the ideal sheaf of the reduced preimage of $E\backslash E^\circ$ in E'', after omitting the components of codimension 1 of the centers of the principalization, as the blow-up along a Cartier divisor is an isomorphism. By [BM97, Theorem 1.6(1)] and [Har77, Corollary II.7.15], $E'' \to E$ is a composition of blow-ups along smooth centers, any of which is contained in the singular locus of the variety being blown up, or in the preimage of the previous centers if that variety is smooth, so by induction on the number of blow-ups, the images of the centers in E are contained in $E_{\mathrm{sing}} \subset E\backslash E^\circ$. Any remaining component of a center of the principalization has codimension at least 2, so the fiber of the principalization over the image of any point of the component is positive-dimensional, and since the principalization is an isomorphism over the complement of the preimage of $E\backslash E^\circ$ [Kol05, Theorem 35(3)], the image of the component in E is contained in $E\backslash E^\circ$. In particular, h is an isomorphism over E°.

Then $F := h^{-1}(E\backslash E^\circ)$ is a simple normal crossing divisor. Write $F = \sum_i F_i$, where F_i are the irreducible components of F. Since $\operatorname{codim}_E(E\backslash E^\circ) \geq 2$, any F_i is h-exceptional, and thus $\operatorname{Exc}(h) = F$. Via h, we regard E° as the open subset $E'\backslash F$ of E'.

Let N° be the preimage of E° in N. As the flow of V at time t multiplies s by e^t, any $\mathbb{C}^*$-orbit in M intersects N in exactly one orbit of the flow of IV. Thus the projection $N^\circ \to E^\circ$ is a principal bundle under the circle group $\{\lambda_{\exp(2it/r)}\}_{t\in\mathbb{R}}$, and the orbits of IV in N° have period πr, as $\mathbb{G}_m$ acts freely on the preimage of E° in M. Moreover, $|IV|_g = 1$ on N, and g_N is invariant under the flow of IV. Let g_E be the Riemannian metric on E° such that $N^\circ \to E^\circ$ is a Riemannian submersion with respect to g_N and g_E. For a function f on E°, we write $\tilde{f}$ for the pullback of f to N°. A direct calculation shows that the Kähler form of g_E is a constant multiple of Θ, that $|d\tilde{f}|_{g_N}$ is the pullback of $|df|_{g_E}$ to N° for any smooth function f on E°, and that

$$\int_{N^\circ} \tilde{f}\, d\operatorname{vol}_{g_N} = \pi r \int_{E^\circ} f\, d\operatorname{vol}_{g_E} \tag{3.1}$$

for any non-negative measurable function f on E°. The preimage of $(M/\mathbb{C}^*)\backslash E^\circ$ in M is a proper closed analytic subset of M invariant under the flow of V, so $N\backslash N^\circ$ has measure zero in N.

Let $\alpha := h^*\omega_E$, which is a smooth semipositive $(1,1)$-form on E'. We also write u for h^*u, which is a bounded α-plurisubharmonic function on E'. Then $\Theta = \alpha + dd^c u$ on E°. For $0 \leq p \leq 2n-1$, we also write Θ^p for the Bedford–Taylor product $(\alpha + dd^c u)^p$ on E' [Dem12, Chapter III, (3.2) and the definition before it], which is a closed positive current on E' and is equal to the smooth form Θ^p on E°. By the definition of the Bedford–Taylor product, $\Theta^{2n-1} - \alpha^{2n-1}$ is equal to dd^c of a current on the compact manifold E', so $\int_{E'} \Theta^{2n-1} = \int_{E'} \alpha^{2n-1}$ by Stokes' theorem. Since $N\backslash N^\circ$ has measure zero and $d\operatorname{vol}_{g_E}$ is a constant multiple of Θ^{2n-1}, (3.1) implies that $\operatorname{vol}(N, g_N) = \pi r \operatorname{vol}(E^\circ, g_E)$ is a constant multiple of

$$\int_{E^\circ} \Theta^{2n-1} \leq \int_{E'} \Theta^{2n-1} = \int_{E'} \alpha^{2n-1} < +\infty.$$

Step 2. In this step we fix a Kähler form θ on E' and show that there exist real numbers $C \geq 1$ and $k > 0$ such that

$$\Theta \geq C^{-1} \prod_i x_i^k \theta \quad \text{on } E^\circ, \tag{3.2}$$

where the functions x_i on E' are defined below. In the rest of the proof, C denotes a positive real number independent of the indices ν, ϵ, and Λ introduced below, which may change from line to line.

For any i, let σ_i be the section of $\mathcal{O}_{E'}(F_i)$ defining F_i, and fix a smooth Hermitian metric on $\mathcal{O}_{E'}(F_i)$ such that $x_i := |\sigma_i|^2 \leq e^{-2}$ on E'. Let θ_i be the curvature form of the Hermitian metric on $\mathcal{O}_{E'}(F_i)$. Then θ_i represents $c_1(\mathcal{O}_{E'}(F_i))$, $dd^c \log x_i = -\theta_i$ on $E'\backslash F_i$, and $-C\theta \leq \theta_i \leq C\theta$. A

direct calculation shows that for any $\epsilon > 0$, the smooth form $dd^c \log(x_i + \epsilon) + \frac{x_i}{x_i+\epsilon}\theta_i$ is a positive constant multiple of $\epsilon x_i^{-1}(x_i+\epsilon)^{-2}\sqrt{-1}\partial x_i \wedge \bar\partial x_i$ on $E'\backslash F_i$. Thus

$$dd^c \log(x_i + \epsilon) + \frac{x_i}{x_i + \epsilon}\theta_i \geq 0 \quad \text{on } E'. \tag{3.3}$$

Since $\Theta = \omega_E + dd^c u$ is a weak Kähler–Einstein metric on $(E, 0)$ [HL20, (3.1)], there exists a positive real multiple A of $-K_E$ such that α represents $c_1(h^*A)$, and A is an ample $\mathbb{R}$-divisor on E. Let a_i be the real numbers with $K_{E'} = h^*K_E + \sum_i a_i F_i$. Then $a_i > -1$, as E is klt. Moreover, by Construction-Proposition 3.7(4), there exists a real number $C_2 > 0$ such that $C_2\Theta$ is a bounded weak solution of [HL20, (3.1)] for $(E, 0)$ with constant weight 1, where the difference between the potential of $C_2\Theta$ and the reference potential may be taken to be $C_2 u$ plus a constant. Thus, on E°, Θ^{2n-1} is equal to $e^{-C_2 u}$ times a positive constant multiple of the measure $d\mu_0$ in [HL20, (1.4) and (3.1)]. Since $\Delta_E = 0$ and $K_{E'} = h^*K_E + \sum_i a_i F_i$, [HL20, (1.8) and (1.9)] imply that the pullback to E' of $d\mu_0$ is locally the product of $\prod_i |z_i|^{2a_i}$, a smooth positive function, and a smooth volume form, where z_i is a local holomorphic function defining F_i. As $x_i = |\sigma_i|^2$ is locally $|z_i|^2$ times a smooth positive function, there exists a smooth positive volume form Ω on E' such that

$$\Theta^{2n-1} = e^{-C_2 u} \prod_i x_i^{a_i} \Omega \quad \text{on } E^\circ.$$

Let φ be a function on E° such that $\varphi = \sum_i t_i \log x_i + \varphi_0$ for some real numbers $t_i > 0$ and a smooth function φ_0 on E', $\varphi \leq 0$, and

$$\alpha + dd^c\varphi \geq C^{-1}\theta \quad \text{on } E^\circ.$$

We show that such a function exists. Write $h = h_1 \circ \cdots \circ h_m$, where $E_0 := E$, $E_m := E'$, and $h_j : E_j \to E_{j-1}$ is the blow-up along a smooth center Z_j. By [Har77, Proposition II.7.13], there exists a Cartier divisor $G_j \geq 0$ on E_j with $\mathcal{O}_{E_j}(-G_j) = h_j^{-1}\mathcal{I}_{Z_j} \cdot \mathcal{O}_{E_j}$, and h_j is an isomorphism over $E_{j-1}\backslash Z_j$. Since $\mathcal{O}_{E_j}(-G_j)$ is the sheaf $\mathcal{O}(1)$ of $E_j = \operatorname{Proj} \bigoplus_{d\geq 0} \mathcal{I}_{Z_j}^d$, the divisor $-G_j$ is h_j-ample [Har77, Proposition II.7.10(b)]. By [KM98, Proposition 1.45], applied to $h_1, \ldots, h_m$ in turn, there exist real numbers $b_j > 0$ such that $h^*A - F'$ is ample, where $F' := \sum_j b_j (h_{j+1} \circ \cdots \circ h_m)^* G_j$. As the image of Z_j in E is contained in $E\backslash E^\circ$ for any j, we have $\operatorname{Supp} F' \subset F$, and as h_j is an isomorphism over $E_{j-1}\backslash Z_j$ for any j, we have $\operatorname{Exc}(h) \subset \operatorname{Supp} F'$. Thus $F' = \sum_i t_i F_i$ with real numbers $t_i > 0$. Let θ' be a Kähler form on E' representing $c_1(h^*A - F')$. Since $\alpha - \sum_i t_i\theta_i$ also represents $c_1(h^*A - F')$, the $\partial\bar\partial$-lemma implies that $\theta' = \alpha - \sum_i t_i\theta_i + dd^c\varphi_0$ for a smooth function φ_0 on E', which we may assume to be non-positive. As $\log x_i \leq -2$ and $dd^c \log x_i = -\theta_i$ on E°, the function $\varphi := \sum_i t_i \log x_i + \varphi_0$ satisfies $\varphi \leq 0$ and $\alpha + dd^c\varphi = \theta' \geq C^{-1}\theta$ on E°.

Let $\epsilon_\nu \in (0, e^{-2})$ be real numbers decreasing to 0, let u_ν^0 be smooth functions on E' decreasing to u with $dd^c u_\nu^0 \geq -C\theta$ [Dem92, Main Theorem 1.1], so that $|u_\nu^0| \leq C$ as u is bounded and $u \leq u_\nu^0 \leq u_1^0$, and let

$$\psi_\nu^+ := \sum_{a_i > 0} a_i \log(x_i + \epsilon_\nu), \qquad \psi_\nu^- := C_2 u_\nu^0 + \sum_{a_i < 0} (-a_i) \log(x_i + \epsilon_\nu).$$

By [HL20, Corollary 3.6, Proposition 3.7, and (3.14)–(3.18)], applied with constant weight, there exist smooth functions u_ν on E' and real numbers $c_\nu > 0$ such that $\Theta_\nu := \alpha + \epsilon_\nu\theta + dd^c u_\nu$ is a Kähler form on E',

$$\Theta_\nu^{2n-1} = c_\nu \exp(\psi_\nu^+ - \psi_\nu^-)\Omega,$$

$|u_\nu| \leq C$, and $C^{-1} \leq c_\nu \leq C$. By (3.3) and $dd^c u_\nu^0 \geq -C\theta$, we have $dd^c\psi_\nu^\pm \geq -C\theta$. As $x_i + \epsilon_\nu \leq 1$, we also have $\psi_\nu^+ \leq 0$ and $\psi_\nu^- \leq C$.

Let $\operatorname{tr}_\theta \Theta_\nu$ and $\operatorname{tr}_{\Theta_\nu} \theta$ be the traces of Θ_ν with respect to θ and of θ with respect to Θ_ν, respectively, and let $\det_\theta \Theta_\nu := \Theta_\nu^{2n-1}/\theta^{2n-1}$. Since

$$-dd^c \log \frac{\Theta_\nu^{2n-1}}{\Omega} = dd^c\psi_\nu^- - dd^c\psi_\nu^+ \leq dd^c\psi_\nu^- + C\theta$$

and $\operatorname{tr}_\theta \Theta_\nu \cdot \operatorname{tr}_{\Theta_\nu} \theta \geq (2n-1)^2$, the Chern–Lu inequality (see [HL20, (3.16)]) implies that

$$\operatorname{tr}_{\Theta_\nu} dd^c \log \operatorname{tr}_\theta \Theta_\nu \geq -\frac{\operatorname{tr}_\theta dd^c \psi_\nu^-}{\operatorname{tr}_\theta \Theta_\nu} - C \operatorname{tr}_{\Theta_\nu} \theta - C \quad \text{on } E'.$$

A direct calculation in local coordinates diagonalizing Θ_ν with respect to θ, together with $dd^c \psi_\nu^- \geq -C\theta$, shows that

$$\operatorname{tr}_{\Theta_\nu} dd^c \psi_\nu^- - \frac{\operatorname{tr}_\theta dd^c \psi_\nu^-}{\operatorname{tr}_\theta \Theta_\nu} \geq -C \operatorname{tr}_{\Theta_\nu} \theta \quad \text{on } E'.$$

Moreover, as $dd^c u_\nu = \Theta_\nu - \alpha - \epsilon_\nu \theta$ and $dd^c \varphi \geq C^{-1}\theta - \alpha$ on E°, we have

$$\operatorname{tr}_{\Theta_\nu} dd^c (u_\nu - \varphi) \leq 2n - 1 - C^{-1} \operatorname{tr}_{\Theta_\nu} \theta \quad \text{on } E^\circ.$$

Thus there exists a real number $C_1 > 0$ independent of ν such that the function

$$H_\nu := \log \operatorname{tr}_\theta \Theta_\nu + \psi_\nu^- - C_1(u_\nu - \varphi)$$

on E° satisfies $\operatorname{tr}_{\Theta_\nu} dd^c H_\nu \geq \operatorname{tr}_{\Theta_\nu} \theta - C$.

For fixed ν, the functions $\operatorname{tr}_\theta \Theta_\nu$, ψ_ν^-, and u_ν are smooth on E', and φ tends to $-\infty$ near F. Thus H_ν attains a maximum at a point $y_\nu \in E^\circ$. At y_ν, we have $\operatorname{tr}_{\Theta_\nu} dd^c H_\nu \leq 0$, so $\operatorname{tr}_{\Theta_\nu} \theta \leq C$, that is, the eigenvalues of Θ_ν with respect to θ are at least C^{-1}. As $\det_\theta \Theta_\nu \leq C e^{-\psi_\nu^-}$ by $\psi_\nu^+ \leq 0$, it follows that $\operatorname{tr}_\theta \Theta_\nu \leq C e^{-\psi_\nu^-}$ at y_ν. Since $|u_\nu| \leq C$ and $\varphi \leq 0$, we get $H_\nu(y_\nu) \leq C$, and thus $H_\nu \leq C$ on E°. Since $|u_\nu^0| \leq C$, $x_i \leq x_i + \epsilon_\nu$, and $x_i \leq 1$, it follows that

$$\operatorname{tr}_\theta \Theta_\nu \leq C \exp(-\psi_\nu^- + C_1 u_\nu - C_1 \varphi) \leq C \prod_i x_i^{-k_1} \quad \text{on } E^\circ$$

for a real number $k_1 > 0$ independent of ν. On the other hand, $\psi_\nu^+ \geq \sum_{a_i > 0} a_i \log x_i$ and $\psi_\nu^- \leq C$, so $\det_\theta \Theta_\nu \geq C^{-1} \prod_{a_i > 0} x_i^{a_i}$. Since $\operatorname{tr}_{\Theta_\nu} \theta \leq (2n-1)(\operatorname{tr}_\theta \Theta_\nu)^{2n-2} / \det_\theta \Theta_\nu$, we get $\operatorname{tr}_{\Theta_\nu} \theta \leq C \prod_i x_i^{-k}$ on E° for $k := (2n-2)k_1 + \sum_{a_i > 0} a_i$. Thus $\Theta_\nu \geq (\operatorname{tr}_{\Theta_\nu} \theta)^{-1} \theta \geq C^{-1} \prod_i x_i^k \theta$ on E°.

Finally, we let $\nu \to +\infty$. On any coordinate ball whose closure is contained in E°, the functions x_i are bounded below by a positive real number, so the bound $\operatorname{tr}_\theta \Theta_\nu \leq C \prod_i x_i^{-k_1}$ implies that the Euclidean Laplacian of u_ν, which is a positive constant multiple of the trace of $dd^c u_\nu = \Theta_\nu - \alpha - \epsilon_\nu \theta$ with respect to the Euclidean Kähler form, is bounded on the ball independently of ν. As $|u_\nu| \leq C$, differentiating the Green–Riesz representation formula [Dem12, Chapter I, (4.5)] under the integral sign shows that the Euclidean norm of du_ν is bounded on the concentric ball of half the radius independently of ν. Thus, by the Arzelà–Ascoli theorem and a diagonal argument, a subsequence of (u_ν) converges locally uniformly on E°. As $C^{-1} \leq c_\nu \leq C$, possibly replacing the sequence of indices ν with a subsequence, we may assume that u_ν converges locally uniformly on E°, and that c_ν converges to a real number $c > 0$. Since $\Theta_\nu > 0$, $\epsilon_\nu \to 0$, and $|u_\nu| \leq C$, the limit of u_ν is a continuous α-plurisubharmonic function on E° with absolute value at most C, so it extends to an α-plurisubharmonic function v on E' [Dem12, Chapter I, Theorem (5.24)], and $|v| \leq C$ on E' by [Dem12, Chapter I, Theorem (5.5)], as F has measure zero.

By [Dem12, Chapter III, Corollary (3.6)], applied to local potentials of Θ_ν, the measures Θ_ν^{2n-1} converge weakly to $(\alpha + dd^c v)^{2n-1}$ on E°. On the other hand, as u_ν^0 decreases to u, the functions $c_\nu \exp(\psi_\nu^+ - \psi_\nu^-)$ converge pointwise on E° to $c e^{-C_2 u} \prod_i x_i^{a_i}$, and they are bounded on any compact subset of E° independently of ν. Thus the dominated convergence theorem and the equality $\Theta^{2n-1} = e^{-C_2 u} \prod_i x_i^{a_i} \Omega$ on E° imply that

$$(\alpha + dd^c v)^{2n-1} = c\Theta^{2n-1} \quad \text{on } E^\circ.$$

By [Dem12, Chapter III, Proposition (3.11)], applied to the plurisubharmonic functions $\log|z_i|^2$ for local holomorphic functions z_i defining F_i, the measures $(\alpha + dd^c v)^{2n-1}$ and Θ^{2n-1} on E' do not charge F, so $(\alpha + dd^c v)^{2n-1} = c\Theta^{2n-1}$ on E'. Since $(\alpha + dd^c v)^{2n-1} - \alpha^{2n-1}$ is equal to dd^c

of a current on E' by the definition of the Bedford–Taylor product, Stokes' theorem and **Step 1** imply that

$$\int_{E'}(\alpha+dd^c v)^{2n-1}=\int_{E'}\alpha^{2n-1}=\int_{E'}\Theta^{2n-1},$$

which is positive as $\Theta>0$ on E°. Thus $c=1$, and [HL20, Proposition 2.14], applied with constant weight, implies that $v-u$ is constant. Hence Θ_ν converges weakly to $\alpha+dd^c v=\Theta$ on E°. As $\Theta_\nu-C^{-1}\prod_i x_i^k\theta\geq 0$ on E° for any ν, the weak limit $\Theta-C^{-1}\prod_i x_i^k\theta$ is a positive current on E°, and since Θ is smooth on E°, we get (3.2).

Step 3. In this step we fix i, write $x:=x_i$, and show that

$$\int_{E'\setminus F_i}\sqrt{-1}\partial\log x\wedge\bar\partial\log x\wedge\Theta^{2n-2}<+\infty. \tag{3.4}$$

For $0<\epsilon<e^{-2}$, let

$$\Xi_\epsilon:=dd^c\log(x+\epsilon)+\frac{x}{x+\epsilon}\theta_i,$$

which is a smooth semipositive $(1,1)$-form on E' by (3.3). By Stokes' theorem,

$$\int_{E'}\Xi_\epsilon\wedge S=\int_{E'}\frac{x}{x+\epsilon}\theta_i\wedge S \tag{3.5}$$

for any closed current S of bidegree $(2n-2,2n-2)$ on E'. Recall that Θ^p is the Bedford–Taylor product $(\alpha+dd^c u)^p$ on E' (**Step 1**). By the definition of the Bedford–Taylor product [Dem12, Chapter III, (3.2) and the definition before it],

$$\Theta^{2n-2}-\alpha^{2n-2}=dd^c\Big(u\sum_{p=0}^{2n-3}\Theta^p\wedge\alpha^{2n-3-p}\Big). \tag{3.6}$$

We first show that

$$\int_{E'}(-\log x)\theta\wedge\Theta^{2n-2}<+\infty. \tag{3.7}$$

Integrating (3.6) against the smooth form $-\log(x+\epsilon)\theta$, and using Stokes' theorem and $dd^c(-\log(x+\epsilon))=\frac{x}{x+\epsilon}\theta_i-\Xi_\epsilon$, we get

$$\int_{E'}-\log(x+\epsilon)\theta\wedge(\Theta^{2n-2}-\alpha^{2n-2})=\sum_{p=0}^{2n-3}\int_{E'}u\Big(\frac{x}{x+\epsilon}\theta_i-\Xi_\epsilon\Big)\wedge\theta\wedge\Theta^p\wedge\alpha^{2n-3-p}.$$

Since $|u|\leq C$, $\Xi_\epsilon\geq 0$, and $-C\theta\leq\theta_i\leq C\theta$, (3.5) implies that the absolute value of the right-hand side is at most $C\sum_p\int_{E'}\theta^2\wedge\Theta^p\wedge\alpha^{2n-3-p}$, which is finite by [Dem12, Chapter III, (3.3)] and independent of ϵ. As $\int_{E'}(-\log x)\theta\wedge\alpha^{2n-2}<+\infty$ and $-\log(x+\epsilon)\leq-\log x$, we get $\int_{E'}-\log(x+\epsilon)\theta\wedge\Theta^{2n-2}\leq C$. Since $-\log(x+\epsilon)$ is positive and increases to $-\log x$ as $\epsilon\to 0$, (3.7) follows from the monotone convergence theorem. In particular, the measure $\theta\wedge\Theta^{2n-2}$ has no mass on F_i.

Next, as θ_i and α represent $c_1(\mathcal{O}_{E'}(F_i))$ and $c_1(h^*A)$, respectively, and $F_i\cdot(h^*A)^{2n-2}=0$ as $h_*F_i=0$, we have $\int_{E'}\theta_i\wedge\alpha^{2n-2}=0$. Since θ_i is closed, (3.6) and Stokes' theorem imply that

$$\int_{E'}\theta_i\wedge\Theta^{2n-2}=\int_{E'}\theta_i\wedge\alpha^{2n-2}=0.$$

Thus (3.5) and $\frac{x}{x+\epsilon}=1-\frac{\epsilon}{x+\epsilon}$ imply

$$0\leq\int_{E'}\Xi_\epsilon\wedge\Theta^{2n-2}=-\int_{E'}\frac{\epsilon}{x+\epsilon}\theta_i\wedge\Theta^{2n-2}.$$

On $E'\backslash F_i$, the function $|\log\epsilon|\frac{\epsilon}{x+\epsilon}$ tends to 0 as $\epsilon\to 0$. If $x\le\epsilon$, then $|\log\epsilon|\le -\log x$, and if $x>\epsilon$, then $|\log\epsilon| = -\log x+\log\frac{x}{\epsilon}$ and $\frac{\epsilon}{x}\log\frac{x}{\epsilon}\le 1$. Thus $|\log\epsilon|\frac{\epsilon}{x+\epsilon}\le 1-\log x$ on $E'\backslash F_i$. Since $|\theta_i\wedge\Theta^{2n-2}|\le C\theta\wedge\Theta^{2n-2}$ as measures, (3.7) and the dominated convergence theorem imply

$$\lim_{\epsilon\to 0}|\log\epsilon|\int_{E'}\Xi_\epsilon\wedge\Theta^{2n-2}=0. \tag{3.8}$$

Finally, let $f_\epsilon:=\log(x+\epsilon)$. Since $\sqrt{-1}\partial\bar\partial$ is a positive constant multiple of dd^c, and

$$\sqrt{-1}\partial\bar\partial(f_\epsilon^2)=2f_\epsilon\sqrt{-1}\partial\bar\partial f_\epsilon+2\sqrt{-1}\partial f_\epsilon\wedge\bar\partial f_\epsilon,$$

Stokes' theorem implies that $\int_{E'}\sqrt{-1}\partial f_\epsilon\wedge\bar\partial f_\epsilon\wedge\Theta^{2n-2}$ is a positive constant multiple of

$$-\int_{E'}f_\epsilon\Xi_\epsilon\wedge\Theta^{2n-2}+\int_{E'}\frac{xf_\epsilon}{x+\epsilon}\theta_i\wedge\Theta^{2n-2}.$$

As $|f_\epsilon|\le|\log\epsilon|$ and $|f_\epsilon|\le-\log x$, the first term tends to 0 by (3.8), and the second term is bounded by (3.7). On $E'\backslash F_i$, we have

$$\sqrt{-1}\partial f_\epsilon\wedge\bar\partial f_\epsilon=\Big(\frac{x}{x+\epsilon}\Big)^2\sqrt{-1}\partial\log x\wedge\bar\partial\log x,$$

and $(\frac{x}{x+\epsilon})^2$ increases to 1 as $\epsilon\to 0$. Thus (3.4) follows from the monotone convergence theorem.

Step 4. In this step we construct the functions η_j.

Recall that $x_i=|\sigma_i|^2$ and that θ is the Kähler form on E' fixed in **Step 2**. Let $\ell:=1-\sum_i\log x_i$, which is a smooth function on E° with $\ell\ge 1$, and $\ell\to+\infty$ near F. A direct calculation shows that $|dx_i|_\theta\le Cx_i^{1/2}$, where $|\cdot|_\theta$ is the norm defined by the Kähler metric with Kähler form θ. Thus $|d\ell|_\theta\le C\prod_i x_i^{-1/2}$, as $x_i\le 1$. Since the Kähler form of g_E is a constant multiple of Θ (**Step 1**), (3.2) implies $|d\ell|_{g_E}\le C\prod_i x_i^{-(k+1)/2}$, and hence

$$\log(e+|d\ell|_{g_E})\le C\ell\quad\text{on } E^\circ.$$

Moreover, a direct calculation shows that $|d\ell|_{g_E}^2 d\operatorname{vol}_{g_E}$ is a constant multiple of $\sqrt{-1}\partial\ell\wedge\bar\partial\ell\wedge\Theta^{2n-2}$ on E°, and $\sqrt{-1}\partial\ell\wedge\bar\partial\ell\le C\sum_i\sqrt{-1}\partial\log x_i\wedge\bar\partial\log x_i$ by the Cauchy–Schwarz inequality. Thus **Step 3** implies

$$\int_{E^\circ}|d\ell|_{g_E}^2 d\operatorname{vol}_{g_E}<+\infty. \tag{3.9}$$

Let $\tau:\mathbb{R}\to[0,1]$ be a smooth function with $\tau=1$ on $(-\infty,0]$ and $\tau=0$ on $[1,+\infty)$. For $\Lambda\ge 1$, let $\beta_\Lambda:=\tau(\log\ell-\Lambda)$, which is a smooth function on E° with compact support. Any compact subset of E° is contained in $\{\beta_\Lambda=1\}$ for $\Lambda\gg 0$. Since $d\beta_\Lambda=\tau'(\log\ell-\Lambda)\ell^{-1}d\ell$ vanishes outside $\{e^\Lambda\le\ell\le e^{\Lambda+1}\}$ and $\log(e+|d\ell|_{g_E})\le C\ell$, we have $\Phi(|d\beta_\Lambda|_{g_E})\le C|d\ell|_{g_E}^2$ on $\{\ell\ge e^\Lambda\}$ and $\Phi(|d\beta_\Lambda|_{g_E})=0$ on $\{\ell<e^\Lambda\}$. Thus (3.9) implies

$$\lim_{\Lambda\to+\infty}\int_{E^\circ}\Phi(|d\beta_\Lambda|_{g_E})d\operatorname{vol}_{g_E}=0.$$

Recall that N° is the preimage of E° in N, and that $\tilde f$ is the pullback to N° of a function f on E° (**Step 1**). We also write β_Λ for the function on N which is equal to $\widetilde{\beta_\Lambda}$ on N° and to 0 on $N\backslash N^\circ$. As the projection $N^\circ\to E^\circ$ is proper, β_Λ is a smooth function on N with compact support in N°. Since $|d\widetilde{\beta_\Lambda}|_{g_N}$ is the pullback of $|d\beta_\Lambda|_{g_E}$ to N° (**Step 1**), (3.1) implies

$$\int_N\Phi(|d\beta_\Lambda|_{g_N})d\operatorname{vol}_{g_N}=\pi r\int_{E^\circ}\Phi(|d\beta_\Lambda|_{g_E})d\operatorname{vol}_{g_E}\to 0\quad\text{as }\Lambda\to+\infty, \tag{3.10}$$

and $\beta_\Lambda\to 1$ pointwise on N° as $\Lambda\to+\infty$.

The functions β_Λ vanish near $N\backslash N^\circ$, which contains the points of N with non-trivial stabilizers in $\mathbb{G}_m$. To obtain cutoff functions equal to 1 near $N\backslash N^\circ$ as well, we use an exhaustion of N. Let $U_1\subset U_2\subset\cdots$ be compact subsets of N such that any compact subset of N is contained in U_j

for $j \gg 0$, and for any j, let $\gamma_j : N \to [0,1]$ be a smooth function with compact support such that $\gamma_j = 1$ on U_j.

For any j and $\Lambda \geq 1$, the function $\gamma_j + (1-\gamma_j)\beta_\Lambda$ on N is smooth, has values in $[0,1]$ and compact support, and is equal to 1 on U_j. We have

$$d\big(\gamma_j + (1-\gamma_j)\beta_\Lambda\big) = (1-\beta_\Lambda)d\gamma_j + (1-\gamma_j)d\beta_\Lambda.$$

Since Φ is increasing and $\Phi(a+b) \leq \Phi(2\max\{a,b\}) \leq 16(\Phi(a)+\Phi(b))$ for $a, b \geq 0$, we get

$$\int_N \Phi\big(|d(\gamma_j + (1-\gamma_j)\beta_\Lambda)|_{g_N}\big)\, d\operatorname{vol}_{g_N} \\ \leq 16\int_N \Phi\big((1-\beta_\Lambda)|d\gamma_j|_{g_N}\big)\, d\operatorname{vol}_{g_N} + 16\int_N \Phi(|d\beta_\Lambda|_{g_N})\, d\operatorname{vol}_{g_N}.$$

For fixed j, the second term on the right-hand side tends to 0 as $\Lambda \to +\infty$ by (3.10). Since $\Phi(|d\gamma_j|_{g_N})$ is bounded with compact support, $\beta_\Lambda \to 1$ on N°, and $N \backslash N^\circ$ has measure zero (**Step 1**), the first term also tends to 0 as $\Lambda \to +\infty$ by the dominated convergence theorem. Thus there exists $\Lambda_j \geq 1$ such that $\eta_j := \gamma_j + (1-\gamma_j)\beta_{\Lambda_j}$ satisfies

$$\int_N \Phi(|d\eta_j|_{g_N})\, d\operatorname{vol}_{g_N} \leq \frac{1}{j}.$$

The functions $\eta_j : N \to [0,1]$ are smooth with compact support, and $\eta_j = 1$ on U_j. Since any compact subset of N is contained in U_j for $j \gg 0$, any compact subset of N is contained in $\{\eta_j = 1\}$ for $j \gg 0$. Together with $\operatorname{vol}(N, g_N) < +\infty$ (**Step 1**), this completes the proof. □

If N is compact, then integrating the identity in Lemma 3.9 over N shows that ω is parallel. In general, $|\omega|_g^2$ is a priori only integrable on N, as ω extends to a resolution of X and s^2 is locally bounded. We therefore work with $\log(1+|\omega|_g^2)$, whose exponential is integrable on N, and the cutoff functions on N (Proposition 3.15) are constructed to absorb exactly such a function.

Proposition 3.16. *Notation and conditions as in Set-up 3.14, and let ∇ be the Levi-Civita connection of g. Then $\nabla\omega = 0$ on M.*

Proof. By [Li21, Definition 2.1], we have $M = (0,+\infty) \times N$, $g = ds^2 + s^2 g_N$, and $\kappa = \frac{\sqrt{-1}}{2}\partial\bar\partial s^2$, where $\kappa := g(I\cdot, \cdot)$. Let $q := |\omega|_g^2$.

We first show that $\int_N q\, d\operatorname{vol}_{g_N} < +\infty$. A direct calculation shows that $\omega \wedge \overline{\omega} \wedge \kappa^{2n-2} = C_n q\, d\operatorname{vol}_g$ on M for a real number $C_n > 0$ depending only on n. By Set-up 3.6(1) and Definition 2.8, for a resolution $h : X' \to X$, the form $h^*\omega$ on $h^{-1}(X_{\mathrm{reg}})$ extends to a regular 2-form ω' on X'. Let s_0 and c be as in Construction-Proposition 3.7(3), and let $Q := h^{-1}(\{(4c)^{-1} \leq s_0 \leq 4c\})$ and $Q' := h^{-1}(\{(2c)^{-1} \leq s_0 \leq 2c\})$, which are compact as s_0 and h are proper. Since s_0 is continuous, Q' is contained in the interior of Q, and since $c^{-1}s_0 \leq s \leq cs_0$ on M, the set Q' contains the preimage of $\{1/2 \leq s \leq 2\} \cap M$. By Construction-Proposition 3.7(2), h^*s^2 is a locally bounded plurisubharmonic function on X', so h^*s^2 is bounded on the compact set Q. Thus, by [Dem12, Chapter III, (3.3)], applied to the compact sets Q' and Q, the Bedford–Taylor product $(\sqrt{-1}\partial\bar\partial h^*s^2)^{2n-2}$ has finite mass on Q'. Since $\omega' \wedge \overline{\omega'}$ is a smooth form on X', $(\sqrt{-1}\partial\bar\partial h^*s^2)^{2n-2} = (2h^*\kappa)^{2n-2}$ on $h^{-1}(M)$, and h is an isomorphism over a dense open subset of X, it follows that

$$C_n \int_{\{1/2\leq s\leq 2\}\cap M} q\, d\operatorname{vol}_g = \int_{\{1/2\leq s\leq 2\}\cap M} \omega \wedge \overline{\omega} \wedge \kappa^{2n-2} < +\infty.$$

Since ω is homogeneous of degree r and the flow of V at time t is $\lambda_{\exp(2t/r)}$ (Construction-Proposition 3.7(1)), we have $L_V\omega = 2\omega$. As $L_V g = 2g$ (Lemma 3.4), the function q is invariant under the flow of V. Since $d\operatorname{vol}_g = s^{4n-1} ds\, d\operatorname{vol}_{g_N}$, we get

$$\int_{1/2}^2 s^{4n-1} ds \cdot \int_N q\, d\operatorname{vol}_{g_N} = \int_{\{1/2\leq s\leq 2\}\cap M} q\, d\operatorname{vol}_g < +\infty.$$

Let $w := \log(1+q)$, which is a non-negative smooth function on M invariant under the flow of V. By Lemma 3.9,

$$\Delta_g w = \frac{\Delta_g q}{1+q} - \frac{|dq|_g^2}{(1+q)^2} \geq \frac{2|\nabla\omega|_g^2}{1+q} - \frac{2q|\nabla\omega|_g^2}{(1+q)^2} = \frac{2|\nabla\omega|_g^2}{(1+q)^2} \quad \text{on } M. \tag{3.11}$$

A direct calculation shows that $\Delta_g f = \Delta_{g_N}(f|_N)$ on N for any smooth function f on M with $Vf = 0$, where Δ_{g_N} is the Laplace–Beltrami operator of g_N. We also write w for $w|_N$. Thus $\Delta_{g_N} w \geq 0$ on N. By Proposition 3.15, $\mathrm{vol}(N, g_N) < +\infty$, so

$$\int_N e^w d\,\mathrm{vol}_{g_N} = \mathrm{vol}(N, g_N) + \int_N q d\,\mathrm{vol}_{g_N} < +\infty.$$

Fix smooth functions $\eta_j : N \to [0,1]$, $j = 1, 2, \ldots$, satisfying the conclusion of Proposition 3.15.

We claim that for any $\delta > 0$, there exists a real number $C_\delta > 0$ such that

$$w^2 z \leq \delta e^w + C_\delta z \log^2(e+z) \tag{3.12}$$

on N for any non-negative function z on N. Since $w \geq 0$ and the maximum of $a^2 e^{-a/2}$ on $[0, +\infty)$ is $16e^{-2} < 3$, we have $w^2 \leq 3e^{w/2}$. If $w^2 z > \delta e^w$ at a point of N, then $\delta e^w < 3e^{w/2} z$, so $0 \leq w < 2\log(3z/\delta)$ at the point, and thus

$$w^2 z < 4\log^2(3z/\delta) z \leq C_\delta z \log^2(e+z)$$

for $C_\delta := 4(1 + |\log(3/\delta)|)^2$, as $0 < \log(3z/\delta) \leq (1 + |\log(3/\delta)|)\log(e+z)$.

Applying (3.12) to $z := |d\eta_j|_{g_N}^2$, and using $\log(e+a^2) \leq 2\log(e+a)$ for $a \geq 0$, we get

$$\int_N w^2 |d\eta_j|_{g_N}^2 d\,\mathrm{vol}_{g_N} \leq \delta \int_N e^w d\,\mathrm{vol}_{g_N} + 4C_\delta \int_N \Phi(|d\eta_j|_{g_N}) d\,\mathrm{vol}_{g_N}.$$

Letting $j \to +\infty$ and then $\delta \to 0$, we get

$$\lim_{j\to+\infty} \int_N w^2 |d\eta_j|_{g_N}^2 d\,\mathrm{vol}_{g_N} = 0.$$

Since $\eta_j^2 w$ is a non-negative smooth function with compact support on N and $\Delta_{g_N} w \geq 0$, integration by parts and the Cauchy–Schwarz inequality imply

$$\begin{aligned}\int_N \eta_j^2 |dw|_{g_N}^2 d\,\mathrm{vol}_{g_N} &\leq -2\int_N \eta_j w \langle d\eta_j, dw\rangle_{g_N} d\,\mathrm{vol}_{g_N} \\ &\leq 2\Big(\int_N \eta_j^2 |dw|_{g_N}^2 d\,\mathrm{vol}_{g_N}\Big)^{1/2} \Big(\int_N w^2 |d\eta_j|_{g_N}^2 d\,\mathrm{vol}_{g_N}\Big)^{1/2}.\end{aligned}$$

Thus $\int_N \eta_j^2 |dw|_{g_N}^2 d\,\mathrm{vol}_{g_N} \leq 4\int_N w^2 |d\eta_j|_{g_N}^2 d\,\mathrm{vol}_{g_N} \to 0$ as $j \to +\infty$. Since any compact subset of N is contained in $\{\eta_j = 1\}$ for $j \gg 0$, we get $dw = 0$ on N. As w is invariant under the flow of V and $M = (0, +\infty) \times N$, it follows that $dw = 0$ on M. Thus $\Delta_g w = 0$ on M, and (3.11) implies $\nabla\omega = 0$ on M. □

3.4. **Extension of Killing flows.** In this subsection, we extend the flows of bounded Killing fields on N to the metric completion L of N, since N need not be complete. In Subsection 3.5, we show that the extended flows preserve N.

To control the integral curves of a Killing field starting at the points of a path, we need an estimate of the supremum of functions on rectangles in $\mathbb{R}^2$. In dimension two, finite Dirichlet energy does not bound the supremum, since the function $\log\log(e/|y|)$ on the punctured unit disk has finite Dirichlet energy and is unbounded. With the additional logarithmic factor in Φ, the corresponding energy controls the supremum.

Lemma 3.17. *Let $Q := (0,1) \times (0, q_0) \subset \mathbb{R}^2$ for a real number $q_0 > 0$, let $\Phi(a) := a^2 \log^2(e+a)$ for $a \geq 0$, and let v_j, $j = 1, 2, \ldots$, be smooth real functions on Q. Assume that*

$$\lim_{j\to+\infty} \int_Q \big(v_j^2 + \Phi(|dv_j|)\big) = 0.$$

Then $\lim_{j\to+\infty}\sup_Q|v_j|=0$.

Proof. We use the Euclidean norm and the Lebesgue measure dy on $\mathbb{R}^2$, and we denote by $|Q|=q_0$ the area of Q. Let v be a smooth real function on Q with $\int_Q v^2<+\infty$, let $v_Q:=|Q|^{-1}\int_Q v$ be the average of v on Q, and let $z\in Q$. We first show that

$$|v(z)-v_Q|\le\frac{(\operatorname{diam}Q)^2}{2|Q|}\int_Q\frac{|dv(y)|}{|y-z|}dy. \tag{3.13}$$

We use polar coordinates $y=z+t\theta$ centered at z, where $t>0$ and θ is in the unit circle $S^1\subset\mathbb{R}^2$, so that $dy=tdtd\theta$, where $d\theta$ is the arc length measure on S^1. Since Q is convex, open, and bounded, for any $\theta\in S^1$ there exists a real number $t_\theta\in(0,\operatorname{diam}Q]$ such that, for $t>0$, we have $z+t\theta\in Q$ if and only if $t<t_\theta$. For $y=z+t\theta\in Q$, the segment from z to y is contained in Q, so the fundamental theorem of calculus implies

$$|v(z)-v(y)|\le\int_0^t|dv(z+\sigma\theta)|d\sigma.$$

Integrating over $y\in Q$ and using the Tonelli theorem, we get

$$\begin{aligned}|Q|\cdot|v(z)-v_Q|&\le\int_Q|v(z)-v(y)|dy\le\int_{S^1}\int_0^{t_\theta}\Big(\int_\sigma^{t_\theta}tdt\Big)|dv(z+\sigma\theta)|d\sigma d\theta\\&\le\frac{(\operatorname{diam}Q)^2}{2}\int_{S^1}\int_0^{t_\theta}|dv(z+\sigma\theta)|d\sigma d\theta=\frac{(\operatorname{diam}Q)^2}{2}\int_Q\frac{|dv(y)|}{|y-z|}dy,\end{aligned}$$

where the last inequality follows from $t_\theta\le\operatorname{diam}Q$, and the last equality follows from $dy=\sigma d\sigma d\theta$ for $y=z+\sigma\theta$. This proves (3.13). Moreover, the Cauchy–Schwarz inequality implies

$$|v_Q|\le|Q|^{-1}\int_Q|v|\le|Q|^{-1/2}\|v\|_{L^2(Q)}.$$

Next, we bound the integral in (3.13). Let $0<\epsilon<1$, let

$$\begin{aligned}Q_1&:=\{y\in Q\mid|y-z|<\epsilon,\ |dv(y)|\le|y-z|^{-1/2}\},\\Q_2&:=\{y\in Q\mid|y-z|<\epsilon,\ |dv(y)|>|y-z|^{-1/2}\},\end{aligned}$$

and let $Q_3:=\{y\in Q\mid|y-z|\ge\epsilon\}$. On Q_1, we have $|dv(y)|\cdot|y-z|^{-1}\le|y-z|^{-3/2}$, so

$$\int_{Q_1}\frac{|dv(y)|}{|y-z|}dy\le\int_{|y-z|<\epsilon}|y-z|^{-3/2}dy=4\pi\sqrt{\epsilon}.$$

On Q_2, we have $|y-z|<1$, hence $\log(e+|dv(y)|)\ge\frac12\log(1/|y-z|)>0$. Thus the Cauchy–Schwarz inequality implies

$$\begin{aligned}\int_{Q_2}\frac{|dv(y)|}{|y-z|}dy&\le\Big(\int_{Q_2}\Phi(|dv(y)|)dy\Big)^{1/2}\Big(\int_{Q_2}\frac{dy}{|y-z|^2\log^2(e+|dv(y)|)}\Big)^{1/2}\\&\le\Big(\int_Q\Phi(|dv|)\Big)^{1/2}\Big(\int_{|y-z|<\epsilon}\frac{4dy}{|y-z|^2\log^2(1/|y-z|)}\Big)^{1/2}\\&=\frac{2\sqrt{2\pi}}{\sqrt{\log(1/\epsilon)}}\Big(\int_Q\Phi(|dv|)\Big)^{1/2}.\end{aligned}$$

On Q_3, we have $\epsilon\le|y-z|\le\operatorname{diam}Q$. Since $\Phi(a)\ge a^2$ for $a\ge0$, the Cauchy–Schwarz inequality implies

$$\begin{aligned}\int_{Q_3}\frac{|dv(y)|}{|y-z|}dy&\le\Big(\int_Q\Phi(|dv|)\Big)^{1/2}\Big(\int_{\epsilon\le|y-z|\le\operatorname{diam}Q}\frac{dy}{|y-z|^2}\Big)^{1/2}\\&=\sqrt{2\pi\log(\operatorname{diam}Q/\epsilon)}\Big(\int_Q\Phi(|dv|)\Big)^{1/2}.\end{aligned}$$

None of the bounds for the integrals over Q_1, Q_2, and Q_3 depends on z. Thus (3.13) and the bound for $|v_Q|$ imply

$$\sup_Q |v| \le \frac{\|v\|_{L^2(Q)}}{|Q|^{1/2}} + \frac{(\operatorname{diam} Q)^2}{2|Q|}\Bigg(4\pi\sqrt{\epsilon} + \Big(\frac{2\sqrt{2\pi}}{\sqrt{\log(1/\epsilon)}} + \sqrt{2\pi\log(\operatorname{diam} Q/\epsilon)}\Big)\Big(\int_Q \Phi(|dv|)\Big)^{1/2}\Bigg). \tag{3.14}$$

Finally, by assumption, $\int_Q v_j^2 < +\infty$ for $j \gg 0$, and both $\|v_j\|_{L^2(Q)}$ and $\int_Q \Phi(|dv_j|)$ tend to 0 as $j \to +\infty$. Applying (3.14) to $v = v_j$ and letting $j \to +\infty$, we get

$$\limsup_{j\to+\infty} \sup_Q |v_j| \le \frac{2\pi(\operatorname{diam} Q)^2\sqrt{\epsilon}}{|Q|}.$$

Letting $\epsilon \to 0$, we get $\lim_{j\to+\infty} \sup_Q |v_j| = 0$. □

Lemma 3.18. *Notation and conditions as in Set-up 3.14. Let L be the metric completion of (N, d_N), with distance d_L, and let W be a smooth Killing vector field on N with $\sup_N |W|_{g_N} < +\infty$. For $t \in \mathbb{R}$, let $N_t \subset N$ be the set of points $x \in N$ such that the integral curve of W starting at x is defined on the closed interval between 0 and t, and for $x \in N_t$, let $F_t(x)$ be the value at time t of the integral curve of W starting at x. Then the following hold.*

(1) For any $t \in \mathbb{R}$, the set N_t is open in N, and $N \setminus N_t$ has measure zero with respect to the Riemannian volume of g_N.

(2) There exists a continuous action $(t, p) \to \bar{F}_t(p)$ of $\mathbb{R}$ on L by isometries of (L, d_L) such that $\bar{F}_t(x) = F_t(x)$ for any $t \in \mathbb{R}$ and $x \in N_t$.

Proof. The proof is long, so we divide it into several steps. In this proof, we write $|\cdot|$ for $|\cdot|_{g_N}$ and $C_W := \sup_N |W|$, and measure zero and almost every refer to the Riemannian volume of g_N.

Step 1. In this step we collect the properties of the local flow of W that we use, and we show that N_t is open. For $x \in N$, let $(t^-(x), t^+(x))$ be the maximal open interval on which the integral curve of W starting at x is defined, where $-\infty \le t^-(x) < 0 < t^+(x) \le +\infty$, and let

$$\mathcal{D} := \{(t, x) \in \mathbb{R} \times N \mid t^-(x) < t < t^+(x)\}.$$

Since $(t^-(x), t^+(x))$ is an interval containing 0, we have $N_t = \{x \in N \mid (t, x) \in \mathcal{D}\}$ for any $t \in \mathbb{R}$. By [Mic08, 3.7 and Lemma 3.6], $\mathcal{D}$ is open in $\mathbb{R} \times N$, the map $\mathcal{D} \to N$, $(t, x) \to F_t(x)$, is smooth, and for $(t', x) \in \mathcal{D}$, the integral curve of W starting at $F_{t'}(x)$ is $t \to F_{t+t'}(x)$, $t^-(x) - t' < t < t^+(x) - t'$. Thus N_t is open in N. Moreover, for $t, t' \in \mathbb{R}$ and $x \in N_{t'}$, we have $F_{t'}(x) \in N_t$ if and only if $x \in N_{t+t'}$, and in this case

$$F_t(F_{t'}(x)) = F_{t+t'}(x).$$

Thus F_t is a bijection from N_t to N_{-t} with inverse F_{-t}. Since W is a Killing field, we have $F_t^* g_N = g_N$ on N_t. Hence $F_t : N_t \to N_{-t}$ is a diffeomorphism preserving g_N, and it preserves the Riemannian volume.

The function t^+ is lower semicontinuous on N: if $t^+(x) > t \ge 0$, then $(t, x) \in \mathcal{D}$, so $(t, x') \in \mathcal{D}$, that is, $t^+(x') > t$, for any x' in a neighborhood of x, since $\mathcal{D}$ is open. Moreover, let $x \in N$ with $t^+(x) < +\infty$. We show that the curve $t \to F_t(x)$, $0 \le t < t^+(x)$, is not contained in any compact subset of N. Let $Z \subset N$ be a compact subset. Since $\mathcal{D}$ is open and contains $\{0\} \times Z$, there exists $\delta > 0$ such that $[-\delta, \delta] \times Z \subset \mathcal{D}$, that is, $t^+ > \delta$ on Z. Since $t^+(F_t(x)) = t^+(x) - t$ for $0 \le t < t^+(x)$, we get $F_t(x) \notin Z$ for $t^+(x) - \delta \le t < t^+(x)$.

Step 2. In this step we fix $t_0 > 0$ and an isotopy φ of N with compact support, and we show that, for almost every $x \in N$, the functions $\eta_j \circ \Gamma_x$ defined below converge to 1 uniformly.

By Proposition 3.15, $\operatorname{vol}(N, g_N) < +\infty$. Fix smooth functions $\eta_j : N \to [0, 1]$, $j = 1, 2, \ldots$, satisfying the conclusion of Proposition 3.15. Since $0 \le 1 - \eta_j \le 1$, $\lim_{j\to+\infty} \eta_j(x) = 1$ for any

$x \in N$, and $\mathrm{vol}(N, g_N) < +\infty$, the dominated convergence theorem implies $\lim_{j\to+\infty} \int_N (1-\eta_j)^2 d\,\mathrm{vol}_{g_N} = 0$. Let $h_j := (1-\eta_j)^2 + \Phi(|d\eta_j|)$. Since $\lim_{j\to+\infty} \int_N h_j d\,\mathrm{vol}_{g_N} = 0$ and any subsequence of (η_j) also satisfies the conclusion of Proposition 3.15, possibly replacing (η_j) with a subsequence, we may assume that

$$\sum_{j=1}^{+\infty} \int_N h_j d\,\mathrm{vol}_{g_N} < +\infty. \tag{3.15}$$

Let $t_0 > 0$, and let $\varphi : [0,1] \times N \to N$, $(\tau, x) \to \varphi_\tau(x)$, be an isotopy of N with compact support, that is, a smooth map such that $\varphi_0 = \mathrm{id}_N$, φ_τ is a diffeomorphism of N for any $\tau \in [0,1]$, and there exists a compact subset $Z_\varphi \subset N$ such that $\varphi_\tau(x) = x$ for any $\tau \in [0,1]$ and $x \in N \backslash Z_\varphi$. Since $[0,1] \times Z_\varphi$ is compact, there exists a real number $C_1 \geq 1$ such that

$$|\partial_\tau \varphi_\tau(x)| \leq C_1 \quad \text{and} \quad \varphi_\tau^* d\,\mathrm{vol}_{g_N} \geq C_1^{-1} d\,\mathrm{vol}_{g_N} \ \text{ at } x$$

for any $\tau \in [0,1]$ and $x \in N$. For $x \in N$, let $t_\varphi(x) := \min_{\tau\in[0,1]} t^+(\varphi_\tau(x))$. Since t^+ is lower semicontinuous by **Step 1** and φ is continuous, the function $(\tau, x) \to t^+(\varphi_\tau(x))$ is lower semicontinuous on $[0,1] \times N$. Thus the minimum is attained, and $0 < t_\varphi(x) \leq +\infty$. Moreover, t_φ is lower semicontinuous: if $t_\varphi(x) > t$, then any point of $[0,1] \times \{x\}$ has a product neighborhood on which $t^+ \circ \varphi > t$, finitely many such neighborhoods cover $[0,1] \times \{x\}$, and $t_\varphi > t$ on the intersection of the corresponding neighborhoods of x. Let $q(x) := \min\{t_\varphi(x), t_0\}$. Then $q : N \to (0, t_0]$ is lower semicontinuous, so the set $\{(x,t) \in N \times \mathbb{R} \mid t < q(x)\}$ is open.

For $x \in N$, let $Q_x := (0,1) \times (0, q(x))$, and let

$$\Gamma_x(\tau, t) := F_t(\varphi_\tau(x)), \qquad (\tau, t) \in [0,1] \times [0, q(x)).$$

Since $t < q(x) \leq t^+(\varphi_\tau(x))$, the map Γ_x is well defined and continuous, and it is smooth on Q_x by **Step 1**. We have $\partial_t \Gamma_x = W(\Gamma_x)$ and $\partial_\tau \Gamma_x(\tau, t) = dF_t(\partial_\tau \varphi_\tau(x))$. Since F_t preserves g_N, we get $|\partial_t \Gamma_x| \leq C_W$ and $|\partial_\tau \Gamma_x| \leq C_1$. For any j, let $v_{j,x} := 1 - \eta_j \circ \Gamma_x$ on Q_x. With respect to the Euclidean norm in the coordinates (τ, t), we have $|dv_{j,x}| \leq C_2 |d\eta_j| \circ \Gamma_x$, where $C_2 := C_1 + C_W$. For a constant $C \geq 1$ and $a \geq 0$, we have

$$\log(e + Ca) \leq \log C + \log(e + a) \leq (1 + \log C) \log(e + a),$$

hence $\Phi(Ca) \leq C^2 (1 + \log C)^2 \Phi(a)$. Since Φ is increasing, we get

$$v_{j,x}^2 + \Phi(|dv_{j,x}|) \leq C_3 h_j \circ \Gamma_x \quad \text{on } Q_x, \qquad \text{where } C_3 := C_2^2 (1 + \log C_2)^2.$$

Let $\beta_j(x) := \int_{Q_x} h_j(\Gamma_x(\tau, t)) d\tau dt$. Since the set $\{t < q(x)\}$ is open, the function $(x, \tau, t) \to h_j(\Gamma_x(\tau, t))$, extended by zero to the set where $t \geq q(x)$, is a non-negative Borel function on $N \times [0,1] \times [0, t_0]$. Thus the Tonelli theorem implies

$$\int_N \beta_j d\,\mathrm{vol}_{g_N} = \int_0^1 \int_0^{t_0} \Big(\int_{\{x\in N | t < q(x)\}} h_j(F_t(\varphi_\tau(x))) d\,\mathrm{vol}_{g_N}(x) \Big) dt d\tau.$$

Fix $\tau \in [0,1]$ and $t \in [0, t_0]$. If $t < q(x)$, then $\varphi_\tau(x) \in N_t$. Since $\varphi_\tau^* d\,\mathrm{vol}_{g_N} \geq C_1^{-1} d\,\mathrm{vol}_{g_N}$, and since $F_t : N_t \to N_{-t}$ is a diffeomorphism preserving the Riemannian volume (**Step 1**), the change of variables formula implies

$$\begin{aligned} \int_{\{x\in N | t < q(x)\}} h_j(F_t(\varphi_\tau(x))) d\,\mathrm{vol}_{g_N}(x) &\leq C_1 \int_{N_t} h_j(F_t(y)) d\,\mathrm{vol}_{g_N}(y) \\ &= C_1 \int_{N_{-t}} h_j d\,\mathrm{vol}_{g_N} \leq C_1 \int_N h_j d\,\mathrm{vol}_{g_N}. \end{aligned}$$

Thus

$$\int_N \beta_j d\,\mathrm{vol}_{g_N} \leq C_1 t_0 \int_N h_j d\,\mathrm{vol}_{g_N},$$

and (3.15) implies $\int_N \sum_j \beta_j d\,\mathrm{vol}_{g_N} < +\infty$. Hence $\sum_j \beta_j(x) < +\infty$, and $\lim_{j\to+\infty} \beta_j(x) = 0$, for almost every $x \in N$. For any such x, we have

$$\lim_{j\to+\infty} \int_{Q_x} \left(v_{j,x}^2 + \Phi(|dv_{j,x}|)\right) \leq C_3 \lim_{j\to+\infty} \beta_j(x) = 0,$$

so Lemma 3.17, applied to Q_x and the functions $v_{j,x}$, implies

$$\lim_{j\to+\infty} \sup_{Q_x} |1 - \eta_j \circ \Gamma_x| = 0. \tag{3.16}$$

Step 3. In this step we show that $\varphi_\tau(x) \in N_{t_0}$ for any $\tau \in [0,1]$ and almost every $x \in N$, and we prove (1). Let $x \in N$ satisfy (3.16). Suppose that $t_\varphi(x) \leq t_0$, and let $\tau_0 \in [0,1]$ with $t^+(\varphi_{\tau_0}(x)) = t_\varphi(x)$. Then $q(x) = t^+(\varphi_{\tau_0}(x)) < +\infty$. By (3.16), there exists j such that $\eta_j \circ \Gamma_x \geq 1/2$ on Q_x. Since Q_x is dense in $[0,1] \times [0, q(x))$ and Γ_x is continuous on $[0,1] \times [0, q(x))$, we get

$$\eta_j(F_t(\varphi_{\tau_0}(x))) = \eta_j(\Gamma_x(\tau_0, t)) \geq 1/2 \quad \text{for any } 0 \leq t < t^+(\varphi_{\tau_0}(x)),$$

also when $\tau_0 \in \{0,1\}$. Thus the curve $t \to F_t(\varphi_{\tau_0}(x))$, $0 \leq t < t^+(\varphi_{\tau_0}(x))$, is contained in the support of η_j, which is compact, contrary to **Step 1**. Hence $t_\varphi(x) > t_0$, that is, $t^+(\varphi_\tau(x)) > t_0$ for any $\tau \in [0,1]$. Thus, for almost every $x \in N$,

$$\varphi_\tau(x) \in N_{t_0} \quad \text{for any } \tau \in [0,1]. \tag{3.17}$$

We prove (1). Since N_t is open by **Step 1** and $N_0 = N$, it suffices to show that $N \backslash N_t$ has measure zero for $t \neq 0$. If $t > 0$, then (3.17), applied to $t_0 = t$ and to the isotopy with $\varphi_\tau = \mathrm{id}_N$ for any $\tau \in [0,1]$, shows that $N \backslash N_t$ has measure zero. The vector field $-W$ is a smooth Killing field with $\sup_N |-W| < +\infty$, and the integral curve of $-W$ starting at x is $t \to F_{-t}(x)$. Thus the case $t > 0$, applied to $-W$, shows that $N \backslash N_t$ has measure zero for $t < 0$.

Step 4. In this step we show that

$$d_N(F_t(x), F_t(x')) = d_N(x, x') \quad \text{for any } t \in \mathbb{R} \text{ and } x, x' \in N_t. \tag{3.18}$$

Let $t_0 > 0$ and $x, x' \in N_{t_0}$. Since any piecewise smooth path can be reparametrized to a smooth path, and the length of a path does not change under reparametrization, $d_N(x, x')$ is the infimum of the lengths of smooth paths $\gamma : [0,1] \to N$ from x to x'. Let γ be such a path. Gluing, by a partition of unity, vector fields that are constant in finitely many charts covering $\gamma([0,1])$, we obtain a smooth time-dependent vector field $(Y_\tau)_{\tau\in[0,1]}$ on N, with supports in a fixed compact subset of N, such that $Y_\tau(\gamma(\tau)) = \gamma'(\tau)$ for any $\tau \in [0,1]$. Let φ_τ be the flow of (Y_τ) from time 0 to time τ. Then φ is an isotopy of N with compact support, and $\varphi_\tau(x) = \gamma(\tau)$ for any $\tau \in [0,1]$ by the uniqueness of integral curves.

By (3.17), applied to t_0 and φ, the set of points $y \in N$ such that $\varphi_\tau(y) \in N_{t_0}$ for any $\tau \in [0,1]$ has complement of measure zero, so it is dense in N. Thus there exist $x_l \in N$, $l = 1, 2, \ldots$, such that $\lim_{l\to+\infty} x_l = x$ and $\varphi_\tau(x_l) \in N_{t_0}$ for any $\tau \in [0,1]$ and any l. By **Step 1**, $\tau \to F_{t_0}(\varphi_\tau(x_l))$ is a smooth path from $F_{t_0}(x_l)$ to $F_{t_0}(\varphi_1(x_l))$, and since F_{t_0} preserves g_N, the length of $\tau \to F_{t_0}(\varphi_\tau(x_l))$ is equal to $\int_0^1 |\partial_\tau \varphi_\tau(x_l)| d\tau$. Since φ is smooth, $\partial_\tau \varphi_\tau(x_l)$ converges to $\partial_\tau \varphi_\tau(x) = \gamma'(\tau)$ uniformly in τ. Moreover, since N_{t_0} is open, F_{t_0} is continuous on N_{t_0}, $\varphi_1(x) = x'$, and d_N induces the topology of N, we have

$$\lim_{l\to+\infty} d_N(F_{t_0}(x_l), F_{t_0}(x)) = 0 \quad \text{and} \quad \lim_{l\to+\infty} d_N(F_{t_0}(\varphi_1(x_l)), F_{t_0}(x')) = 0.$$

Thus the triangle inequality implies

$$d_N(F_{t_0}(x), F_{t_0}(x')) \leq \lim_{l\to+\infty} \int_0^1 |\partial_\tau \varphi_\tau(x_l)| d\tau = \int_0^1 |\gamma'(\tau)| d\tau.$$

Taking the infimum over γ, we get

$$d_N(F_{t_0}(x), F_{t_0}(x')) \leq d_N(x, x'). \tag{3.19}$$

The vector field $-W$ satisfies the assumptions on W, and the integral curve of $-W$ starting at $y \in N$ is $t \to F_{-t}(y)$. By **Step 1**, $F_{t_0}(x), F_{t_0}(x') \in N_{-t_0}$, $F_{-t_0}(F_{t_0}(x)) = x$, and $F_{-t_0}(F_{t_0}(x')) = x'$. Thus (3.19) for $-W$ and the points $F_{t_0}(x), F_{t_0}(x')$ implies $d_N(x, x') \leq d_N(F_{t_0}(x), F_{t_0}(x'))$. Hence (3.18) holds for $t > 0$. Applying the case $t > 0$ to $-W$, we get (3.18) for $t < 0$, and (3.18) holds for $t = 0$ since $F_0 = \mathrm{id}_N$.

Step 5. In this step we prove (2). We regard N as a dense subset of L, so that d_L restricts to d_N on N. Since any non-empty open subset of N has positive volume, (1) implies that, for any $t, t' \in \mathbb{R}$, the sets N_t and $N_t \cap N_{t'}$ are dense in N, hence dense in L.

Let $t \in \mathbb{R}$. By **Step 1** and (3.18), $F_t : N_t \to N_{-t}$ is a bijection preserving d_L. Since L is complete, F_t extends uniquely to a map $\bar{F}_t : L \to L$ preserving d_L. The image of $\bar{F}_t$ is complete, hence closed in L, and contains the dense subset N_{-t} of L. Thus $\bar{F}_t$ is a surjective isometry of (L, d_L), and $\bar{F}_t(x) = F_t(x)$ for any $x \in N_t$.

Let $t, t' \in \mathbb{R}$. By **Step 1**, for any $x \in N_{t'} \cap N_{t+t'}$, we have $F_{t'}(x) \in N_t$ and

$$\bar{F}_t(\bar{F}_{t'}(x)) = F_t(F_{t'}(x)) = F_{t+t'}(x) = \bar{F}_{t+t'}(x).$$

Since $N_{t'} \cap N_{t+t'}$ is dense in L, and $\bar{F}_t \circ \bar{F}_{t'}$ and $\bar{F}_{t+t'}$ are continuous, we get $\bar{F}_t \circ \bar{F}_{t'} = \bar{F}_{t+t'}$. Moreover, $\bar{F}_0 = \mathrm{id}_L$ since $F_0 = \mathrm{id}_N$. Thus $(t, p) \to \bar{F}_t(p)$ is an action of $\mathbb{R}$ on L by isometries of (L, d_L).

It remains to show that the action is continuous. Let $t, t' \in \mathbb{R}$ and $x \in N_t \cap N_{t'}$. The interval $(t^-(x), t^+(x))$ contains t and t', and $|W| \leq C_W$ on N, where we recall that $C_W = \sup_N |W|$. Thus the integral curve of W starting at x joins $F_{t'}(x)$ to $F_t(x)$ by a path of length at most $C_W|t - t'|$, and $d_L(\bar{F}_t(x), \bar{F}_{t'}(x)) \leq C_W|t - t'|$. Since $N_t \cap N_{t'}$ is dense in L, and $\bar{F}_t$ and $\bar{F}_{t'}$ are continuous, we get $d_L(\bar{F}_t(p), \bar{F}_{t'}(p)) \leq C_W|t - t'|$ for any $p \in L$. Hence, for $t, t' \in \mathbb{R}$ and $p, p' \in L$,

$$d_L(\bar{F}_t(p), \bar{F}_{t'}(p')) \leq d_L(\bar{F}_t(p), \bar{F}_{t'}(p)) + d_L(\bar{F}_{t'}(p), \bar{F}_{t'}(p')) \leq C_W|t - t'| + d_L(p, p').$$

Thus the map $\mathbb{R} \times L \to L$, $(t, p) \to \bar{F}_t(p)$, is continuous. □

3.5. **Completeness of Killing fields.** In this subsection, we show that the extended flows of Subsection 3.4 preserve N, prove Theorem 3.20, and then prove Theorem 3.1.

By Lemma 3.18, the flow of a bounded Killing field on N extends to isometries of the metric completion L of N, and Proposition 3.19(1) below characterizes the points of N among the points of L by a condition on neighborhoods that isometries of L preserve, so the extended flow preserves N and Proposition 3.19(2) follows. Proposition 3.19(1) excludes the behavior of a constant vector field on the punctured plane, whose flow extends to translations of the plane that do not preserve the punctured plane.

Proposition 3.19. *Notation and conditions as in Set-up 3.14, and let L be the metric completion of (N, d_N), with distance d_L. Then the following hold.*

(1) A point $p \in L$ belongs to N if and only if p has an open neighborhood U in L such that $(U, d_L|_U)$ is isometric to an open subset of a smooth Riemannian manifold of dimension $4n - 1$, endowed with the restriction of the Riemannian distance of the manifold.

(2) Any smooth Killing vector field W on N with $\sup_N |W|_{g_N} < +\infty$ is complete.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that N is open in L and that any point of N has an open neighborhood as in (1).

By [Li21, Definition 2.1], $M = (0, +\infty) \times N$ and $g = ds^2 + s^2 g_N$, where s is the coordinate of $(0, +\infty)$, and $V = s\partial_s$. In particular, N is a smooth manifold of dimension $4n - 1$. We regard N as a dense subset of L. Thus $d_L(x, x') = d_N(x, x')$ for any $x, x' \in N$.

Let $x \in N$. Since d_N induces the topology of the manifold N and N is locally compact, there exists $\epsilon > 0$ such that the set $\{x' \in N \mid d_N(x, x') \leq 2\epsilon\}$ is compact. Let $p \in L$ with $d_L(x, p) < \epsilon$,

and let (z_l) be a sequence in N converging to p in L. Then $d_N(x, z_l) \leq 2\epsilon$ for any $l \gg 0$, so a subsequence of (z_l) converges in N to a point $z \in N$. Thus $p = z \in N$. Hence the open ball

$$U_x := \{p \in L \mid d_L(x, p) < \epsilon\}$$

is contained in N. In particular, N is open in L, and $(U_x, d_L|_{U_x})$ is the open subset U_x of the Riemannian manifold (N, g_N), endowed with the restriction of d_N.

Step 2. In this step we fix a point $p \in L$ with an open neighborhood U as in (1), and we construct the product $U' = (1/2, 2) \times U$ with a cone metric, together with cutoff functions on U and on U'.

Let $p \in L$, and let U be an open neighborhood of p in L as in (1). Since any open neighborhood of p contained in U satisfies the condition in (1) as well, possibly replacing U with a connected open neighborhood of p contained in U, we may assume that U is connected. We identify $(U, d_L|_U)$ with an open subset of a smooth Riemannian manifold of dimension $4n-1$ as in (1), and let g_U be the restriction of the Riemannian metric to U. By **Step 1**, $U \cap N$ is open in U, and $U \cap N$ is dense in U because N is dense in L. The identity map of $U \cap N$ is locally distance preserving for g_N and for g_U, hence a smooth isometry by the Myers–Steenrod theorem [MS39]. Thus the smooth structures on $U \cap N$ induced from U and from N coincide, and $g_U = g_N$ on $U \cap N$.

Fix smooth functions $\eta_j : N \to [0, 1]$, $j = 1, 2, \ldots$, satisfying the conclusion of Proposition 3.15. Since $\Phi(a) \geq a^2$ for any $a \geq 0$,

$$\lim_{j \to +\infty} \int_N |d\eta_j|^2_{g_N} d\operatorname{vol}_{g_N} = 0.$$

We still denote by η_j the function on U which equals η_j on $U \cap N$ and 0 on $U \backslash N$. It is smooth on U, because $\operatorname{Supp} \eta_j$ is a compact subset of N, hence closed in L, and U is the union of the open subsets $U \cap N$ and $U \backslash \operatorname{Supp} \eta_j$. Since $g_U = g_N$ on $U \cap N$ and $d\eta_j = 0$ on $U \backslash \operatorname{Supp} \eta_j$,

$$\lim_{j \to +\infty} \int_U |d\eta_j|^2_{g_U} d\operatorname{vol}_{g_U} = 0.$$

Moreover, η_j converges pointwise on U to the characteristic function $1_{U \cap N}$ of $U \cap N$. Thus for any smooth vector field ξ with compact support in U, by dominated convergence,

$$\int_U 1_{U \cap N} \operatorname{div} \xi d\operatorname{vol}_{g_U} = \lim_{j \to +\infty} \int_U \eta_j \operatorname{div} \xi d\operatorname{vol}_{g_U} = - \lim_{j \to +\infty} \int_U d\eta_j(\xi) d\operatorname{vol}_{g_U} = 0,$$

where div is the divergence with respect to g_U. Hence $1_{U \cap N}$ is almost everywhere equal to a constant on the connected manifold U. Since $U \cap N$ is a non-empty open subset of U, the set $U \backslash N$ has measure zero with respect to $d\operatorname{vol}_{g_U}$.

Let $U' := (1/2, 2) \times U$, let $g_{U'} := ds^2 + s^2 g_U$, where s is the coordinate of $(1/2, 2)$, and let ∇' be the Levi-Civita connection of $g_{U'}$. We regard the open subset $(1/2, 2) \times (U \cap N)$ of U' as an open subset of $M = (0, +\infty) \times N$, denote it by $U' \cap M$, and denote by $U' \backslash M$ the complement of $U' \cap M$ in U'. Then $g_{U'} = g$ on $U' \cap M$, the function s on $U' \cap M$ is the radius function of g, the set $U' \cap M$ is dense in U', and $U' \backslash M$ has measure zero. We still denote by η_j the smooth function $(s, x) \to \eta_j(x)$ on U'. Then $\eta_j \to 1$ pointwise on $U' \cap M$, and for any smooth form or tensor φ with compact support in U', the product $\eta_j \varphi$ has compact support in $U' \cap M$. Since $|d\eta_j|_{g_{U'}}(s, x) = s^{-1} |d\eta_j|_{g_U}(x)$ and $d\operatorname{vol}_{g_{U'}} = s^{4n-1} ds d\operatorname{vol}_{g_U}$,

$$\int_{U'} |d\eta_j|^2_{g_{U'}} d\operatorname{vol}_{g_{U'}} \leq \int_{1/2}^2 s^{4n-3} ds \cdot \int_U |d\eta_j|^2_{g_U} d\operatorname{vol}_{g_U} \to 0.$$

Thus by the Cauchy–Schwarz inequality, the integral of $|d\eta_j|_{g_{U'}}$ with respect to $d\operatorname{vol}_{g_{U'}}$ over any compact subset of U' tends to 0 as $j \to +\infty$.

Step 3. In this step we extend I to a complex structure on U' which is parallel with respect to ∇'.

Let ∇ be the Levi-Civita connection of g. In this step and the next, pointwise norms and inner products on U' are taken with respect to $g_{U'}$, and integrals on U' with respect to $d\operatorname{vol}_{g_{U'}}$. Since g is a Kähler metric with complex structure I, we have $\nabla' I = \nabla I = 0$ and $|I| = 2\sqrt{n}$ on $U' \cap M$. Let Ξ be a smooth section of $T^*U' \otimes \operatorname{End}(TU')$ with compact support, and let ∇'^* be the formal adjoint of ∇'. Since $\eta_j \Xi$ has compact support in $U' \cap M$, integration by parts implies

$$\left| \int_{U' \cap M} \eta_j \langle I, \nabla'^* \Xi \rangle \right| = \left| \int_{U' \cap M} \langle I, \Xi(\operatorname{grad} \eta_j) \rangle \right| \leq 2\sqrt{n} \sup_{U'} |\Xi| \int_{\operatorname{Supp} \Xi} |d\eta_j|,$$

where $\Xi(\operatorname{grad} \eta_j)$ is the section of $\operatorname{End}(TU')$ obtained by inserting the gradient of η_j into Ξ. Let $\tilde{I}$ be the bounded measurable section of $\operatorname{End}(TU')$ which equals I on $U' \cap M$ and 0 on $U' \backslash M$. Letting $j \to +\infty$, **Step 2** and dominated convergence imply that $\int_{U'} \langle \tilde{I}, \nabla'^* \Xi \rangle = 0$. Thus $\nabla' \tilde{I} = 0$ in the sense of distributions.

In local coordinates, $\nabla' \tilde{I} = 0$ reads

$$\partial_k \tilde{I}^a_b = \sum_c \big(\Gamma^c_{kb} \tilde{I}^a_c - \Gamma^a_{kc} \tilde{I}^c_b \big),$$

where Γ^a_{bc} are the Christoffel symbols of $g_{U'}$. Since the right-hand side is locally bounded, the distributional first derivatives of $\tilde{I}$ are locally bounded, so $\tilde{I}$ is almost everywhere equal to a locally Lipschitz section [EG15, Theorem 4.5]. Differentiating the formula for $\partial_k \tilde{I}^a_b$, we see that for any $m \geq 0$, the distributional derivatives of $\tilde{I}$ of order $m+1$ are linear combinations, with smooth coefficients, of the distributional derivatives of $\tilde{I}$ of order at most m. Thus by induction on m, the distributional derivatives of $\tilde{I}$ of any order are locally Lipschitz, and in particular continuous. Hence $\tilde{I}$ is almost everywhere equal to a smooth section I' of $\operatorname{End}(TU')$, and $\nabla' I' = 0$.

Since I' and I are continuous on $U' \cap M$ and equal almost everywhere, $I' = I$ on $U' \cap M$. Since $U' \cap M$ is dense in U', we have $I'^2 = -\operatorname{id}_{TU'}$ and $g_{U'}(I'v, I'v') = g_{U'}(v, v')$ for any $(s, x) \in U'$ and $v, v' \in T_{(s,x)}U'$. Since I' is parallel with respect to the torsion-free connection ∇', the Nijenhuis tensor of I' vanishes. By the Newlander–Nirenberg theorem [Dem12, Chapter VIII, Theorem (11.8)], there exists a unique structure of complex manifold on U' with complex structure I'. From now on, we denote I' by I. Thus $(U', g_{U'}, I)$ is a Kähler manifold, and the structure of complex manifold on $U' \cap M$ induced from U' coincides with the one induced from M.

Step 4. In this step we construct a holomorphic map $\psi : U' \to X^{\mathrm{an}}$ which is the inclusion on $U' \cap M$.

By Construction-Proposition 3.7(3), $s_0 \leq cs < 2c$ on $U' \cap M$. Since s_0 is proper, the closure of $U' \cap M$ in X^{an} is compact. Thus any $f \in R$ is bounded on $U' \cap M$.

Let $f \in R$, and let F be the bounded measurable function on U' which equals f on $U' \cap M$ and 0 on $U' \backslash M$. Let φ be a smooth $(2n, 2n-1)$-form with compact support in U'. Since $\eta_j \varphi$ has compact support in $U' \cap M$ and f is holomorphic on $U' \cap M$, Stokes' theorem implies

$$\int_{U' \cap M} f \bar{\partial}(\eta_j \varphi) = \int_{U' \cap M} d(f \eta_j \varphi) = 0.$$

Thus by dominated convergence and **Step 2**,

$$\begin{aligned} \left| \int_{U'} F \bar{\partial} \varphi \right| &= \lim_{j \to +\infty} \left| \int_{U' \cap M} \eta_j f \bar{\partial} \varphi \right| = \lim_{j \to +\infty} \left| \int_{U' \cap M} f \bar{\partial} \eta_j \wedge \varphi \right| \\ &\leq C_n \sup_{U' \cap M} |f| \sup_{U'} |\varphi| \lim_{j \to +\infty} \int_{\operatorname{Supp} \varphi} |d\eta_j| = 0, \end{aligned}$$

where C_n is a constant depending only on n. Hence $\bar{\partial} F = 0$ in the sense of currents, and by [Dem12, Chapter I, Corollary (3.30)], F is almost everywhere equal to a holomorphic function $\tilde{f}$ on U'. Since $\tilde{f}$ and f are continuous on $U' \cap M$, we have $\tilde{f} = f$ on $U' \cap M$.

Let $f_1, \ldots, f_k$ be generators of the $\mathbb{C}$-algebra R, and let $\iota : X^{\mathrm{an}} \to \mathbb{C}^k$ be the closed embedding defined by $f_1, \ldots, f_k$. The holomorphic functions $\tilde{f}_1, \ldots, \tilde{f}_k$ define a holomorphic map $U' \to \mathbb{C}^k$,

which maps the dense subset $U' \cap M$ of U' into the closed subset $\iota(X^{\mathrm{an}})$ of $\mathbb{C}^k$. Thus we obtain a holomorphic map $\psi : U' \to X^{\mathrm{an}}$ with $\iota \circ \psi = (\tilde{f}_1, \dots, \tilde{f}_k)$ and $\psi(s, x) = (s, x)$ for any $(s, x) \in U' \cap M$.

Step 5. In this step we show that $y := \psi(1, p)$ is different from o and that any fiber of ψ is discrete.

By Construction-Proposition 3.7(2), there exists a locally bounded plurisubharmonic function w on X with $w = s^2$ on M and $w(o) = 0$. Near any point of X^{an}, the function w is the composite of a local embedding of X^{an} into some $\mathbb{C}^{k'}$ and a plurisubharmonic function on an open subset of $\mathbb{C}^{k'}$. Thus by [Dem12, Chapter I, Theorem (5.11)], $w \circ \psi$ is plurisubharmonic on U'.

By [Li21, Definition 2.1], $g(I\cdot, \cdot) = \frac{\sqrt{-1}}{2}\partial\bar{\partial}s^2$ on M. Since $g_{U'} = g$ on $U' \cap M$ and the structures of complex manifold on $U' \cap M$ induced from U' and from M coincide (**Step 3**), the equality $g_{U'}(I\cdot, \cdot) = \frac{\sqrt{-1}}{2}\partial\bar{\partial}s^2$ holds on the dense subset $U' \cap M$ of U', hence on U' by continuity. Thus s^2 is strictly plurisubharmonic on U'.

Since $w \circ \psi = s^2$ on $U' \cap M$ and $U' \backslash M$ has measure zero, $w \circ \psi = s^2$ almost everywhere on U'. Both functions are plurisubharmonic, so $w \circ \psi = s^2$ on U' by [Dem12, Chapter I, Theorem (5.5)]. In particular, $w(y) = 1 \neq 0 = w(o)$, so $y \neq o$.

Suppose that for some $z \in X^{\mathrm{an}}$, the fiber $\psi^{-1}(z)$ is not discrete. Then $\psi^{-1}(z)$ contains a connected complex submanifold Σ of U' with $\dim \Sigma > 0$. Since $s^2 = w \circ \psi$ is constant on Σ, the restriction $s^2|_\Sigma$ is not strictly plurisubharmonic, a contradiction.

Step 6. In this step we show that $y \in X_{\mathrm{reg}}$.

By **Step 5**, there exists an open neighborhood $\mathbb{B}$ of $(1, p)$ in U' which a holomorphic chart of U' maps onto an open ball in $\mathbb{C}^{2n}$, such that the closure $\bar{\mathbb{B}}$ of $\mathbb{B}$ is compact and contained in the domain of the chart, and $\bar{\mathbb{B}} \cap \psi^{-1}(y) = \{(1, p)\}$. Then $\psi(\partial\mathbb{B})$ is a compact subset of X^{an} with $y \notin \psi(\partial\mathbb{B})$. Let Ω_y be an open neighborhood of y in X^{an} disjoint from $\psi(\partial\mathbb{B})$, and let $\mathbb{B}_y := \mathbb{B} \cap \psi^{-1}(\Omega_y)$. Then $\psi|_{\mathbb{B}_y} : \mathbb{B}_y \to \Omega_y$ is proper, because the preimage in $\mathbb{B}_y$ of a compact subset of Ω_y is a closed subset of the compact set $\bar{\mathbb{B}}$. Any fiber of $\psi|_{\mathbb{B}_y}$ is compact because $\psi|_{\mathbb{B}_y}$ is proper, and discrete by **Step 5**, hence finite. By Remmert's proper mapping theorem [Dem12, Chapter II, (8.8)], $\psi(\mathbb{B}_y)$ is an analytic subset of Ω_y.

Since X is normal, X^{an} is normal [Gro03, Exposé XII, Proposition 2.1], and in particular locally irreducible at y [Dem12, Chapter II, Definition (7.4)]. For any open neighborhood $\Omega \subset \Omega_y$ of y, the set $\mathbb{B}_y \cap \psi^{-1}(\Omega)$ is an open neighborhood of $(1, p)$ in U', so it contains a non-empty open subset of $U' \cap M$, which ψ maps onto an open subset of X^{an} contained in $\psi(\mathbb{B}_y) \cap \Omega$. Thus by [Dem12, Chapter II, Proposition (4.26)], $\psi(\mathbb{B}_y)$ contains an open neighborhood of y. Replacing Ω_y with a smaller open neighborhood of y preserves the properties of Ω_y and $\mathbb{B}_y$ above. Thus, possibly replacing Ω_y with an open neighborhood of y contained in $\psi(\mathbb{B}_y)$, we may assume that $\psi(\mathbb{B}_y) = \Omega_y$.

Let $C \subset \mathbb{B}_y$ be the set of points at which the differential of $\iota \circ \psi$ has rank less than $2n$. Then C is an analytic subset of $\mathbb{B}_y$, and C is disjoint from $U' \cap M$ because ψ is the inclusion on $U' \cap M$. Since $U' \cap M$ is dense in U', we have $\dim C < 2n$. By [Dem12, Chapter II, (8.8)], $\psi(C)$ is an analytic subset of Ω_y, and $\dim \psi(C) < 2n$ by [Dem12, Chapter II, Lemma (8.1)(b)]. Thus

$$\Omega_y^\circ := \Omega_y \backslash (\psi(C) \cup X_{\mathrm{sing}})$$

is a dense open subset of Ω_y. Moreover, $\mathbb{B}_y \cap \psi^{-1}(\Omega_y^\circ)$ is dense in $\mathbb{B}_y$, because ψ maps any non-empty open subset of $\mathbb{B}_y$ onto a set containing a non-empty open subset of X^{an}, while $\Omega_y \backslash \Omega_y^\circ$ is an analytic subset of Ω_y of dimension less than $2n$.

Suppose that a point $z \in \Omega_y^\circ$ has two distinct preimages $w_1, w_2 \in \mathbb{B}_y$. Since $w_1, w_2 \notin C$ and $z \in X_{\mathrm{reg}}$, there exist an open neighborhood Ω_z of z and disjoint open neighborhoods G_1 of w_1 and G_2 of w_2 in $\mathbb{B}_y$ such that $\psi|_{G_1} : G_1 \to \Omega_z$ and $\psi|_{G_2} : G_2 \to \Omega_z$ are biholomorphic. For $i = 1, 2$, the set $G_i \cap U' \cap M$ is dense in G_i, so $\psi(G_i \cap U' \cap M)$ is a dense open subset of Ω_z. Thus $\psi(G_1 \cap U' \cap M) \cap \psi(G_2 \cap U' \cap M) \neq \emptyset$, which contradicts the injectivity of ψ on $U' \cap M$.

Hence ψ maps $\mathbb{B}_y \cap \psi^{-1}(\Omega_y^\circ)$ bijectively onto Ω_y°, and the inverse map is holomorphic. Composing with the holomorphic chart of U' around $\bar{\mathbb{B}}$, we regard the inverse map as a bounded holomorphic map from Ω_y° to $\mathbb{C}^{2n}$. Since $\Omega_y \backslash \Omega_y^\circ$ is a nowhere dense analytic subset of Ω_y, the inverse map extends holomorphically over $\psi(C) \cap X_{\mathrm{reg}}$ by the Riemann extension theorem [Dem12, Chapter II, Remark (4.2)], and then, since X^{an} is normal, to a holomorphic map $\psi' : \Omega_y \to \mathbb{C}^{2n}$ [Dem12, Chapter II, Definitions (7.1) and (7.4)]. For $z \in \Omega_y$ and a sequence (z_l) in Ω_y° converging to z, the properness of $\psi|_{\mathbb{B}_y}$ implies that a subsequence of $(\psi'(z_l))$ converges in $\mathbb{B}_y$. Thus $\psi'(\Omega_y) \subset \mathbb{B}_y$ and $\psi \circ \psi' = \mathrm{id}_{\Omega_y}$. Since $\psi'(\psi(w)) = w$ for any w in the dense subset $\mathbb{B}_y \cap \psi^{-1}(\Omega_y^\circ)$ of $\mathbb{B}_y$, we have $\psi' \circ \psi = \mathrm{id}_{\mathbb{B}_y}$. Thus $\psi|_{\mathbb{B}_y} : \mathbb{B}_y \to \Omega_y$ is biholomorphic, and y is a regular point of X^{an}, that is, $y \in X_{\mathrm{reg}}$.

Step 7. In this step we show that $p \in N$. Together with **Step 1**, this step proves (1).

Choose a sequence (s_l, x_l) in $U' \cap M$ converging to $(1, p)$ in U'. Then x_l converges to p in U, hence in L. On the other hand, $\psi(s_l, x_l) = (s_l, x_l)$ converges to y in X^{an}. Since $y \in X_{\mathrm{reg}}$ and $y \neq o$ by **Steps 5 and 6**, we have $y \in M$. Let $x_y \in N$ be the image of y under the projection $M = (0, +\infty) \times N \to N$. Since M is open in X^{an} and the projection is continuous, x_l converges to x_y in N. Since $d_L = d_N$ on N, the sequence (x_l) also converges to x_y in L. Thus $p = x_y \in N$ by the uniqueness of limits in L.

Step 8. In this step we prove (2).

Let W be a smooth Killing vector field on N with $\sup_N |W|_{g_N} < +\infty$. By Lemma 3.18, there exists a continuous action $(t, p) \to \bar{F}_t(p)$ of $\mathbb{R}$ on L by isometries of (L, d_L) such that for any $t \in \mathbb{R}$ and $x \in N$, if the integral curve of W starting at x is defined on the closed interval between 0 and t, then $\bar{F}_t(x)$ is the value at time t of the integral curve. For $t \in \mathbb{R}$, the isometry $\bar{F}_t$ of L maps a point with an open neighborhood as in (1) to a point with such a neighborhood. Thus $\bar{F}_t(N) \subset N$ by (1), and $\bar{F}_t(N) = N$ because $\bar{F}_{-t}(N) \subset N$.

Let $x \in N$, and let $c_x : \mathbb{R} \to N$ be the curve $t \to \bar{F}_t(x)$. Let $t_0 \in \mathbb{R}$, and let $z := c_x(t_0)$. There exist $\epsilon > 0$ and an integral curve $\gamma : (-\epsilon, \epsilon) \to N$ of W with $\gamma(0) = z$. For $|\tau| < \epsilon$, we have

$$c_x(t_0 + \tau) = \bar{F}_\tau(z) = \gamma(\tau).$$

Thus c_x is smooth near t_0, and $c_x'(t_0) = W(c_x(t_0))$. Since t_0 is arbitrary, c_x is an integral curve of W defined on $\mathbb{R}$ with $c_x(0) = x$. Hence W is complete. In particular, by the uniqueness of integral curves, $\bar{F}_t|_N$ is the flow of W at time t for any $t \in \mathbb{R}$. □

Theorem 3.20. *Notation and conditions as in Set-up 3.14. Let ∇ be the Levi-Civita connection of g, and let J and K be as in Construction-Proposition 3.3 for (M, g, I) and ω. Then the following hold.*

(1) $\nabla\omega = 0$.
(2) Any real linear combination of IV, JV, and KV is a complete vector field on M.
(3) The vector fields IV, JV, and KV integrate to a smooth action of $\mathrm{SU}(2)$ on M such that $\exp(t\mathbf{i})$, $\exp(t\mathbf{j})$, and $\exp(t\mathbf{k})$ act by the flows of IV, JV, and KV, respectively, where $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are the standard unit quaternions.

In particular, g and the action of $\mathrm{SU}(2)$ in (3) satisfy the conditions of Proposition 3.5.

Proof. By Proposition 3.16, (1) holds.

We prove (2). Since g is a Kähler metric with complex structure I, the endomorphism I is parallel and orthogonal with respect to g. By (1) and Construction-Proposition 3.3(2) and (4), J and K are parallel and orthogonal with respect to g. By [Li21, Definition 2.1], $M = (0, +\infty) \times N$ and $g = ds^2 + s^2 g_N$, where s is the coordinate of $(0, +\infty)$, and $V = s\partial_s$. By Lemma 3.4, $\nabla V = \mathrm{id}_{TM}$, $|V|_g = s$, and the gradient of s is V/s. Since I, J, and K are parallel,

$$\nabla(IV) = I, \quad \nabla(JV) = J, \quad \nabla(KV) = K.$$

Since I, J, and K are orthogonal with respect to g and square to $-\mathrm{id}_{TM}$, they are skew-adjoint with respect to g. Let $a, b, c \in \mathbb{R}$, and let $W := aIV + bJV + cKV$. Then $\nabla W = aI + bJ + cK$ is

skew-adjoint with respect to g, so W is a Killing vector field. Moreover, $ds(W) = s^{-1}g(V,W) = 0$ by the skew-adjointness, and $[V,W] = \nabla_V W - \nabla_W V = W - W = 0$. Thus W is tangent to the hypersurfaces $\{s\} \times N$ and invariant under the flow $(s,x) \to (e^t s, x)$ of V. Hence there exists a vector field W_N on N such that $W(s,x) = (0, W_N(x))$ in $T_{(s,x)}M = \mathbb{R} \oplus T_x N$ for any $(s,x) \in M$. The vector field W_N is the restriction of W to N, so W_N is a smooth Killing vector field on (N, g_N) and

$$\sup_N |W_N|_{g_N} \leq \big(|a| + |b| + |c|\big) \sup_N |V|_g = |a| + |b| + |c|.$$

By Proposition 3.19(2), W_N is complete. For $(s,x) \in M$, if γ is the integral curve of W_N with $\gamma(0) = x$, then $t \to (s, \gamma(t))$, $t \in \mathbb{R}$, is the integral curve of W starting at (s,x). Thus W is complete.

By (1), (2), and Lemma 3.10, (3) holds, and g and the action of SU(2) in (3) satisfy the conditions of Proposition 3.5. □

Proof of Theorem 3.1. The conditions of Theorem 3.1 are the conditions of Set-up 3.2 together with the conditions of Set-up 3.6. If $n = 1$, then by Theorem 3.11, there exist a Kähler cone metric on M and a smooth action of SU(2) on M satisfying the conditions of Proposition 3.5. If $n \geq 2$, then the conditions of Set-up 3.14 hold, and by Theorem 3.20, there exist a Kähler cone metric on M and a smooth action of SU(2) on M satisfying the conditions of Proposition 3.5. Thus by Proposition 3.5(2) and (3), there exists a ring automorphism $\rho : R \to R$ satisfying (1), (2), and (3) of Theorem 3.1, and $\rho(R_{d,\chi}) = R_{d,-\chi}$ for any $d \geq 0$ and $\chi \in X^*(T)$ by Proposition 3.5(4). □

Remark 3.21. Notation and conditions as in Set-up 3.2. Assume that (X, ω) is a symplectic singularity, and that $R = \bigoplus_{d \geq 0} R_d$ is a minimizing grading of (X, ω). Then the conditions of Set-up 3.6 hold by Theorem 2.17. Let g and V be as in Construction-Proposition 3.7, and let J and K be as in Construction-Proposition 3.3 for (M, g, I) and ω. Then g is a Ricci-flat Kähler cone metric on M invariant under T_c and $\lambda(S^1)$, the form ω is parallel with respect to g, and the vector fields IV, JV, and KV integrate to an action of SU(2) on M by Theorems 3.11 and 3.20. Moreover, (g, I, J, K) is a hyperkähler structure on M by Construction-Proposition 3.3.

The conditions of Set-up 3.6 hold for the following reason: (X, ω) is conical by Definition 2.8, because $\dim X = 2n > 0$ and Set-up 2.10 provides a grading with $R_0 = \mathbb{C}$ for which ω is homogeneous of degree $r > 0$. Thus (X, ω) is a conical symplectic singularity with a minimizing grading in the sense of Definition 2.14, and Theorem 2.17 implies that the log Fano cone $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$. By Construction-Proposition 3.7, T_c' contains the images of T_c and $\lambda(S^1)$, so g is invariant under T_c and $\lambda(S^1)$. The quadruple (g, I, J, K) is a hyperkähler structure because I, J, and K satisfy the quaternion relations, are orthogonal with respect to g, and are parallel, by Construction-Proposition 3.3(1)(2)(4) and the parallelism of ω. By Construction-Proposition 3.3(3), $\omega = g(PJ\cdot, \cdot) + ig(PK\cdot, \cdot)$ for the endomorphism P of Construction-Proposition 3.3, and P need not be a constant multiple of id_{TM} (Example 3.8).

4. Graded Poisson automorphism groups

In this section, we prove the properties of $\mathrm{Aut}_{\mathrm{gr}}(R)$, $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$, and G_R which we use to prove Theorem 1.8. The K-polystability of (E, Δ_E) controls the automorphisms of E preserving Δ_E, so we first prove that the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive, and then deduce that G_R is reductive in Subsection 4.3.

4.1. Algebraicity and Hamiltonian derivations.

In this subsection, we prove that $\mathrm{Aut}_{\mathrm{gr}}(R)$ and $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ are linear algebraic groups, and we describe the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ by the Poisson bracket on R_r.

Definition-Lemma 4.1. Notation and conditions as in Set-up 2.10. Let N_0 be a positive integer such that $W := \bigoplus_{1 \leq d \leq N_0} R_d$ generates R as a $\mathbb{C}$-algebra. Then the following hold.

(1) The restriction to W embeds $\mathrm{Aut}_{\mathrm{gr}}(R)$ and $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ as closed subgroups of
$$\prod_{1\le d\le N_0}\mathrm{GL}(R_d),$$
and the induced structures of linear algebraic groups on $\mathrm{Aut}_{\mathrm{gr}}(R)$ and $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ do not depend on N_0. Moreover, R_d is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$ for any $d\ge 0$.
(2) The Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is the space of $\mathbb{C}$-linear derivations δ of R with $\delta(R_d)\subset R_d$ for any $d\ge 0$, and the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ consists of the derivations δ in the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ with $\delta(\{f,h\})=\{\delta(f),h\}+\{f,\delta(h)\}$ for any $f,h\in R$. In particular, the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ contains the derivation eu of R with $\mathrm{eu}(f)=jf$ for any $j\ge 0$ and $f\in R_j$.
(3) Any algebraic torus $T\subset\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ is contained in the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$, and in a maximal torus of the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$.

We denote by G_R the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$.

Proof. Since R is generated by finitely many homogeneous elements of positive degree, $\dim R_j<+\infty$ for any $j\ge 0$. We grade the symmetric algebra $\mathrm{Sym}(W)$ so that $R_j\subset W$ has degree j. Let $q:\mathrm{Sym}(W)\to R$ be the homomorphism of graded $\mathbb{C}$-algebras which is the identity on W, and let $I:=\operatorname{Ker} q$. Then q is surjective, and I is generated by finitely many homogeneous elements $b_1,\dots,b_s$. Any $g\in\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$ extends to a graded $\mathbb{C}$-algebra automorphism of $\mathrm{Sym}(W)$, which we also denote by g. We call the entries of the matrices of g on $R_1,\dots,R_{N_0}$ the *entries* of g. For any $j\ge 0$, the entries of the matrix of g on $\mathrm{Sym}(W)_j$ are polynomials in the entries of g.

(1) Let $g\in\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$. Then $g(I)$ is the ideal generated by $g(b_1),\dots,g(b_s)$, so $g(I)\subset I$ if and only if $q(g(b_i))=0$ for any $1\le i\le s$. If g is the restriction to W of an element $g'\in\mathrm{Aut}_{\mathrm{gr}}(R)$, then $q\circ g=g'\circ q$, since both are homomorphisms of $\mathbb{C}$-algebras which coincide on W. Hence $g(I)\subset I$. Conversely, if $g(I)\subset I$, then $g(I_j)=I_j$ for any $j\ge 0$, as g is injective on $\mathrm{Sym}(W)_j$ and $\dim I_j<+\infty$. Thus g induces a graded automorphism of $R=\mathrm{Sym}(W)/I$ whose restriction to W is g. Since W generates R, an element of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is determined by the restriction to W. Therefore, the restriction to W embeds $\mathrm{Aut}_{\mathrm{gr}}(R)$ into $\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$ as the subgroup defined by the equations
$$q(g(b_i))=0,\quad 1\le i\le s.$$
For each i, the coordinates of $q(g(b_i))\in R_{\deg b_i}$ with respect to a basis of $R_{\deg b_i}$ are polynomials in the entries of g, so $\mathrm{Aut}_{\mathrm{gr}}(R)$ is a closed subgroup.

For any $j\ge 0$ and $g'\in\mathrm{Aut}_{\mathrm{gr}}(R)$ with restriction g to W, the action of g' on R_j is induced by the action of g on $\mathrm{Sym}(W)_j$ through the surjection q. Thus the entries of the matrix of g' on R_j are polynomials in the entries of g, and R_j is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$.

Let $w_1,\dots,w_p$ be a basis of W consisting of homogeneous elements. For $g\in\mathrm{Aut}_{\mathrm{gr}}(R)$ and $h\in R$, the set of $f\in R$ with $g(\{f,h\})=\{g(f),g(h)\}$ is a $\mathbb{C}$-subalgebra of R by the Leibniz rule. Since the Poisson bracket is antisymmetric and W generates R, $g\in\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ if and only if
$$g(\{w_a,w_b\})=\{g(w_a),g(w_b)\},\quad 1\le a,b\le p.$$
By Lemma 2.11(2) and the previous paragraph, the coordinates of both sides are polynomials in the entries of g. Thus $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ is a closed subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R)$.

Let $N_1\ge N_0$ be an integer. The projection
$$\prod_{1\le j\le N_1}\mathrm{GL}(R_j)\to\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$$
maps the image of $\mathrm{Aut}_{\mathrm{gr}}(R)$ defined by N_1 bijectively onto the image defined by N_0, and the inverse map is given by the rational representations R_j, $N_0<j\le N_1$, of the group defined by N_0. Thus the identity maps of $\mathrm{Aut}_{\mathrm{gr}}(R)$ and of the closed subgroup $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ are isomorphisms of linear algebraic groups from the structures defined by N_1 to the structures defined by N_0. Since any two choices of N_0 are bounded by a third one, (1) follows.

(2) To compute the Lie algebras by [Mil17, 10.6], we first show that the equations of (1) define algebraic subgroups of $\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$. Let e be a positive integer with $\deg b_i \le e$ for any $1\le i\le s$, let B be a $\mathbb{C}$-algebra, and let $g\in\prod_{1\le j\le N_0}\mathrm{GL}(R_j\otimes_{\mathbb{C}}B)$. As $I\otimes_{\mathbb{C}}B$ is the ideal of $\mathrm{Sym}(W)\otimes_{\mathbb{C}}B$ generated by $b_1,\dots,b_s$, the equations of (1) defining $\mathrm{Aut}_{\mathrm{gr}}(R)$ hold at g if and only if the extension of g to a graded B-algebra automorphism of $\mathrm{Sym}(W)\otimes_{\mathbb{C}}B$ maps $I_j\otimes_{\mathbb{C}}B$ into itself for any $j\le e$. Thus the equations define the stabilizer of the subspace $\bigoplus_{j\le e}I_j$ of the representation $\bigoplus_{j\le e}\mathrm{Sym}(W)_j$ of $\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$, which is an algebraic subgroup [Mil17, Proposition 4.3]. Similarly, for g in the stabilizer, the equations of (1) defining $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ hold at g if and only if the automorphism of $R\otimes_{\mathbb{C}}B$ induced by g preserves the Poisson bracket, so they define an algebraic subgroup of the stabilizer.

Let $\mathbb{C}[\epsilon]$ be the ring of dual numbers. By [Mil17, Theorem 3.23], the algebraic subgroup of $\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$ defined by the equations of (1) is smooth. Thus by [Mil17, 10.6], the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ consists of the elements $\delta\in\prod_{1\le j\le N_0}\mathrm{End}(R_j)$ such that $1+\epsilon\delta$ satisfies the equations of (1) over $\mathbb{C}[\epsilon]$. For such δ, let $\tilde\delta$ be the derivation of $\mathrm{Sym}(W)$ extending δ. Then $1+\epsilon\tilde\delta$ is the automorphism of $\mathrm{Sym}(W)\otimes_{\mathbb{C}}\mathbb{C}[\epsilon]$ extending $1+\epsilon\delta$. Thus $1+\epsilon\delta$ satisfies the equations of (1) if and only if $q(\tilde\delta(b_i))=0$ for any $1\le i\le s$, that is, $\tilde\delta(I)\subset I$, as $\tilde\delta$ is a derivation. The condition $\tilde\delta(I)\subset I$ holds if and only if $\tilde\delta$ induces a derivation of R, which then maps R_j into R_j for any $j\ge 0$. Conversely, let δ' be a derivation of R with $\delta'(R_j)\subset R_j$ for any $j\ge 0$, and let $\delta:=\delta'|_W$. Then $q\circ\tilde\delta=\delta'\circ q$, since both are derivations along q which coincide on W. Hence $\tilde\delta(I)\subset I$, and δ' is the derivation induced by $\tilde\delta$. This proves the description of the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ in (2).

For $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$, we have

$$(1+\epsilon\delta)(\{f,h\})-\{(1+\epsilon\delta)(f),(1+\epsilon\delta)(h)\}=\epsilon\big(\delta(\{f,h\})-\{\delta(f),h\}-\{f,\delta(h)\}\big)$$

for any $f,h\in R$. By [Mil17, Theorem 3.23] and [Mil17, 10.6] again, the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ consists of the derivations δ in the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$ with

$$\delta(\{w_a,w_b\})=\{\delta(w_a),w_b\}+\{w_a,\delta(w_b)\}\quad\text{for any } 1\le a,b\le p.$$

By the Leibniz rule and the antisymmetry of the Poisson bracket, this holds if and only if $\delta(\{f,h\})=\{\delta(f),h\}+\{f,\delta(h)\}$ for any $f,h\in R$.

Finally, for $f\in R_j$ and $h\in R_k$, we have

$$\mathrm{eu}(fh)=(j+k)fh=\mathrm{eu}(f)h+f\,\mathrm{eu}(h).$$

Thus eu is a derivation of R with $\mathrm{eu}(R_j)\subset R_j$ for any $j\ge 0$, and it is the derivative of λ_t^* at $t=1$.

(3) By Definition 2.13, R_j is a rational representation of T for any $j\ge 0$, so

$$T\to\prod_{1\le j\le N_0}\mathrm{GL}(R_j)$$

is a homomorphism of algebraic groups. Thus the image of T is a closed subgroup [Mil17, Proposition 1.68]. It is a torus, as it is a quotient of T [Mil17, Definition 12.14 and the paragraph after it]. The image of T is connected, so it is contained in G_R. The dimension of any torus in G_R is at most $\dim G_R$. Thus there exists a torus $A\subset G_R$ containing T whose dimension is maximal among the tori in G_R containing T. Any torus $A'\subset G_R$ containing A contains T, so $\dim A'\le\dim A$, and $A'=A$ as A' is irreducible. Hence A is a maximal torus of G_R. □

Lemma 4.2(2) below is known for conical symplectic singularities (cf. [Los16, Proposition 2.14]), and the contraction with eu in the proof is also used in [XZ26b, Lemma 7.8] for torus actions. We include the proof, since in Set-up 2.10, we only assume that ω is a closed non-degenerate 2-form on X_{reg}.

Lemma 4.2. *Notation and conditions as in Set-up 2.10. Then the following hold.*

(1) If $f\in R$ satisfies $\{f,h\}=0$ for any $h\in R$, then $f\in\mathbb{C}$.

(2) The map $R_r \to \operatorname{Lie} G_R$, $a \to \{a, -\}$, is an isomorphism of Lie algebras.

Proof. By Lemma 2.11(1), for any $f \in R$, the derivation $\{f, -\}$ of R is the vector field on X_{reg} with $\iota_{\{f,-\}}\omega = -df$. Since X is affine, the differentials of the elements of R span the cotangent space of X_{reg} at any point. As ω is non-degenerate, the vector fields $\{f, -\}$, $f \in R$, span the tangent space of X_{reg} at any point. Moreover, for any $f, h \in R$,

$$\omega(\{f, -\}, \{h, -\}) = -df(\{h, -\}) = -\{h, f\} = \{f, h\}.$$

We regard eu as a vector field on X_{reg}. Since eu is the derivative of λ_t^* at $t = 1$ and $\lambda_t^*\omega = t^r\omega$, we have $L_{\mathrm{eu}}\omega = r\omega$.

(1) For any $h \in R$, we have $df(\{h, -\}) = \{h, f\} = -\{f, h\} = 0$. Thus $df = 0$ on X_{reg}. Write $f = \sum_j f_j$ with $f_j \in R_j$. Then

$$\sum_j j f_j = \mathrm{eu}(f) = \iota_{\mathrm{eu}} df = 0$$

on X_{reg}, hence in R. Thus $f_j = 0$ for any $j > 0$, and $f = f_0 \in R_0 = \mathbb{C}$.

(2) Let $a \in R_r$. By Lemma 2.11(2), the derivation $\{a, -\}$ maps R_j into R_j for any $j \geq 0$, and by the Jacobi identity,

$$\{a, \{f, h\}\} = \{\{a, f\}, h\} + \{f, \{a, h\}\}$$

for any $f, h \in R$. Thus by Definition-Lemma 4.1(2), $\{a, -\}$ belongs to the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$. Since G_R is an open subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$, the tangent space of G_R at the identity is the tangent space of $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ at the identity, that is, $\operatorname{Lie} G_R = \operatorname{Lie} \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$. By the Jacobi identity again, $[\{a, -\}, \{b, -\}] = \{\{a, b\}, -\}$ for any $a, b \in R_r$. Thus the map $a \to \{a, -\}$ is a homomorphism of Lie algebras $R_r \to \operatorname{Lie} G_R$. If $\{a, -\} = 0$, then $a \in \mathbb{C}$ by (1), so $a = 0$ as $r > 0$. Thus the map is injective.

It remains to prove that the map is surjective. Let $\delta \in \operatorname{Lie} G_R$. By Definition-Lemma 4.1(2), δ is a derivation of R with $\delta(R_j) \subset R_j$ for any $j \geq 0$ and

$$\delta(\{f, h\}) = \{\delta(f), h\} + \{f, \delta(h)\}$$

for any $f, h \in R$. Thus $[\delta, \{f, -\}] = \{\delta(f), -\}$ for any $f \in R$. We regard δ as a vector field on X_{reg}. A direct calculation shows that for any $f, h \in R$,

$$(L_\delta\omega)(\{f, -\}, \{h, -\}) = \delta(\{f, h\}) - \omega(\{\delta(f), -\}, \{h, -\}) - \omega(\{f, -\}, \{\delta(h), -\}),$$

which is equal to $\delta(\{f, h\}) - \{\delta(f), h\} - \{f, \delta(h)\} = 0$. Since the vector fields $\{f, -\}$ span the tangent space of X_{reg} at any point, $L_\delta\omega = 0$. Let $\alpha := \iota_\delta\omega$, which is a regular 1-form on X_{reg}. By the Cartan formula [Mic08, Theorem 9.9(2)],

$$d\alpha = L_\delta\omega - \iota_\delta d\omega = 0.$$

Since δ commutes with λ_t^*, we have $\lambda_t^*\alpha = \iota_\delta(\lambda_t^*\omega) = t^r\alpha$ for any $t \in \mathbb{G}_m$, hence $L_{\mathrm{eu}}\alpha = r\alpha$. By the Cartan formula again,

$$r\alpha = L_{\mathrm{eu}}\alpha = d(\iota_{\mathrm{eu}}\alpha) + \iota_{\mathrm{eu}} d\alpha = d(\iota_{\mathrm{eu}}\alpha).$$

Let $F := -\frac{1}{r}\iota_{\mathrm{eu}}\alpha$, which is a regular function on X_{reg} with $\lambda_t^* F = t^r F$ for any $t \in \mathbb{G}_m$. Since X is normal, $F \in R$ [Har77, Proposition II.6.3A], and hence $F \in R_r$. Moreover, $dF = -\alpha$, that is,

$$\iota_\delta\omega = -dF = \iota_{\{F,-\}}\omega.$$

As ω is non-degenerate, $\delta = \{F, -\}$ on X_{reg}. Thus $\delta(h) = \{F, h\}$ on X_{reg} for any $h \in R$, hence in R, and $\delta = \{F, -\}$. □

4.2. **Seifert quotients.** In this subsection, we prove that the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive if the grading of R is primitive, $(X,0)$ is klt, and (E,Δ_E) is K-polystable.

To lift a vector field on E tangent to the components of Δ_E to a derivation of R, we write R as the section ring of a $\mathbb{Q}$-divisor on E which is $\mathbb{Q}$-linearly equivalent to a rational multiple of $-(K_E+\Delta_E)$.

Construction-Proposition 4.3. Notation and conditions as in Definition-Proposition 2.15, and assume that $(X,0)$ is klt. Let $\pi : X\backslash\{o\} \to E$ be the quotient morphism, and regard the function field $\mathbb{C}(E)$ of E as the subfield of the field of fractions of R consisting of the fractions f/f' with $f, f' \in R_j$ for some $j \geq 0$ and $f' \neq 0$. Then there exist an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor D on E and an element u of the field of fractions of R with $\lambda_t^* u = tu$ for any $t \in \mathbb{G}_m$, such that the following hold.

(1) $R_j = H^0(E, \mathcal{O}_E(\lfloor jD \rfloor))u^j$ for any $j \geq 0$.
(2) $K_E + \Delta_E + nrD \sim_{\mathbb{Q}} 0$.
(3) For any prime divisor P on E, the coefficient of D along P is a rational number whose denominator in lowest terms is m_P.
(4) The field of fractions of R is $\mathbb{C}(E)(u)$, and the elements of the field of fractions of R fixed by λ_t^* for any $t \in \mathbb{G}_m$ are the elements of $\mathbb{C}(E)$.

Proof. By Definition-Proposition 2.15(1), π is a Seifert $\mathbb{G}_m$-bundle. We use [Kol04, Definition 5 and Theorem 7] with the $\mathbb{G}_m$-action λ or with $t \to \lambda_{t^{-1}}$ on $X\backslash\{o\}$, so that λ_t^* acts on the summand of index j in [Kol04, Definition 5] by t^j for any $j \in \mathbb{Z}$. By [Kol04, Theorem 7(1)], there exist a rank one reflexive sheaf $\mathcal{L}$ on E, distinct prime divisors P_i on E, and rational numbers $s_i \in (0,1)$ such that $\pi_*\mathcal{O}_{X\backslash\{o\}}$ is the graded $\mathcal{O}_E$-algebra defined by $\mathcal{L}$ and $\sum_i s_i P_i$ in [Kol04, Definition 5]. As $0 < s_i < 1$, the summand of index 1 is $\mathcal{L}$, so a non-zero rational section σ of $\mathcal{L}$ is a rational function on $X\backslash\{o\}$, that is, an element of the field of fractions of R with $\lambda_t^*\sigma = t\sigma$ for any $t \in \mathbb{G}_m$. Let $u := \sigma$ and $D := \mathrm{div}(\sigma) + \sum_i s_i P_i$. Since $\mathrm{div}(\sigma)$ is a divisor, the summand of index j is $\mathcal{O}_E(\lfloor jD \rfloor)u^j$ for any $j \in \mathbb{Z}$, that is,

$$\pi_*\mathcal{O}_{X\backslash\{o\}} = \bigoplus_{j\in\mathbb{Z}} \mathcal{O}_E(\lfloor jD \rfloor)u^j, \tag{4.1}$$

where the multiplication on the right-hand side is the multiplication in the field of fractions of R.

We prove (3). For a prime divisor P on E and a general point x of P, the stabilizer in $\mathbb{G}_m$ of any point of $\pi^{-1}(x)$ is $\mu_{m(x)}$, where $m(x)$ is the multiplicity of the fiber of π over x [Kol04, Definition 3], so $m_P = m(x)$. As E is normal, E is smooth at x. Hence the rank one reflexive sheaves in [Kol04, Theorem 7(3)] are locally free at x. Moreover, $x \notin P_i$ for any $P_i \neq P$. Thus by [Kol04, Theorem 7(3)], $m(x)$ is the denominator in lowest terms of the coefficient of $\sum_i s_i P_i$ along P. Since $D - \sum_i s_i P_i = \mathrm{div}(\sigma)$ is a divisor, (3) follows.

Since X is normal and $\dim X \geq 2$, any regular function on $X\backslash\{o\}$ extends to X [Har77, Proposition II.6.3A]. As λ_t^* acts on the summand of index j in (4.1) by t^j, taking global sections of the summands with $j \geq 0$ implies (1). In particular, the field of fractions of R is $\mathbb{C}(E)(u)$. Comparing the coefficients of the Laurent expansions in u, we see that the elements of $\mathbb{C}(E)(u)$ fixed by λ_t^* for any $t \in \mathbb{G}_m$ are the elements of $\mathbb{C}(E)$. This proves (4).

It remains to prove (2). Let κ_1 be a non-zero rational differential form of degree $2n-1$ on E. Then $\omega^n = F\frac{du}{u} \wedge \pi^*\kappa_1$ for a unique $F \in \mathbb{C}(E)(u)$. Since $\lambda_t^*\omega^n = t^{nr}\omega^n$, $\lambda_t^* u = tu$, and $\pi \circ \lambda_t = \pi$, we have $\lambda_t^* F = t^{nr} F$, hence $Fu^{-nr} \in \mathbb{C}(E)$. Thus there exists a unique rational differential form κ of degree $2n-1$ on E such that

$$\omega^n = u^{nr}\frac{du}{u} \wedge \pi^*\kappa.$$

Then $\mathrm{div}(\kappa)$ is a canonical divisor of E, and it suffices to prove that

$$\mathrm{div}(\kappa) + \Delta_E + nrD = 0.$$

Let P be a prime divisor on E, let k be the coefficient of $\operatorname{div}(\kappa)$ along P, and let $m := m_P$. By (3), the coefficient of D along P is b/m for an integer b coprime to m. Since E is normal, E is smooth at the generic point of P. Thus there exist an open subset $U \subset E$ with $U \cap P \neq \emptyset$, regular functions $z, y_2, \dots, y_{2n-1}$ on U, and a nowhere vanishing regular function a on U, such that Ω_U is free with basis $dz, dy_2, \dots, dy_{2n-1}$, $\operatorname{div}(z) = P \cap U$, $U \cap \operatorname{Supp} D \subset P$, and

$$\kappa = az^k dz \wedge dy_2 \wedge \cdots \wedge dy_{2n-1}$$

on U. In particular, $\mathcal{O}_E(\lfloor jD \rfloor)|_U = z^{-\lfloor jb/m \rfloor}\mathcal{O}_U$ for any $j \in \mathbb{Z}$.

Let $U' \subset U \times \mathbb{A}^1$ be the closed subvariety defined by $v^m = z$, where v is the coordinate of $\mathbb{A}^1$. Then U' is smooth, and $\Omega_{U'}$ is free with basis $dv, dy_2, \dots, dy_{2n-1}$. Let w be the coordinate of $\mathbb{G}_m$, and let μ_m act on $U' \times \mathbb{G}_m$ over U by $(v, w) \to (\zeta v, \zeta^{-b} w)$ for $\zeta \in \mu_m$. The pushforward of $\mathcal{O}_{U' \times \mathbb{G}_m}$ to U is $\bigoplus_{0 \le i < m, j \in \mathbb{Z}} \mathcal{O}_U v^i w^j$, and $\zeta \in \mu_m$ acts on $v^i w^j$ by ζ^{i-bj}. Thus the μ_m-invariant part is

$$\bigoplus_{j \in \mathbb{Z}} \mathcal{O}_U v^{jb - m\lfloor jb/m \rfloor} w^j = \bigoplus_{j \in \mathbb{Z}} \mathcal{O}_U z^{-\lfloor jb/m \rfloor} (wv^b)^j.$$

By (4.1), there exists a morphism $q : U' \times \mathbb{G}_m \to \pi^{-1}(U)$ over U with $q^* u = wv^b$, which is the quotient morphism by μ_m. Since b and m are coprime, μ_m acts freely on $U' \times \mathbb{G}_m$: if $(\zeta v, \zeta^{-b} w) = (v, w)$, then $\zeta^{-b} = 1$ as $w \neq 0$, so $\zeta = 1$. Thus q is étale, and $\pi^{-1}(U)$ is smooth, that is, $\pi^{-1}(U) \subset X_{\mathrm{reg}}$. As ω is non-degenerate, ω^n is a nowhere vanishing regular $2n$-form on X_{reg}. Hence $q^*\omega^n$ is a nowhere vanishing regular $2n$-form on $U' \times \mathbb{G}_m$. On the other hand, since $z = v^m$ on U',

$$q^* u^{nr} = w^{nr} v^{nrb}, \quad q^* \frac{du}{u} = \frac{dw}{w} + b\frac{dv}{v}, \quad (\pi \circ q)^* \kappa = amv^{mk+m-1} dv \wedge dy_2 \wedge \cdots \wedge dy_{2n-1}.$$

Thus

$$q^*\omega^n = amw^{nr-1} v^{nrb+mk+m-1} dw \wedge dv \wedge dy_2 \wedge \cdots \wedge dy_{2n-1},$$

and $nrb + mk + m - 1 = 0$, that is,

$$k + \left(1 - \frac{1}{m}\right) + nr\frac{b}{m} = 0.$$

The left-hand side is the coefficient of $\operatorname{div}(\kappa) + \Delta_E + nrD$ along P. Since P is arbitrary, $\operatorname{div}(\kappa) + \Delta_E + nrD = 0$, and (2) follows.

By (2) and Definition-Proposition 2.15(2), $nrD \sim_{\mathbb{Q}} -(K_E + \Delta_E)$, which is ample. Since $K_E + \Delta_E$ is $\mathbb{Q}$-Cartier and $n, r > 0$, D is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor. □

By [ABHLX19, Theorem 1.3], the group of automorphisms of E preserving Δ_E is reductive if (E, Δ_E) is a K-polystable log Fano pair. In the following proposition, we deduce from it that the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R)$ is reductive (cf. [XZ26b, second paragraph after Lemma 7.1]).

Proposition 4.4. *Notation and conditions as in Definition-Proposition 2.15. Let H be the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R)$, and let $\operatorname{Aut}(E, \Delta_E)$ be the group of automorphisms g of E with $g^*\Delta_E = \Delta_E$ [ABHLX19, Section 4.1]. Assume that $(X, 0)$ is klt and (E, Δ_E) is a K-polystable log Fano pair. Then the action of H on E induces a surjective homomorphism $\varphi : H \to \operatorname{Aut}(E, \Delta_E)^\circ$ of algebraic groups with $\operatorname{Ker}\varphi = \lambda(\mathbb{G}_m)$, and $\lambda(\mathbb{G}_m)$ is contained in the center of H. In particular, H is reductive.*

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we construct φ and prove that φ is a homomorphism of algebraic groups.

Let π, D, and u be as in Construction-Proposition 4.3, and regard $\mathbb{C}(E)$ as a subfield of the field of fractions of R as there.

Let $g \in H$, and let $\varphi(g)$ be the automorphism of $E = \operatorname{Proj} R$ induced by the graded automorphism g^{-1} of R, that is, the automorphism of E whose pullback on $\mathbb{C}(E)$ is the restriction of g^{-1}. For $g, g' \in H$, the pullback of $\varphi(g) \circ \varphi(g')$ on $\mathbb{C}(E)$ is $g'^{-1} \circ g^{-1} = (gg')^{-1}$, so $\varphi(gg') = \varphi(g) \circ \varphi(g')$. Let $\gamma := \gamma_{g^{-1}}$ (Definition 2.13), the automorphism of X whose pullback

on R is g^{-1}. Since g preserves each R_j, γ commutes with λ_t for any $t \in \mathbb{G}_m$, $\gamma(o) = o$, and $\pi \circ \gamma = \varphi(g) \circ \pi$ on $X \backslash \{o\}$. Thus for any $x \in X \backslash \{o\}$, the stabilizer of $\gamma(x)$ in $\mathbb{G}_m$ is the stabilizer of x in $\mathbb{G}_m$, and for any prime divisor P on E, γ maps the preimage of P onto the preimage of $\varphi(g)(P)$. Hence $m_{\varphi(g)(P)} = m_P$, and $\varphi(g)^* \Delta_E = \Delta_E$. Therefore, φ is a group homomorphism from H to $\mathrm{Aut}(E, \Delta_E)$.

Since D is an ample $\mathbb{Q}$-Cartier $\mathbb{Q}$-divisor, by Construction-Proposition 4.3(2), there exists a positive integer N divisible by nr such that ND is a very ample Cartier divisor and $\frac{N}{nr}(K_E + \Delta_E + nrD) \sim 0$. Thus $\frac{N}{nr}(K_E + \Delta_E) \sim -ND$, so $\frac{N}{nr}(K_E + \Delta_E)$ is a Cartier divisor, and

$$\mathcal{O}_E(ND) \cong \mathcal{O}_E\Big(-\frac{N}{nr}(K_E + \Delta_E)\Big).$$

By Construction-Proposition 4.3(1), $R_N = H^0(E, \mathcal{O}_E(ND))u^N$. Thus the complete linear system $|ND|$ defines a closed embedding of E into the projective space $\mathbb{P}(R_N^\vee)$ of hyperplanes in R_N, which maps $x \in E$ to the hyperplane consisting of the elements of R_N vanishing on $\pi^{-1}(x)$. For $x \in E$ and $f \in R_N$, f vanishes on $\pi^{-1}(\varphi(g)(x)) = \gamma(\pi^{-1}(x))$ if and only if $\gamma^* f = g^{-1}(f)$ vanishes on $\pi^{-1}(x)$. Thus $\varphi(g)$ is the restriction to E of the automorphism of $\mathbb{P}(R_N^\vee)$ which maps any hyperplane of R_N to the image of the hyperplane under g. Hence φ is the composite of the homomorphism $H \to \mathrm{GL}(R_N)$, $g \to g|_{R_N}$, and the quotient homomorphism $\mathrm{GL}(R_N) \to \mathrm{PGL}(R_N)$, where $\mathrm{PGL}(R_N)$ acts on $\mathbb{P}(R_N^\vee)$. Since $\frac{N}{nr}$ is a positive integer and $\mathcal{O}_E(ND) \cong \mathcal{O}_E(-\frac{N}{nr}(K_E + \Delta_E))$, $\mathrm{Aut}(E, \Delta_E)$ is a closed subgroup of $\mathrm{PGL}(R_N)$ [ABHLX19, Section 4.1]. As R_N is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$ (Definition-Lemma 4.1(1)), $\varphi : H \to \mathrm{Aut}(E, \Delta_E)$ is a homomorphism of algebraic groups. Since H is connected, $\varphi(H) \subset \mathrm{Aut}(E, \Delta_E)^\circ$.

Step 2. In this step we prove that $\mathrm{Ker}\, \varphi = \lambda(\mathbb{G}_m)$ and that $\lambda(\mathbb{G}_m)$ is contained in the center of H.

Here $\lambda(\mathbb{G}_m) \subset \mathrm{Aut}_{\mathrm{gr}}(R)$ consists of the automorphisms λ_t^*, $t \in \mathbb{G}_m$. For $t \in \mathbb{G}_m$, λ_t^* acts on R_j as the multiplication by t^j for any $j \geq 0$. Thus λ_t^* commutes with any element of $\mathrm{Aut}_{\mathrm{gr}}(R)$, and λ_t^* is the identity on $\mathbb{C}(E)$. As $\lambda(\mathbb{G}_m)$ is connected, $\lambda(\mathbb{G}_m) \subset H$. Hence $\lambda(\mathbb{G}_m)$ is contained in the center of H and in $\mathrm{Ker}\, \varphi$.

Conversely, let $g \in \mathrm{Ker}\, \varphi$. Then g is the identity on $\mathbb{C}(E)$. Since g commutes with λ_t^* for any $t \in \mathbb{G}_m$, we have $\lambda_t^*(g(u)) = g(\lambda_t^* u) = tg(u)$, so $g(u)/u$ is fixed by λ_t^* for any $t \in \mathbb{G}_m$. Thus by Construction-Proposition 4.3(4), $g(u) = cu$ for some $c \in \mathbb{C}(E)^*$. By Construction-Proposition 4.3(1), any $f \in R_j$ is of the form su^j with $s \in \mathbb{C}(E)$, so

$$g(f) = g(s)g(u)^j = c^j f.$$

Since $\dim X \geq 2$, there exists $j > 0$ with $R_j \neq 0$. Let $f \in R_j$ be an eigenvector of $g|_{R_j}$ with eigenvalue $a \in \mathbb{C}^*$. Then $c^j f = af$, so $c^j = a$ as R is a domain. Thus c is algebraic over $\mathbb{C}$, and $c \in \mathbb{C}^*$. Hence $g(f) = c^j f = \lambda_c^* f$ for any $j \geq 0$ and $f \in R_j$, that is, $g = \lambda_c^* \in \lambda(\mathbb{G}_m)$.

Step 3. In this step we prove that the differential $d\varphi$ of φ is surjective.

Let $\mathbb{C}[\epsilon]$ be the ring of dual numbers. By [Mil17, 10.6], an element of the Lie algebra of $\mathrm{Aut}(E, \Delta_E)^\circ$ is a $\mathbb{C}[\epsilon]$-point of $\mathrm{Aut}(E, \Delta_E)^\circ \subset \mathrm{PGL}(R_N)$ which is the identity modulo ϵ. As $\mathbb{C}[\epsilon]$ is a local ring, the point is the image of $1 + \epsilon A$ for some $A \in \mathrm{End}(R_N)$. The action of $\mathrm{PGL}(R_N)$ on $\mathbb{P}(R_N^\vee)$ restricts to an action of $\mathrm{Aut}(E, \Delta_E)$ on E, so $1 + \epsilon A$ induces an automorphism of $E \times \mathrm{Spec}\, \mathbb{C}[\epsilon]$ over $\mathbb{C}[\epsilon]$ which is the identity modulo ϵ. As in **Step 1**, since $(1 + \epsilon A)^{-1} = 1 - \epsilon A$, the pullback of the automorphism maps f/f' to $(f - \epsilon Af)/(f' - \epsilon Af')$ for any $f, f' \in R_N$ with $f' \neq 0$. Thus the pullback is $1 + \epsilon\delta$ for the vector field δ on E with

$$\delta\Big(\frac{f}{f'}\Big) = \frac{f \cdot Af' - f' \cdot Af}{f'^2} \quad \text{for any } f, f' \in R_N \text{ with } f' \neq 0.$$

If $\delta = 0$, then $Af/f = Af'/f'$ in $\mathbb{C}(E)$ for any non-zero $f, f' \in R_N$. Taking f to be an eigenvector of A, we see that A is a scalar multiple of the identity, so the image of $1 + \epsilon A$ in $\mathrm{PGL}(R_N)$ is the

identity. Hence we identify the element of the Lie algebra with δ, and regard δ as a derivation of $\mathbb{C}(E)$. Let δ be an element of the Lie algebra of $\mathrm{Aut}(E,\Delta_E)^\circ$.

Since $\mathrm{Aut}(E,\Delta_E)$ permutes the finitely many components of Δ_E, for any component P of Δ_E, the stabilizer of P in $\mathrm{Aut}(E,\Delta_E)$ is a closed subgroup of finite index, and it contains $\mathrm{Aut}(E,\Delta_E)^\circ$. Thus δ belongs to the Lie algebra of the stabilizer of P, which is smooth [Mil17, Theorem 3.23]. Hence by [Mil17, 10.6], the automorphism corresponding to δ preserves $P\times \operatorname{Spec}\mathbb{C}[\epsilon]$, that is, δ is tangent to P. In particular, for a local parameter z at the generic point of P, $\delta(z)/z$ is regular at the generic point of P.

Recall that D and u are given by Construction-Proposition 4.3. Let κ be a non-zero rational differential form of degree $2n-1$ on E. As $\mathrm{div}(\kappa)$ is a canonical divisor of E, by Construction-Proposition 4.3(2), there exist a positive integer q and $h\in\mathbb{C}(E)^*$ such that

$$q(\mathrm{div}(\kappa)+\Delta_E+nrD)=\mathrm{div}(h).$$

Let $b_\delta\in\mathbb{C}(E)$ be the divergence of δ with respect to κ, that is, $L_\delta\kappa=b_\delta\kappa$, and let

$$c_\delta:=\frac{1}{qnr}\Big(\frac{\delta(h)}{h}-qb_\delta\Big).$$

Let $\tilde\delta$ be the derivation of $\mathbb{C}(E)(u)$ with $\tilde\delta|_{\mathbb{C}(E)}=\delta$ and $\tilde\delta(u)=c_\delta u$, that is,

$$\tilde\delta(su^j)=(\delta(s)+jc_\delta s)u^j$$

for any $s\in\mathbb{C}(E)$ and $j\in\mathbb{Z}$. We prove that $\tilde\delta(R_j)\subset R_j$ for any $j\geq 0$. By Construction-Proposition 4.3(1), it suffices to prove that for any $j\geq 0$, any $s\in\mathbb{C}(E)$ with $\mathrm{div}(s)+\lfloor jD\rfloor\geq 0$, and any prime divisor P on E, the coefficient of $\mathrm{div}(\delta(s)+jc_\delta s)+\lfloor jD\rfloor$ along P is non-negative.

Let d_P, e_P, and k_P be the coefficients of D, Δ_E, and $\mathrm{div}(\kappa)$ along P respectively, and let z be a local parameter at the generic point of P. Write

$$\kappa=z^{k_P}\kappa_0\quad\text{and}\quad h=z^{q(k_P+e_P+nrd_P)}h_0,$$

where κ_0 is a rational differential form of degree $2n-1$ and $h_0\in\mathbb{C}(E)^*$, both regular and non-vanishing at the generic point of P. Since δ is regular at the generic point of P, so are $L_\delta\kappa_0/\kappa_0$ and $\delta(h_0)/h_0$. Thus

$$b_\delta-k_P\frac{\delta(z)}{z}=\frac{L_\delta\kappa_0}{\kappa_0}\quad\text{and}\quad\frac{\delta(h)}{h}-q(k_P+e_P+nrd_P)\frac{\delta(z)}{z}=\frac{\delta(h_0)}{h_0}$$

are regular at the generic point of P. By the definition of c_δ,

$$c_\delta-d_P\frac{\delta(z)}{z}-\frac{e_P}{nr}\cdot\frac{\delta(z)}{z}$$

is regular at the generic point of P. If $e_P\neq 0$, then P is a component of Δ_E, so $\delta(z)/z$ is regular at the generic point of P. Thus in any case, $c_\delta-d_P\delta(z)/z$ is regular at the generic point of P.

Now let $s\in\mathbb{C}(E)$ with $\mathrm{div}(s)+\lfloor jD\rfloor\geq 0$, and write $s=z^{-\lfloor jd_P\rfloor}s_0$, where $s_0\in\mathbb{C}(E)$ is regular at the generic point of P. Then

$$\delta(s)+jc_\delta s=z^{-\lfloor jd_P\rfloor}\Big(\delta(s_0)+j\Big(c_\delta-d_P\frac{\delta(z)}{z}\Big)s_0+(jd_P-\lfloor jd_P\rfloor)\frac{\delta(z)}{z}s_0\Big).$$

If $d_P\in\mathbb{Z}$, then $jd_P-\lfloor jd_P\rfloor=0$. If $d_P\notin\mathbb{Z}$, then $m_P>1$ by Construction-Proposition 4.3(3), so $e_P=1-\frac{1}{m_P}>0$, and $\delta(z)/z$ is regular at the generic point of P. In both cases, the function in the parentheses is regular at the generic point of P. Thus the coefficient of $\mathrm{div}(\delta(s)+jc_\delta s)+\lfloor jD\rfloor$ along P is non-negative, and $\tilde\delta(R_j)\subset R_j$ for any $j\geq 0$.

Therefore, $\tilde\delta|_R$ is a derivation of R with $\tilde\delta(R_j)\subset R_j$ for any $j\geq 0$. By Definition-Lemma 4.1(2), $\tilde\delta|_R$ belongs to the Lie algebra of $\mathrm{Aut}_{\mathrm{gr}}(R)$, which is the Lie algebra of H as H is an open subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R)$. Since the pullback of $\varphi(g)$ on $\mathbb{C}(E)$ is the restriction of g^{-1} for any $g\in H$, $d\varphi$ maps $\tilde\delta|_R$ to $-\delta$. As δ is arbitrary, $d\varphi$ is surjective.

Step 4. In this step we prove that φ is surjective and H is reductive.

By [Mil17, Proposition 1.68], $\varphi(H)$ is a closed subgroup of $\mathrm{Aut}(E,\Delta_E)^\circ$. Since $d\varphi$ is surjective and factors through the Lie algebra of $\varphi(H)$, the Lie algebra of $\varphi(H)$ is the Lie algebra of $\mathrm{Aut}(E,\Delta_E)^\circ$. As $\varphi(H)$ is smooth [Mil17, Theorem 3.23] and $\mathrm{Aut}(E,\Delta_E)^\circ$ is connected, $\varphi(H)=\mathrm{Aut}(E,\Delta_E)^\circ$ [Mil17, Proposition 10.15], that is, φ is surjective. Together with **Step 2**, this proves the conclusions on φ and $\lambda(\mathbb{G}_m)$.

The group $\mathrm{Aut}(E,\Delta_E)$ is a linear algebraic group [ABHLX19, Section 4.1], and it is reductive, as (E,Δ_E) is a K-polystable log Fano pair [ABHLX19, Theorem 1.3]. Since φ is surjective, φ maps the unipotent radical of H onto a connected normal unipotent subgroup of the reductive group $\mathrm{Aut}(E,\Delta_E)^\circ$, which is trivial [Mil17, 6.45 and 6.46]. Thus the unipotent radical of H is contained in $\operatorname{Ker}\varphi=\lambda(\mathbb{G}_m)$. As $\lambda(\mathbb{G}_m)$ is a torus, the unipotent radical of H is trivial, and H is reductive. □

In general, the homomorphism φ in Proposition 4.4 does not have a section which is a group homomorphism.

Example 4.5. Let $R=\mathbb{C}[x,y]$ with $\deg x=\deg y=1$, and let $\omega=dx\wedge dy$. Then $X=\mathbb{C}^2$, $n=1$, $r=2$, the grading is primitive, $E=\mathbb{P}^1$, $\Delta_E=0$, and the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is $\mathrm{Aut}_{\mathrm{gr}}(R)=\mathrm{GL}_2$. The homomorphism $\mathrm{GL}_2\to\mathrm{Aut}(E)$ induced by the action of GL_2 on E is the quotient homomorphism $\mathrm{GL}_2\to\mathrm{PGL}_2$, and it has no section which is a group homomorphism. Indeed, let $\sigma:\mathrm{PGL}_2\to\mathrm{GL}_2$ be such a section. Then $\sigma(\mathrm{PGL}_2)$ is isomorphic to PGL_2, and the derived subgroup of PGL_2 is PGL_2. Thus $\sigma(\mathrm{PGL}_2)$ is contained in the derived subgroup SL_2 of GL_2. As $\dim\mathrm{SL}_2=\dim\mathrm{PGL}_2$ and SL_2 is connected, $\sigma(\mathrm{PGL}_2)=\mathrm{SL}_2$. This contradicts the fact that the center of SL_2 is μ_2 while the center of PGL_2 is trivial.

4.3. **Reductivity of the Poisson automorphism group.** In this subsection, we show that G_R is the identity component of the stabilizer of ω in the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$, and we deduce that G_R is reductive if the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive. The identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive under the conditions of Proposition 4.4, that is, if the grading is primitive, $(X,0)$ is klt, and (E,Δ_E) is a K-polystable log Fano pair.

The stabilizer of a vector in a representation of a reductive group need not be reductive. Moreover, the image of the $\mathbb{G}_m$-action λ is contained in the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ and acts on the closed 2-forms of degree r on X_{reg} by multiplication by non-zero scalars, so the closure of the orbit of ω contains the zero form, and the form of Matsushima's criterion for closed orbits cannot be used. Instead, we use Matsushima's criterion (see [Arz05, Theorem 2.1]): a closed subgroup of a reductive group is reductive if and only if the quotient of the group by the subgroup is affine. Lemma 4.6 shows that the orbit of ω is the principal open subset defined by the non-vanishing of a polynomial, hence affine.

Lemma 4.6. *Notation and conditions as in Set-up 2.10. Let H be the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$, and let V be the $\mathbb{C}$-vector space of closed algebraic 2-forms η on X_{reg} with $\lambda_t^*\eta=t^r\eta$ for any $t\in\mathbb{G}_m$. Then the following hold.*

(1) $\dim V<+\infty$, and there exists a homogeneous polynomial function p on V with $p(\omega)=1$ such that $\eta^n=p(\eta)\omega^n$ for any $\eta\in V$.
(2) The open subset $V^\times:=\{\eta\in V\mid p(\eta)\neq 0\}$ consists of the non-degenerate forms in V.
(3) $V^\times$ is a single orbit of the natural action of H on V, and $H/\operatorname{Stab}_H(\omega)\cong V^\times$. In particular, $H/\operatorname{Stab}_H(\omega)$ is affine.

Proof. For $g\in\mathrm{Aut}_{\mathrm{gr}}(R)$, let γ_g and $g\cdot\eta=\gamma_g^*\eta$ be as in Definition 2.13. Since $g(R_d)=R_d$ for any $d\geq 0$, γ_g commutes with λ_t for any $t\in\mathbb{G}_m$, so the action of $\mathrm{Aut}_{\mathrm{gr}}(R)$ preserves V, and the natural action of H on V is the restriction of the action of $\mathrm{Aut}_{\mathrm{gr}}(R)$ to H.

(1) Let N_0 be a positive integer such that $W:=\bigoplus_{1\leq d\leq N_0}R_d$ generates R as a $\mathbb{C}$-algebra. For $a\in\mathbb{Z}$, let $\mathrm{Der}(R)_a$ be the space of the $\mathbb{C}$-linear derivations δ of R with $\delta(R_d)\subset R_{d+a}$ for any $d\geq 0$, where $R_m:=0$ for $m<0$. The group $\mathrm{Aut}_{\mathrm{gr}}(R)$ acts on $\mathrm{Der}(R)_a$ by $g\cdot\delta:=g\circ\delta\circ g^{-1}$.

Since W generates R, the restriction to W is an injective $\mathrm{Aut}_{\mathrm{gr}}(R)$-equivariant linear map

$$\mathrm{Der}(R)_a \to \bigoplus_{1\le d\le N_0} \mathrm{Hom}(R_d, R_{d+a}).$$

By Definition-Lemma 4.1(1), the right-hand side is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$. Thus $\mathrm{Der}(R)_a$ is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$ with $\dim \mathrm{Der}(R)_a < +\infty$.

Let $w \in R_e$ with $1 \le e \le N_0$. By Lemma 2.11(1) and (2), $\{w, -\} \in \mathrm{Der}(R)_{e-r}$, and on X_{reg}, $\{w, -\}$ is the vector field ζ_w with $\iota_{\zeta_w}\omega = -dw$. Since W generates R, the differentials of the elements of $R_1, \dots, R_{N_0}$ span the cotangent space of X_{reg} at any point. Since ω is non-degenerate, the derivations $\{w, -\}$ with $w \in R_e$ and $1 \le e \le N_0$ span the tangent space of X_{reg} at any point.

Let $a, b \in \mathbb{Z}$, let $\eta \in V$, let $\delta \in \mathrm{Der}(R)_a$, and let $\delta' \in \mathrm{Der}(R)_b$. Then $\eta(\delta, \delta')$ is a regular function on X_{reg}, so $\eta(\delta, \delta') \in R$ because X is normal. Since $\lambda_t^*\eta = t^r\eta$, $\lambda_t^* \circ \delta \circ (\lambda_t^*)^{-1} = t^a\delta$, and $\lambda_t^* \circ \delta' \circ (\lambda_t^*)^{-1} = t^b\delta'$ for any $t \in \mathbb{G}_m$, we have $\eta(\delta, \delta') \in R_{a+b+r}$. Thus we obtain a linear map

$$V \to \bigoplus_{a,b} \mathrm{Hom}\big(\mathrm{Der}(R)_a \otimes \mathrm{Der}(R)_b, R_{a+b+r}\big), \quad \eta \to \big(\delta \otimes \delta' \to \eta(\delta, \delta')\big),$$

where a and b run over the integers $d - r$ with $1 \le d \le N_0$. The map is $\mathrm{Aut}_{\mathrm{gr}}(R)$-equivariant because pullback by γ_g commutes with contraction, and the map is injective because the derivations $\{w, -\}$ above span the tangent space of X_{reg} at any point. Thus $\dim V < +\infty$, and V is a rational representation of $\mathrm{Aut}_{\mathrm{gr}}(R)$.

Let $\eta \in V$. Since ω^n vanishes nowhere on X_{reg}, the quotient η^n/ω^n is a regular function on X_{reg}. Since X is normal and λ_t^* multiplies both η^n and ω^n by t^{nr}, we have $\eta^n/\omega^n \in R_0 = \mathbb{C}$. Let $p(\eta) := \eta^n/\omega^n$. For any $x \in X_{\mathrm{reg}}$, $p(\eta)$ is the quotient of the values of η^n and ω^n at x, and the value of η^n at x is a homogeneous polynomial of degree n in the value of η at x, which is linear in η. Thus p is a homogeneous polynomial function of degree n on V, $p(\omega) = 1$, and $\eta^n = p(\eta)\omega^n$ for any $\eta \in V$.

(2) A skew-symmetric bilinear form on a $\mathbb{C}$-vector space of dimension $2n$ is non-degenerate if and only if the n-th exterior power of the form is non-zero. Since ω^n vanishes nowhere on X_{reg} and $\eta^n = p(\eta)\omega^n$, for any $\eta \in V$ and any $x \in X_{\mathrm{reg}}$, η is non-degenerate at x if and only if $p(\eta) \ne 0$. Thus $V^\times$ consists of the non-degenerate forms in V.

(3) For $g \in H$, γ_g is an automorphism of X_{reg}, so the action of H preserves $V^\times$ by (2). Set-up 2.10 is a special case of the setting of [Nam11, Theorem (3.1)]: a normal affine variety whose coordinate ring is graded by non-negative integers with degree-0 part $\mathbb{C}$, and a closed non-degenerate algebraic 2-form on the regular locus which is homogeneous of non-zero degree. By [Nam11, Theorem (3.1)], the closed non-degenerate algebraic 2-forms η on X_{reg} with $\lambda_t^*\eta = t^r\eta$ for any $t \in \mathbb{G}_m$ form a single orbit of the group of automorphisms of X commuting with λ_t for any $t \in \mathbb{G}_m$. Since R_d is the subspace of R on which λ_t^* acts by t^d for any $t \in \mathbb{G}_m$, an automorphism of X commutes with λ_t for any $t \in \mathbb{G}_m$ if and only if it is γ_g for some $g \in \mathrm{Aut}_{\mathrm{gr}}(R)$. Thus by (2), $V^\times \subset \mathrm{Aut}_{\mathrm{gr}}(R) \cdot \omega$.

By Definition-Lemma 4.1(1), $\mathrm{Aut}_{\mathrm{gr}}(R)$ is a linear algebraic group, so $\mathrm{Aut}_{\mathrm{gr}}(R)$ is smooth [Mil17, Theorem 3.23]. Thus H is a normal subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R)$ [Mil17, Proposition 1.52] of finite index [Mil17, Definition 6.5 and the paragraph after it]. Let $g_1, \dots, g_m \in \mathrm{Aut}_{\mathrm{gr}}(R)$ be representatives of the cosets of H with $g_1 = \mathrm{id}_R$. Since $g_iH = Hg_i$ for $1 \le i \le m$, we have

$$V^\times \subset \bigcup_{i=1}^m g_i \cdot (H \cdot \omega) = \bigcup_{i=1}^m H \cdot (g_i \cdot \omega).$$

By [Mil17, Proposition 1.65(b), Proposition 7.17, and Corollary 7.13], for any $\eta \in V$, the orbit $H \cdot \eta$ is a locally closed subvariety of V, and the orbit map induces an isomorphism $H/\mathrm{Stab}_H(\eta) \cong H \cdot \eta$. Since g_i acts on V by a linear automorphism, $\dim H \cdot (g_i \cdot \omega) = \dim H \cdot \omega$ for $1 \le i \le m$. Since $V^\times$ is a non-empty open subset of V, $V^\times$ is irreducible, so $V^\times$ is contained in the closure of $H \cdot (g_j \cdot \omega)$ for some j. Thus $\dim H \cdot (g_i \cdot \omega) = \dim H \cdot (g_j \cdot \omega) = \dim V$ for $1 \le i \le m$, so the closure of $H \cdot (g_i \cdot \omega)$ in V is a closed subset of V with dimension $\dim V$,

that is, V. Since $H \cdot (g_i \cdot \omega)$ is locally closed, $H \cdot (g_i \cdot \omega)$ is open in V. Since V is irreducible, the non-empty open subsets $H \cdot (g_i \cdot \omega)$ and $H \cdot \omega$ of V intersect, so $H \cdot (g_i \cdot \omega) = H \cdot \omega$ for $1 \leq i \leq m$. Thus $V^\times \subset H \cdot \omega$. Since the action of H preserves $V^\times$ and $\omega \in V^\times$, $V^\times = H \cdot \omega$ is a single orbit, and $H/\operatorname{Stab}_H(\omega) \cong V^\times$. Since $V^\times$ is the principal open subset of V defined by p, $V^\times \cong \operatorname{Spec} \mathbb{C}[V][p^{-1}]$ is affine, where $\mathbb{C}[V]$ is the coordinate ring of V. □

Lemma 4.6(3) is the version of [Nam11, Theorem (3.1)] for the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R)$.

Proposition 4.7. *Notation and conditions as in Set-up 2.10. Let $T \subset G_R$ be a maximal torus, and let $\mathfrak{h} \subset R_r$ be the subalgebra corresponding to $\operatorname{Lie} T$ under the isomorphism of Lemma 4.2. Assume that the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R)$ is reductive. Then the following hold.*

(1) G_R is a connected reductive group.
(2) There exists an algebraic group involution c_0 of G_R with $c_0(t) = t^{-1}$ for any $t \in T$, and under the isomorphism of Lemma 4.2, the differential of c_0 is a Lie algebra involution of R_r whose restriction to $\mathfrak{h}$ is $-\mathrm{id}_{\mathfrak{h}}$.

Proof. Let H be the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R)$.

(1) Let $g \in \operatorname{Aut}_{\mathrm{gr}}(R)$. By Definition 2.13, g acts on 2-forms on X_{reg} by the pullback by γ_g. By Lemma 2.11(1), the Poisson bracket on R is the extension of the bracket on X_{reg} defined by ω, that is, $\{f, h\} = \zeta_f(h)$ on X_{reg} for any $f, h \in R$. Thus Lemma 2.12, applied to γ_g and the scalar 1, implies that g preserves the Poisson bracket if and only if $\gamma_g^* \omega = \omega$. Hence

$$\operatorname{Stab}_H(\omega) = H \cap \operatorname{Aut}_{\mathrm{gr}}(R, \omega).$$

Since G_R is a connected closed subgroup of $\operatorname{Aut}_{\mathrm{gr}}(R)$, we have $G_R \subset H$, so

$$G_R \subset \operatorname{Stab}_H(\omega) \subset \operatorname{Aut}_{\mathrm{gr}}(R, \omega).$$

Since G_R is the identity component of $\operatorname{Aut}_{\mathrm{gr}}(R, \omega)$, G_R is the identity component of $\operatorname{Stab}_H(\omega)$. By Lemma 4.6(3), $H/\operatorname{Stab}_H(\omega)$ is affine. Since H is reductive, Matsushima's criterion (see [Arz05, Theorem 2.1]) implies that $\operatorname{Stab}_H(\omega)$ is reductive. Thus G_R is a connected reductive group.

(2) Since T is contained in a Borel subgroup of G_R [Mil17, Definition 17.12 and the sentence after it], there exists a pinning of G_R with maximal torus T [Mil17, Definition 23.36]. By [Ada12, Section 2], the pinning determines an algebraic group involution c_0 of G_R with $c_0(t) = t^{-1}$ for any $t \in T$. Since $c_0 \circ c_0 = \mathrm{id}_{G_R}$, the differential dc_0 of c_0 is a Lie algebra involution of $\operatorname{Lie} G_R$. Since the differential of the morphism $T \to T$, $t \to t^{-1}$, is $-\mathrm{id}_{\operatorname{Lie} T}$, the restriction of dc_0 to $\operatorname{Lie} T$ is $-\mathrm{id}_{\operatorname{Lie} T}$. Since $\mathfrak{h}$ corresponds to $\operatorname{Lie} T$ under the isomorphism $R_r \to \operatorname{Lie} G_R$, $a \to \{a, -\}$, of Lemma 4.2(2), (2) follows. □

4.4. **Chevalley involutions.** In this subsection, for a connected reductive group G and a maximal torus $T \subset G$, we study the Lie algebra involutions of $\operatorname{Lie} G$ whose restriction to $\operatorname{Lie} T$ is $-\mathrm{id}_{\operatorname{Lie} T}$, and we integrate such involutions to algebraic group involutions of G. For $G = G_R$, under the isomorphism of Lemma 4.2, such involutions are the Chevalley involutions in the sense of Question 1.5. Lemma 4.8 reduces Theorem 1.8(3) to the condition that, for any d, the weights of a maximal torus of G_R on R_d, counted with multiplicity, are symmetric under negation, and the condition in Lemma 4.8(2) does not involve C.

Lemma 4.8. *Let G be a connected reductive group, let $T \subset G$ be a maximal torus, let c_0 be an algebraic group involution of G with $c_0(t) = t^{-1}$ for any $t \in T$, and let C be a Lie algebra involution of $\operatorname{Lie} G$ whose restriction to $\operatorname{Lie} T$ is $-\mathrm{id}_{\operatorname{Lie} T}$. Let W be a rational representation of G with $\dim W < +\infty$, and for $\chi \in X^*(T)$, let $W_\chi \subset W$ be the subspace on which T acts by χ. Then the following hold.*

(1) There exists a unique algebraic group involution c of G whose differential is C, and $c(t) = t^{-1}$ for any $t \in T$.
(2) $W \cong W^c$ as representations of G if and only if $\dim W_\chi = \dim W_{-\chi}$ for any $\chi \in X^*(T)$, where W^c is the representation of G on the vector space W given by $g \cdot_c w := c(g)w$.

Proof. (1) Since $c_0 \circ c_0 = \mathrm{id}_G$, we have $dc_0 \circ dc_0 = \mathrm{id}_{\mathrm{Lie}\, G}$. Since $c_0(t) = t^{-1}$ for any $t \in T$, the restriction of dc_0 to $\mathrm{Lie}\, T$ is $-\mathrm{id}_{\mathrm{Lie}\, T}$. Thus the Lie algebra automorphism $C \circ dc_0$ of $\mathrm{Lie}\, G$ is the identity on $\mathrm{Lie}\, T$.

Let $\Phi \subset X^*(T)$ be the set of roots of G with respect to T, and for $\alpha \in \Phi$, let $(\mathrm{Lie}\, G)_\alpha$ be the root space of α. Recall that

$$\mathrm{Lie}\, G = \mathrm{Lie}\, T \oplus \bigoplus_{\alpha \in \Phi} (\mathrm{Lie}\, G)_\alpha,$$

that $\dim(\mathrm{Lie}\, G)_\alpha = 1$ for any $\alpha \in \Phi$, and that, via $\alpha \to d\alpha$, Φ is the root system of the semisimple Lie algebra $[\mathrm{Lie}\, G, \mathrm{Lie}\, G]$ with respect to the Cartan subalgebra $[\mathrm{Lie}\, G, \mathrm{Lie}\, G] \cap \mathrm{Lie}\, T$, with root spaces $(\mathrm{Lie}\, G)_\alpha$ (see [Mil17, 21.56, Proposition 21.60(c), and Theorem 21.11(b)]).

Let $\alpha \in \Phi$. Since $C \circ dc_0$ is the identity on $\mathrm{Lie}\, T$, $C \circ dc_0$ preserves $(\mathrm{Lie}\, G)_\alpha$. Let $b_\alpha \in \mathbb{C}^*$ be the scalar by which $C \circ dc_0$ acts on $(\mathrm{Lie}\, G)_\alpha$. Since the restriction of dc_0 to $\mathrm{Lie}\, T$ is $-\mathrm{id}_{\mathrm{Lie}\, T}$, dc_0 maps $(\mathrm{Lie}\, G)_\alpha$ onto $(\mathrm{Lie}\, G)_{-\alpha}$. Since C and dc_0 are involutions, we have $dc_0 \circ (C \circ dc_0) \circ dc_0 = dc_0 \circ C = (C \circ dc_0)^{-1}$, so $b_{-\alpha} = b_\alpha^{-1}$.

Fix a system of positive roots $\Phi^+ \subset \Phi$, so that Φ is the disjoint union of Φ^+ and $-\Phi^+$, and let $\alpha_1, \dots, \alpha_l$ be the simple roots, that is, the elements of Φ^+ that are not sums of two elements of Φ^+ [Eti24, Definition 21.13]. Since $\alpha_1, \dots, \alpha_l$ are linearly independent [Eti24, Theorem 21.16], the homomorphism $T \to \mathbb{G}_m^l$, $s \to (\alpha_1(s), \dots, \alpha_l(s))$, is surjective [Mil17, Theorem 12.9]. Let $t \in T$ with $\alpha_j(t) = b_{\alpha_j}$ for $1 \le j \le l$.

We show that $\alpha(t) = b_\alpha$ for any $\alpha \in \Phi^+$. By [Eti24, Lemma 21.14 and Theorem 21.16], any $\alpha \in \Phi^+$ can be written uniquely as $\alpha = \sum_{j=1}^l m_j \alpha_j$ with non-negative integers $m_1, \dots, m_l$, and we argue by induction on $\sum_{j=1}^l m_j$. If α is simple, then $\alpha(t) = b_\alpha$ by the choice of t. If α is not simple, then $\alpha = \beta + \gamma$ for some $\beta, \gamma \in \Phi^+$, and $\beta \neq \gamma$ because Φ is reduced [Eti24, Proposition 21.3]. By the uniqueness of the coefficients, the coefficient sums of β and γ are smaller than the coefficient sum of α, so $\beta(t) = b_\beta$ and $\gamma(t) = b_\gamma$ by induction. By [Eti24, Corollary 19.18(i) and Theorem 19.19(iv)], $[(\mathrm{Lie}\, G)_\beta, (\mathrm{Lie}\, G)_\gamma] = (\mathrm{Lie}\, G)_\alpha$. Since $C \circ dc_0$ is a Lie algebra automorphism, we have $b_\alpha = b_\beta b_\gamma = \beta(t)\gamma(t) = \alpha(t)$.

Moreover, $(-\alpha)(t) = \alpha(t)^{-1} = b_\alpha^{-1} = b_{-\alpha}$ for any $\alpha \in \Phi^+$, so $\alpha(t) = b_\alpha$ for any $\alpha \in \Phi$. Since $\mathrm{Ad}(t)$ is the identity on $\mathrm{Lie}\, T$ and acts on $(\mathrm{Lie}\, G)_\alpha$ by $\alpha(t)$ for any $\alpha \in \Phi$, we have $C \circ dc_0 = \mathrm{Ad}(t)$, that is, $C = \mathrm{Ad}(t) \circ dc_0$. Let $c := \mathrm{Int}(t) \circ c_0$, where $\mathrm{Int}(t)(g) := tgt^{-1}$ for $g \in G$. Then c is an algebraic group automorphism of G whose differential is $\mathrm{Ad}(t) \circ dc_0 = C$, and $c(s) = ts^{-1}t^{-1} = s^{-1}$ for any $s \in T$. Since $c_0 \circ \mathrm{Int}(t) = \mathrm{Int}(c_0(t)) \circ c_0$ and $c_0(t) = t^{-1}$, we have

$$c \circ c = \mathrm{Int}(t) \circ \mathrm{Int}(c_0(t)) \circ c_0 \circ c_0 = \mathrm{Int}(tt^{-1}) = \mathrm{id}_G\,.$$

Let c' be an algebraic group endomorphism of G whose differential is C, and let

$$G_1 := \{g \in G \mid c(g) = c'(g)\}.$$

Since G_1 is the preimage of the diagonal under $(c, c') : G \to G \times G$, $\mathrm{Lie}\, G_1$ is the subspace of $\mathrm{Lie}\, G$ on which the differentials of c and c' coincide [Mil17, 10.14], that is, $\mathrm{Lie}\, G_1 = \mathrm{Lie}\, G$. Since G_1 is smooth [Mil17, Theorem 3.23] and G is connected, $G_1 = G$ [Mil17, Proposition 10.15]. Thus $c' = c$.

(2) Since $c(s) = s^{-1}$ for any $s \in T$, the subspace of W^c on which T acts by χ is $W_{-\chi}$ for any $\chi \in X^*(T)$. Since G is connected and reductive, two rational representations W' and W'' of G with $\dim W' < +\infty$ and $\dim W'' < +\infty$ are isomorphic if and only if $\dim W'_\chi = \dim W''_\chi$ for any $\chi \in X^*(T)$, where $W'_\chi \subset W'$ and $W''_\chi \subset W''$ are the subspaces on which T acts by χ [Mil17, Theorems 22.38 and 22.42]. Thus $W \cong W^c$ if and only if $\dim W_\chi = \dim W_{-\chi}$ for any $\chi \in X^*(T)$. □

5. Moment maps and Hamiltonian reduction

In this section, we prove Theorem 5.6, which, together with Theorem 1.6, implies Theorem 1.9.

5.1. **Moment maps.** In this subsection, we define the moment map of a torus acting on R and the Hamiltonian reduction of X by a torus.

Definition-Lemma 5.1. Notation and conditions as in Set-up 2.10, and let T be an algebraic torus acting on R by graded Poisson automorphisms. Then there exists a unique $\mathbb{C}$-linear map $\mu : \mathfrak{t} \to R_r$ such that $\{\mu(a), f\} = a \cdot f$ for any $a \in \mathfrak{t}$ and $f \in R$. We call μ the *moment map* of the T-action, and we also denote by $\mu : X \to \mathfrak{t}^*$ the morphism with $\langle \mu(x), a\rangle = \mu(a)(x)$ for any closed point $x \in X$ and $a \in \mathfrak{t}$. Moreover, the following hold.

(1) $g(\mu(a)) = \mu(a)$ for any $a \in \mathfrak{t}$ and any $g \in \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ commuting with the image of T.
(2) $\mu(a) = 0$ for any a in the Lie algebra of the kernel of the homomorphism $T \to \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$.
(3) $\mu(o) = 0$.

Proof. Since R_d is a rational representation of T for any $d \geq 0$, Definition-Lemma 4.1(1) implies that the homomorphism $T \to \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ defining the action is a homomorphism of algebraic groups. Since T is connected, the image of T is contained in G_R. By Definition-Lemma 4.1(2), the differential of $T \to G_R$ maps any $a \in \mathfrak{t}$ to the derivation $f \to a \cdot f$ of R, which is thus an element of $\mathrm{Lie}\, G_R$. By Lemma 4.2(2), for any $a \in \mathfrak{t}$, there exists a unique $\mu(a) \in R_r$ such that $\{\mu(a), f\} = a \cdot f$ for any $f \in R$. The map $\mu : \mathfrak{t} \to R_r$ is $\mathbb{C}$-linear, since the differential of $T \to G_R$ and the inverse of the isomorphism of Lemma 4.2(2) are $\mathbb{C}$-linear. The morphism $\mu : X \to \mathfrak{t}^*$ is defined by the homomorphism of $\mathbb{C}$-algebras $\mathbb{C}[\mathfrak{t}^*] \to R$ extending $\mu : \mathfrak{t} \to R_r$, where $\mathbb{C}[\mathfrak{t}^*] = \mathrm{Sym}(\mathfrak{t})$ is the coordinate ring of $\mathfrak{t}^*$.

(1) Let $g \in \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ commute with the image of T, and let $a \in \mathfrak{t}$. Since $g(R_d) = R_d$ for any $d \geq 0$, and $g(t \cdot h) = t \cdot g(h)$ for any $t \in T$ and $h \in R$, differentiating at the identity of T in the direction of a implies that $g(a \cdot h) = a \cdot g(h)$ for any $h \in R$. Since g preserves the Poisson bracket, and $\{\mu(a), f\} = a \cdot f$ for any $f \in R$, we have

$$\{g(\mu(a)), -\} = g \circ \{\mu(a), -\} \circ g^{-1} = \{\mu(a), -\}.$$

Thus $\{g(\mu(a)) - \mu(a), f\} = 0$ for any $f \in R$, and hence $g(\mu(a)) - \mu(a) \in \mathbb{C}$ by Lemma 4.2(1). Since $g(\mu(a)) - \mu(a) \in R_r$ and $r > 0$, we have $g(\mu(a)) = \mu(a)$.

(2) Let a be an element of the Lie algebra of the kernel of $T \to \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$. Since the kernel acts trivially on R, we have $a \cdot f = 0$, and hence $\{\mu(a), f\} = 0$, for any $f \in R$. By Lemma 4.2(1), $\mu(a) \in \mathbb{C}$. Since $\mu(a) \in R_r$ and $r > 0$, we have $\mu(a) = 0$.

(3) Since $r > 0$, we have $\mu(a) \in \bigoplus_{d>0} R_d$ for any $a \in \mathfrak{t}$. By Definition 2.5, $\langle \mu(o), a\rangle = \mu(a)(o) = 0$ for any $a \in \mathfrak{t}$. Thus $\mu(o) = 0$. □

See Remark 1.10(2) for moment maps with the opposite sign convention.

Definition-Proposition 5.2. Notation and conditions as in Set-up 2.10, let T be an algebraic torus acting on R by graded Poisson automorphisms, and let μ be the moment map of the T-action. Let $I := R\mu(\mathfrak{t})$, $S := (R/I)^T$, and $Y := \mathrm{Spec}\, S$. We call Y the *Hamiltonian reduction* of X by T. Then the following hold.

(1) I is a homogeneous ideal with $I \cap R_r = \mu(\mathfrak{t})$.
(2) $S = \bigoplus_{d \geq 0} S_d$ with $S_d := (R_d/(I \cap R_d))^T$ is a grading with $S_0 = \mathbb{C}$.
(3) The formula $\{f + I, h + I\} := \{f, h\} + I$ for $f, h \in R^T$ defines a Poisson bracket on S with $\{S_d, S_e\} \subset S_{d+e-r}$ for any $d, e \geq 0$, where $S_m := 0$ for $m < 0$.
(4) Any $g \in \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ commuting with the image of T satisfies $g(I) = I$ and induces a graded automorphism of S preserving the Poisson bracket of S.

Proof. For any $t \in T$, the image of t in $\mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ commutes with the image of T since T is commutative. Thus $t \cdot \mu(a) = \mu(a)$ for any $t \in T$ and $a \in \mathfrak{t}$ by Definition-Lemma 5.1(1). Hence $\mu(\mathfrak{t}) \subset R_r^T$, the ideal I is T-stable, and T acts on R/I.

(1) Since I is generated by $\mu(\mathfrak{t}) \subset R_r$, the ideal I is homogeneous. Let $a_1, \ldots, a_k$ be a basis of $\mathfrak{t}$, where $k = \dim T$. Any element of I is of the form $\sum_{j=1}^k f_j \mu(a_j)$ with $f_j \in R$, and the degree-r component of $\sum_{j=1}^k f_j \mu(a_j)$ is $\sum_{j=1}^k c_j \mu(a_j) \in \mu(\mathfrak{t})$, where $c_j \in R_0 = \mathbb{C}$ is the degree-0

component of f_j. Any $x \in I \cap R_r$ is equal to the degree-r component of x, so $I \cap R_r \subset \mu(\mathfrak{t})$. Since $\mu(\mathfrak{t}) \subset I \cap R_r$, we have $I \cap R_r = \mu(\mathfrak{t})$.

(2) Since I is homogeneous and T-stable, $R/I = \bigoplus_{d\geq 0} R_d/(I \cap R_d)$ as representations of T. Thus $S = \bigoplus_{d\geq 0} S_d$, and $S_d S_e \subset S_{d+e}$ for any $d, e \geq 0$. Since I is generated by elements of degree $r > 0$, we have $I \cap R_0 = 0$. Since T acts trivially on $R_0 = \mathbb{C}$, we have $S_0 = \mathbb{C}$.

(3) Since R_d is a rational representation of T for any $d \geq 0$, taking T-invariants is exact on the sequence $0 \to I \cap R_d \to R_d \to R_d/(I \cap R_d) \to 0$ [Mil17, Theorem 12.12]. Thus the homomorphism $R^T \to S$ is surjective and maps R_d^T onto S_d for any $d \geq 0$. Let $f \in R^T$. For any $a \in \mathfrak{t}$, we have $\{\mu(a), f\} = a \cdot f = 0$, and thus the Leibniz rule implies that

$$\{f'\mu(a), f\} = f'\{\mu(a), f\} + \mu(a)\{f', f\} = \mu(a)\{f', f\} \in I$$

for any $f' \in R$. Thus $\{I, f\} \subset I$ for any $f \in R^T$. Moreover, $\{f, h\} \in R^T$ for any $f, h \in R^T$, since T acts by Poisson automorphisms. Therefore, for $f, f', h, h' \in R^T$ with $f - f' \in I$ and $h - h' \in I$, we have

$$\{f, h\} - \{f', h'\} = \{f - f', h\} - \{h - h', f'\} \in I.$$

Thus the formula defines a bilinear map $S \times S \to S$. Since $R^T \to S$ is a surjective homomorphism of $\mathbb{C}$-algebras mapping $\{f, h\}$ to $\{f + I, h + I\}$ for any $f, h \in R^T$, the skew-symmetry, the Jacobi identity, and the Leibniz rule for the bracket of S follow from those for the bracket of R. Hence the formula defines a Poisson bracket on S. For $f \in R_d^T$ and $h \in R_e^T$, we have $\{f, h\} \in R_{d+e-r}$ by Lemma 2.11(2), where $R_m = 0$ for $m < 0$. Thus $\{S_d, S_e\} \subset S_{d+e-r}$ for any $d, e \geq 0$.

(4) Let $g \in \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ commute with the image of T. By Definition-Lemma 5.1(1), $g(\mu(a)) = \mu(a)$ for any $a \in \mathfrak{t}$, and thus $g(I) = g(R)\mu(\mathfrak{t}) = I$. Hence g induces a graded automorphism $\bar{g}$ of R/I, whose inverse is induced by g^{-1}, and $\bar{g}$ commutes with the action of T on R/I. Thus $\bar{g}$ preserves S_d for any $d \geq 0$, and induces a graded automorphism of S, which we also denote by $\bar{g}$. For $f, h \in R^T$, we have $g(f), g(h) \in R^T$, and

$$\bar{g}(\{f + I, h + I\}) = g(\{f, h\}) + I = \{g(f), g(h)\} + I = \{\bar{g}(f + I), \bar{g}(h + I)\}.$$

Thus $\bar{g}$ preserves the Poisson bracket of S. □

5.2. **Flatness of moment maps.** In this subsection, we show that the moment map of an algebraic torus $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ is faithfully flat if (X, ω) is a symplectic singularity.

In Subsection 5.4, we compute the dimensions of the weight spaces of R/I from the dimensions of the weight spaces of R, using the Koszul complex of $\mu(a_1), \dots, \mu(a_k)$ for a basis $a_1, \dots, a_k$ of $\mathfrak{t}$. The computation uses that $\mu(a_1), \dots, \mu(a_k)$ is an R-regular sequence (Proposition 5.3(2)). Otherwise, the alternating sums of the dimensions of the weight spaces of the terms of the Koszul complex only compute the Euler characteristics of the Koszul homology.

Proposition 5.3. *Notation and conditions as in Set-up 2.10. Let $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ be an algebraic torus of dimension k, and let μ be the moment map of the T-action. Assume that (X, ω) is a symplectic singularity. Then the following hold.*

(1) The morphism $\mu : X \to \mathfrak{t}^*$ is faithfully flat, and any fiber of μ is of pure dimension $2n - k$.
(2) For any basis $a_1, \dots, a_k$ of $\mathfrak{t}$, the sequence $\mu(a_1), \dots, \mu(a_k)$ is R-regular, and the Koszul complex of $\mu(a_1), \dots, \mu(a_k)$ is a free resolution of the R-module $R/R\mu(\mathfrak{t})$.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we show that $\dim X^{T_1} \leq 2n - s$ for any subtorus $T_1 \subset T$ of dimension s, where X^{T_1} is the set of closed points of X fixed by T_1.

For $\psi \in X^*(T_1)$, we call the subspace $\{f \in R \mid t \cdot f = \psi(t)f \text{ for any } t \in T_1\}$ the *eigenspace* of T_1 with character ψ. Since R_d is a rational representation of T_1 for any $d \geq 0$, R is the direct sum of the eigenspaces of T_1 [Mil17, Theorem 12.12]. We first show that there exist non-zero eigenvectors $f_1, \dots, f_s \in R$ of T_1 whose characters $\psi_1, \dots, \psi_s$ are linearly independent in $X^*(T_1)$. Otherwise, the characters of the non-zero eigenvectors of T_1 in R span a subgroup of $X^*(T_1)$ of rank less than s, and hence there exists a non-zero cocharacter $\nu \in X_*(T_1)$ with $\langle \psi, \nu \rangle = 0$ for

any such character ψ. Then $\psi(\nu(z)) = z^{\langle\psi,\nu\rangle} = 1$ for any such ψ and any $z \in \mathbb{G}_m$, and hence the non-trivial subgroup $\nu(\mathbb{G}_m)$ of T_1 acts trivially on R, which contradicts $T \subset \mathrm{Aut}_{\mathrm{gr}}(R,\omega)$.

Next, we show that $f_1,\dots,f_s$ are algebraically independent over the fraction field $\mathrm{Frac}(R^{T_1})$ of R^{T_1}. Assume that $\sum_\alpha c_\alpha f^\alpha = 0$, where $\alpha = (\alpha_1,\dots,\alpha_s)$ runs over a finite subset of $\mathbb{Z}^s_{\geq 0}$, $c_\alpha \in \mathrm{Frac}(R^{T_1})$, and $f^\alpha := f_1^{\alpha_1}\cdots f_s^{\alpha_s}$. Multiplying all c_α by a common denominator in $R^{T_1}\setminus\{0\}$, we may assume that $c_\alpha \in R^{T_1}$ for any α. Then $c_\alpha f^\alpha$ is an eigenvector of T_1 with character $\sum_{j=1}^s \alpha_j\psi_j$, and

$$\sum_{j=1}^s \alpha_j\psi_j \neq \sum_{j=1}^s \alpha'_j\psi_j \quad \text{for any } \alpha \neq \alpha'$$

since $\psi_1,\dots,\psi_s$ are linearly independent. Since R is the direct sum of the eigenspaces of T_1, we have $c_\alpha f^\alpha = 0$ for any α. Since R is a domain, $c_\alpha = 0$ for any α.

Let $W \subset R$ be a T-stable subspace with $\dim W < +\infty$ which generates R as a $\mathbb{C}$-algebra, for example, $W = \bigoplus_{1\leq d\leq N_0} R_d$ for a large integer N_0. Then $\mathrm{Sym}(W)^{T_1}$ is a finitely generated $\mathbb{C}$-algebra [Mil17, Proposition 12.54 and Theorem 12.57], and the homomorphism $\mathrm{Sym}(W)^{T_1} \to R^{T_1}$ is surjective [Mil17, Theorem 12.12]. Thus R^{T_1} is a finitely generated domain. By [Har77, Theorem I.1.8A(a)], we have

$$\dim R^{T_1} = \mathrm{trdeg}_{\mathbb{C}}\,\mathrm{Frac}(R^{T_1}) \leq \mathrm{trdeg}_{\mathbb{C}}\,\mathrm{Frac}(R) - s = 2n - s.$$

Let $J \subset R$ be the ideal generated by the eigenvectors of T_1 with non-trivial characters. Let $x \in X$ be a closed point. For any eigenvector $f \in R$ of T_1 with character ψ and any $t \in T_1$, we have $f(t(x)) = (t\cdot f)(x) = \psi(t)f(x)$. Since R is spanned by the eigenvectors of T_1, the point x is fixed by T_1 if and only if $f(x) = 0$ for any eigenvector f of T_1 with non-trivial character, that is, x is a zero of J. Thus X^{T_1} is the closed subset of X defined by J. Since any $f \in R$ is congruent modulo J to the component of f in the eigenspace of T_1 with the trivial character, the homomorphism $R^{T_1} \to R/J$ is surjective. Thus

$$\dim X^{T_1} = \dim R/J \leq \dim R^{T_1} \leq 2n - s.$$

Step 2. In this step we cover X by finitely many T-stable smooth irreducible locally closed subvarieties V, such that the identity component of the stabilizer in T of a closed point $x \in V$ is a subtorus $T_1 \subset T$ independent of x, $\dim V \leq 2n - \dim T_1$, and the Poisson bracket induces a non-degenerate 2-form on V.

A *Poisson subscheme* of X is a closed subscheme defined by an ideal $\mathfrak{a} \subset R$ with $\{R,\mathfrak{a}\} \subset \mathfrak{a}$ [Kal03, Section 1]. Since (X,ω) is a symplectic singularity, by [Kal03, Theorem 2.3], there exist finitely many smooth locally closed subvarieties $X_1,\dots,X_m$ of X such that X is the disjoint union of $X_1,\dots,X_m$, the closures $\overline{X_1},\dots,\overline{X_m}$ are exactly the integral Poisson subschemes of X, and for any i, the subvariety X_i is open in $\overline{X_i}$, and the Poisson bracket of $\mathcal{O}_{\overline{X_i}}$ is non-degenerate on X_i. For any i, let ω_i be the symplectic form on X_i induced by the Poisson bracket with the sign convention of Lemma 2.11, that is, $\{f,h\}|_{X_i} = \zeta(h|_{X_i})$ for any $f,h \in R$, where ζ is the vector field on X_i with $\iota_\zeta\omega_i = -d(f|_{X_i})$. Since T is connected and acts on X by Poisson automorphisms, T permutes the finitely many integral Poisson subschemes $\overline{X_1},\dots,\overline{X_m}$ trivially. Thus T preserves $\overline{X_i}$, X_i, and ω_i for any i.

Choose generators $h_1,\dots,h_q$ of the $\mathbb{C}$-algebra R such that h_j is an eigenvector of T with character $\chi_j \in X^*(T)$ for any j [Mil17, Theorem 12.12]. For a closed point $x \in X$, let $\mathrm{Stab}_T(x)$ be the stabilizer of x in T. Since $h_j(t(x)) = (t\cdot h_j)(x) = \chi_j(t)h_j(x)$ for any $t \in T$ and any j, we have

$$\mathrm{Stab}_T(x) = \bigcap_{j:h_j(x)\neq 0} \mathrm{Ker}\,\chi_j.$$

In particular, $\mathrm{Stab}_T(x)$ runs over finitely many subgroups of T when x runs over the closed points of X.

Let $T_1 \subset T$ be a subtorus of dimension s, and let $1 \leq i \leq m$. Let $X_i^{T_1} := X_i \cap X^{T_1}$, which is a closed subset of X_i by **Step 1**. Since T_1 is linearly reductive [Mil17, Proposition 12.54] and X_i is smooth, $X_i^{T_1}$ is smooth [Mil17, Theorems 7.1, 13.1, and 13.7]. Let $x \in X_i^{T_1}$ be a closed point, and let $\mathfrak{m}_x$ be the maximal ideal of $\mathcal{O}_{X_i,x}$. By [Mil17, Theorem 12.12], the vectors $v - t \cdot v$ with $v \in \mathfrak{m}_x/\mathfrak{m}_x^2$ and $t \in T_1$ span the sum of the subspaces of $\mathfrak{m}_x/\mathfrak{m}_x^2$ on which T_1 acts by non-trivial characters. Thus by [Mil17, Lemma 13.4], the cotangent space of $X_i^{T_1}$ at x is the quotient of $\mathfrak{m}_x/\mathfrak{m}_x^2$ by the sum of the subspaces on which T_1 acts by non-trivial characters, and the tangent space of $X_i^{T_1}$ at x is $(T_xX_i)^{T_1}$. Then T_1 acts on T_xX_i preserving $\omega_i(x)$, and T_xX_i is the direct sum of the subspaces on which T_1 acts by characters [Mil17, Theorem 12.12]. If $v, w \in T_xX_i$ are vectors on which T_1 acts by characters ψ and ψ', respectively, then $\omega_i(x)(v,w) = \psi(t)\psi'(t)\omega_i(x)(v,w)$ for any $t \in T_1$, and thus $\omega_i(x)(v,w) = 0$ unless $\psi + \psi' = 0$. Hence $(T_xX_i)^{T_1}$, on which T_1 acts by the trivial character, is orthogonal with respect to $\omega_i(x)$ to the subspaces on which T_1 acts by non-trivial characters. Since $\omega_i(x)$ is non-degenerate, the restriction of $\omega_i(x)$ to $(T_xX_i)^{T_1}$ is non-degenerate. Hence the restriction of ω_i to $X_i^{T_1}$ is non-degenerate.

The set $\{x \in X_i \mid \mathrm{Stab}_T(x)^\circ = T_1\}$ is the complement in $X_i^{T_1}$ of the union of the closed subsets $X^{T'}$, where T' runs over the finitely many subtori of T of the form $\mathrm{Stab}_T(y)^\circ$ with $T_1 \subsetneq T'$. Thus $\{x \in X_i \mid \mathrm{Stab}_T(x)^\circ = T_1\}$ is open in $X_i^{T_1}$. Let V be a connected component of $\{x \in X_i \mid \mathrm{Stab}_T(x)^\circ = T_1\}$. Then V is a smooth irreducible locally closed subvariety of X, the restriction $\omega_i|_V$ is non-degenerate, and $\dim V \leq \dim X^{T_1} \leq 2n - s$ by **Step 1**. Since T is connected and commutative and preserves X_i, the subvariety V is T-stable. Any closed point $x \in X$ is contained in such a V for $T_1 = \mathrm{Stab}_T(x)^\circ$ and the index i with $x \in X_i$. Since the subtori of the form $\mathrm{Stab}_T(x)^\circ$ are finite in number, and a scheme of finite type over $\mathbb{C}$ has finitely many connected components, X is the union of finitely many such V.

Step 3. In this step we show that for any closed point $\beta \in \mathfrak{t}^*$, the fiber $\mu^{-1}(\beta)$ is either empty or of pure dimension $2n - k$.

Let i, T_1, and $V \subset X_i$ be as in **Step 2**, and let $s := \dim T_1$. For $a \in \mathfrak{t}$, since $\overline{X_i}$ is a Poisson subscheme of X, the derivation $\{\mu(a), -\}$ of R induces a vector field η_a on X_i. Since $\eta_a(h|_{X_i}) = \{\mu(a), h\}|_{X_i}$ for any $h \in R$, the definition of ω_i in **Step 2** implies $\iota_{\eta_a}\omega_i = -d(\mu(a)|_{X_i})$. Let $x \in V$ be a closed point. Since $\eta_a(h|_{X_i}) = (a \cdot h)|_{X_i}$ for any $h \in R$, the tangent vector $\eta_a(x)$ is the image of a under the differential at the identity of the orbit map $T \to V$, $t \to t(x)$. Thus η_a is tangent to V, and $\iota_{\eta_a}(\omega_i|_V) = -d(\mu(a)|_V)$.

Recall that $h_1, \ldots, h_q$ are generators of R such that h_j is an eigenvector of T with character χ_j. Let $d\chi_j \in \mathfrak{t}^*$ be the differential of χ_j. Since $a \cdot h_j = d\chi_j(a)h_j$ for any j, and a tangent vector of X at x is determined by the values at $h_1, \ldots, h_q$, we have $\eta_a(x) = 0$ if and only if $d\chi_j(a) = 0$ for any j with $h_j(x) \neq 0$. By **Step 2**, $\mathrm{Stab}_T(x)$ is the kernel of the homomorphism from T to a product of copies of $\mathbb{G}_m$ defined by the characters χ_j with $h_j(x) \neq 0$. Thus by [Mil17, 10.14], the kernel of the linear map $\mathfrak{t} \to T_xV$, $a \to \eta_a(x)$, is

$$\bigcap_{j:h_j(x)\neq 0} \mathrm{Ker}(d\chi_j) = \mathrm{Lie}\,\mathrm{Stab}_T(x) = \mathrm{Lie}\,T_1,$$

which is of dimension s. Since $\omega_i|_V$ is non-degenerate, the linear map

$$\mathfrak{t} \to T_x^*V, \quad a \to d(\mu(a)|_V)_x = -\iota_{\eta_a(x)}(\omega_i|_V)(x),$$

is of rank $k - s$. Since $a \to d(\mu(a)|_V)_x$ is the transpose of the differential $d(\mu|_V)_x : T_xV \to \mathfrak{t}^*$, the kernel of $d(\mu|_V)_x$ is of dimension $\dim V - (k - s)$. Let F_x be the fiber of $\mu|_V : V \to \mathfrak{t}^*$ over $\mu(x)$. Since the Zariski tangent space of F_x at x is contained in the kernel of $d(\mu|_V)_x$, [Har77, Proposition I.5.2A] and **Step 2** imply that

$$\dim \mathcal{O}_{F_x,x} \leq \dim V - (k - s) \leq (2n - s) - (k - s) = 2n - k.$$

Let $\beta \in \mathfrak{t}^*$ be a closed point. By **Step 2**, $\mu^{-1}(\beta)$ is the union of the finitely many locally closed subsets $\mu^{-1}(\beta) \cap V$, and $\mu^{-1}(\beta) \cap V$ is the underlying set of F_x for any closed point

$x \in \mu^{-1}(\beta) \cap V$. Thus any irreducible component of $\mu^{-1}(\beta)$ is of dimension at most $2n-k$. On the other hand, let $X \subset \mathbb{A}^N$ be a closed embedding. Then $\mu^{-1}(\beta)$ is isomorphic to the intersection of the graph of μ, which is a closed subvariety of $\mathbb{A}^N \times \mathfrak{t}^* \cong \mathbb{A}^{N+k}$ of dimension $2n$, and the linear subvariety $\mathbb{A}^N \times \{\beta\}$ of dimension N. By [Har77, Proposition I.7.1], any irreducible component of $\mu^{-1}(\beta)$ is of dimension at least $2n+N-(N+k) = 2n-k$. Thus $\mu^{-1}(\beta)$ is either empty or of pure dimension $2n-k$.

Step 4. In this step we prove (1).

Let $x \in X$ be a closed point, let $\beta := \mu(x)$, and let $a_1, \dots, a_k$ be a basis of $\mathfrak{t}$. For $1 \leq j \leq k$, let t_j be the function $\beta' \to \langle \beta' - \beta, a_j \rangle$ on $\mathfrak{t}^*$. Then $t_1, \dots, t_k$ is a regular system of parameters of the regular local ring $\mathcal{O}_{\mathfrak{t}^*,\beta}$ of dimension k. We also denote by t_j the image of t_j in $\mathcal{O}_{X,x}$, that is, $t_j = \mu(a_j) - \langle \beta, a_j \rangle$. By Lemma 2.9, $\mathcal{O}_{X,x}$ is a Cohen–Macaulay local ring of dimension $2n$. Since $\mathcal{O}_{X,x}/(t_1, \dots, t_k)$ is the local ring of $\mu^{-1}(\beta)$ at x, it is of dimension $2n-k$ by **Step 3**. Thus $t_1, \dots, t_k$ is an $\mathcal{O}_{X,x}$-regular sequence by [Har77, Theorem II.8.21A(c)].

We show that $\mathcal{O}_{X,x}/(t_1, \dots, t_i)$ is flat over $\mathcal{O}_{\mathfrak{t}^*,\beta}/(t_1, \dots, t_i)$ for any $0 \leq i \leq k$ by descending induction on i. If $i = k$, then $\mathcal{O}_{\mathfrak{t}^*,\beta}/(t_1, \dots, t_k) = \mathbb{C}$, and any $\mathbb{C}$-module is flat. If $i < k$ and the claim holds for $i+1$, then the claim holds for i by [Har77, Lemma III.10.3A], since t_{i+1} is a non-zero-divisor in both $\mathcal{O}_{\mathfrak{t}^*,\beta}/(t_1, \dots, t_i)$ and $\mathcal{O}_{X,x}/(t_1, \dots, t_i)$. The case $i = 0$ of the claim implies that μ is flat at x. Since the local ring of X at any point is a localization of the local ring of X at a closed point, μ is flat by [Har77, Proposition III.9.1A].

By [Har77, Exercise III.9.1], $\mu(X)$ is an open subset of $\mathfrak{t}^*$, and $0 = \mu(o) \in \mu(X)$ by Definition-Lemma 5.1(3). Since $\mu(\mathfrak{t}) \subset R_r$, we have $\mu(\lambda_z(y)) = z^r \mu(y)$ for any $z \in \mathbb{C}^*$ and any closed point $y \in X$. Since any element of $\mathbb{C}^*$ is of the form z^r with $z \in \mathbb{C}^*$, the open subset $\mu(X)$, and hence the closed subset $\mathfrak{t}^* \backslash \mu(X)$, is stable under the multiplication by $\mathbb{C}^*$. If $\mathfrak{t}^* \backslash \mu(X)$ contained a point $\beta' \neq 0$, then $\mathfrak{t}^* \backslash \mu(X)$ would contain the closure of $\mathbb{C}^* \beta'$, which contains 0, a contradiction. Thus μ is surjective, and hence faithfully flat. By [Har77, Corollary III.9.6], since μ is flat and X is irreducible of dimension $2n$, any fiber of μ over a point of $\mathfrak{t}^*$, closed or not, is of pure dimension $2n-k$. This proves (1).

Step 5. In this step we prove (2).

Let $a_1, \dots, a_k$ be a basis of $\mathfrak{t}$, let $\mathbb{C}[\mathfrak{t}^*] = \mathrm{Sym}(\mathfrak{t})$ be the coordinate ring of $\mathfrak{t}^*$, and for $1 \leq j \leq k$, let $u_j \in \mathbb{C}[\mathfrak{t}^*]$ be the function $\beta' \to \langle \beta', a_j \rangle$. Then $\mathbb{C}[\mathfrak{t}^*] = \mathbb{C}[u_1, \dots, u_k]$ is a polynomial ring, and $\mu^* u_j = \mu(a_j)$ for any j. By **Step 4**, R is a flat $\mathbb{C}[\mathfrak{t}^*]$-module via μ^*. For $1 \leq j \leq k$, the multiplication by u_j on $\mathbb{C}[\mathfrak{t}^*]/(u_1, \dots, u_{j-1})$ is injective. Tensoring the multiplication by u_j with the flat $\mathbb{C}[\mathfrak{t}^*]$-module R, we see that the multiplication by $\mu(a_j)$ on $R/(\mu(a_1), \dots, \mu(a_{j-1}))$ is injective. Moreover, $R/(\mu(a_1), \dots, \mu(a_k)) = R/R\mu(\mathfrak{t})$ is non-zero, since $R\mu(\mathfrak{t})$ is contained in the maximal ideal of o by Definition-Lemma 5.1(3). Thus $\mu(a_1), \dots, \mu(a_k)$ is an R-regular sequence. By [Har77, Proposition III.7.10A], the Koszul complex of $\mu(a_1), \dots, \mu(a_k)$ is a free resolution of $R/R\mu(\mathfrak{t})$. □

5.3. **Maximal tori of the reduction.** In this subsection, we compare maximal tori of G_R with maximal tori of the graded Poisson automorphism group of the Hamiltonian reduction.

Set-up 5.4. Notation and conditions as in Set-up 2.10. Let T, μ, I, S, Y, ω_Y, o_Y, λ_S, G_S, H, and φ be as follows.

(1) $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ is an algebraic torus, and μ is the moment map of the T-action.
(2) $I := R\mu(\mathfrak{t})$, $S = \bigoplus_{d \geq 0} S_d := (R/I)^T$ with the grading and the Poisson bracket of Definition-Proposition 5.2, and $Y := \operatorname{Spec} S$.
(3) S is a normal domain, and $\dim Y > 0$.
(4) ω_Y is a closed non-degenerate algebraic 2-form on Y_{reg} which induces the Poisson bracket of S, that is, $\{f, h\} = \zeta_f(h)$ on Y_{reg} for any $f, h \in S$, where ζ_f is the vector field on Y_{reg} with $\iota_{\zeta_f} \omega_Y = -df$.
(5) $o_Y \in Y$ is the vertex of the grading of S, and λ_S is the $\mathbb{G}_m$-action on Y defined by the grading of S.

(6) G_S is the identity component of $\mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$, which is a linear algebraic group by Proposition 5.5(1) and Definition-Lemma 4.1(1).
(7) H is the identity component of the centralizer of T in G_R, and $\varphi : H \to \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ is the group homomorphism induced by Definition-Proposition 5.2(4).

The proof of Proposition 5.5(1) only uses Set-up 5.4(1)–(5).

The hypothesis of Theorem 5.6 concerns maximal tori of G_R containing T, while the argument of Subsection 5.4 uses maximal tori of G_S. Proposition 5.5 shows that any maximal torus of G_S is the image under φ of a maximal torus of G_R containing T.

Proposition 5.5. *Notation and conditions as in Set-up 5.4. Then the following hold.*

(1) The form ω_Y is homogeneous of degree r with respect to the grading of S. In particular, S and ω_Y satisfy the conditions of Set-up 2.10, with $\dim Y$ in place of $2n$.
(2) The map φ is a homomorphism of algebraic groups with $\varphi(H) = G_S$, and the identity component of $\operatorname{Ker}\varphi$ is T.
(3) Under the isomorphisms $R_r \cong \operatorname{Lie} G_R$ and $S_r \cong \operatorname{Lie} G_S$ of Lemma 4.2(2), $\operatorname{Lie} H$ corresponds to R_r^T, and the differential of φ is the quotient map $R_r^T \to R_r^T/\mu(\mathfrak{t}) \cong S_r$.
(4) For any maximal torus $A_S \subset G_S$, the identity component A of $\varphi^{-1}(A_S)$ is a maximal torus of G_R with $T \subset A$ and $\varphi(A) = A_S$, and the induced homomorphism $A/T \to A_S$ is an isogeny.

Proof. (1) Let N_0 be a positive integer such that $W := \bigoplus_{1 \le d \le N_0} R_d$ generates R as a $\mathbb{C}$-algebra. Since W is T-stable and $\dim W < +\infty$, the $\mathbb{C}$-algebra $\mathrm{Sym}(W)^T$ is finitely generated [Mil17, Proposition 12.54 and Theorem 12.57]. Since the surjection $\mathrm{Sym}(W) \to R/I$ is T-equivariant, the homomorphism $\mathrm{Sym}(W)^T \to (R/I)^T = S$ is surjective [Mil17, Theorem 12.12]. Thus S is finitely generated. By Definition-Proposition 5.2(2), $S = \bigoplus_{d \ge 0} S_d$ is a grading with $S_0 = \mathbb{C}$, and S is a normal domain by Set-up 5.4(3). Since ω_Y is non-degenerate on the non-empty open subset Y_{reg} of Y, the dimension of Y is even. Thus $\dim Y \ge 2$ by Set-up 5.4(3).

Let $t \in \mathbb{G}_m$, and let $\lambda_{S,t}$ be the automorphism of Y defined by t under λ_S. For $f \in S_d$ and $h \in S_e$, we have $\{f, h\} \in S_{d+e-r}$ by Definition-Proposition 5.2(3), and thus

$$\lambda_{S,t}^*\{f, h\} = t^{d+e-r}\{f, h\} = t^{-r}\{\lambda_{S,t}^* f, \lambda_{S,t}^* h\}.$$

Since the Poisson bracket of S is bilinear, $\lambda_{S,t}^*\{f, h\} = t^{-r}\{\lambda_{S,t}^* f, \lambda_{S,t}^* h\}$ for any $f, h \in S$. By Set-up 5.4(4) and Lemma 2.12 applied to Y, ω_Y, $\gamma = \lambda_{S,t}$, and $c = t^{-r}$, we have $\lambda_{S,t}^* \omega_Y = t^r \omega_Y$ on Y_{reg}. Therefore ω_Y is homogeneous of degree r. Since $r > 0$, S and ω_Y satisfy the conditions of Set-up 2.10 with $\dim Y$ in place of $2n$. Moreover, by Set-up 5.4(4), the Poisson bracket of Lemma 2.11 for S and ω_Y is the Poisson bracket of S.

(2) and (3) We first show that φ is a homomorphism of algebraic groups with $\varphi(H) \subset G_S$. By Definition-Lemma 4.1(1), R_d is a rational representation of H for any $d \ge 0$. Since any element of H commutes with T and preserves I (Definition-Proposition 5.2(4)), $S_d = (R_d/(I \cap R_d))^T$ is an H-stable subquotient of R_d, and hence a rational representation of H. By (1), Definition-Lemma 4.1(1) for S and ω_Y embeds $\mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ as a closed subgroup of $\prod_{1 \le d \le N_1} \mathrm{GL}(S_d)$ for some positive integer N_1, and $H \to \prod_{1 \le d \le N_1} \mathrm{GL}(S_d)$ is a homomorphism of algebraic groups. Thus φ is a homomorphism of algebraic groups. Since H is connected, $\varphi(H) \subset G_S$.

Next, we prove (3). Since H is the identity component of the centralizer of T in G_R, the Lie algebra $\operatorname{Lie} H$ is the subspace of $\operatorname{Lie} G_R$ fixed by the adjoint action of T [Mil17, Proposition 10.34]. For $t \in T$ and $f \in R_r$, we have $t \circ \{f, -\} \circ t^{-1} = \{t \cdot f, -\}$ since t preserves the Poisson bracket. Thus the isomorphism $R_r \to \operatorname{Lie} G_R$, $f \to \{f, -\}$, of Lemma 4.2(2) maps R_r^T onto $\operatorname{Lie} H$. Since φ is defined by passing to the subquotients S_d of R_d, the differential of φ maps any derivation in $\operatorname{Lie} H$ to the induced derivation of S (Definition-Lemma 4.1(2)). For $f \in R_r^T$ and $h \in R^T$, we have $\{f, h\} + I = \{f + I, h + I\}$ by Definition-Proposition 5.2(3), and hence the derivation of S induced by $\{f, -\}$ is $\{f + I, -\}$. Since S and ω_Y satisfy the conditions of Set-up 2.10 by (1), Lemma 4.2(2) implies that $S_r \to \operatorname{Lie} G_S$, $f \to \{f, -\}$, is an isomorphism. Under the

isomorphisms of Lemma 4.2(2) for R and S, the differential of φ is thus the homomorphism $R_r^T \to S_r$, $f \to f + I$. Since $S_r = (R_r/(I \cap R_r))^T$, the homomorphism $R_r^T \to S_r$ is surjective by the exactness of taking T-invariants [Mil17, Theorem 12.12]. The kernel of $R_r^T \to S_r$ is $(I \cap R_r)^T = \mu(\mathfrak{t})$ by Definition-Proposition 5.2(1) and Definition-Lemma 5.1(1). This proves (3).

Finally, we prove the rest of (2). By [Mil17, Proposition 1.68], $\varphi(H)$ is a closed connected subgroup of G_S. Since the differential of φ is surjective onto $\operatorname{Lie} G_S$ by (3), we have $\operatorname{Lie}\varphi(H) = \operatorname{Lie} G_S$. Since $\varphi(H)$ is smooth [Mil17, Theorem 3.23] and G_S is connected, $\varphi(H) = G_S$ [Mil17, Proposition 10.15]. Since T is connected and commutative, and $T \subset G_R$ by Definition-Lemma 4.1(3), we have $T \subset H$. Since T acts trivially on $S = (R/I)^T$, we have $T \subset \operatorname{Ker}\varphi$. If $a \in \mathfrak{t}$ satisfies $\mu(a) = 0$, then $a \cdot f = \{\mu(a), f\} = 0$ for any $f \in R$, that is, a is contained in the kernel of the differential of the inclusion $T \to G_R$ (Definition-Lemma 4.1(2)), and thus $a = 0$ [Mil17, 10.14]. Hence μ is injective. By (3) and [Mil17, 10.14], $\operatorname{Lie}\operatorname{Ker}\varphi$ corresponds to $\mu(\mathfrak{t})$ under the isomorphism of Lemma 4.2(2). Thus $\dim\operatorname{Ker}\varphi = \dim\mu(\mathfrak{t}) = \dim T$ [Mil17, Theorem 3.23 and Proposition 1.37]. Since $T \subset (\operatorname{Ker}\varphi)^\circ$ and $(\operatorname{Ker}\varphi)^\circ$ is irreducible [Mil17, Corollary 1.35], we have $(\operatorname{Ker}\varphi)^\circ = T$. This proves (2).

(4) By (2) and [Mil17, Theorem 3.23 and Proposition 1.70], $\varphi : H \to G_S$ is a quotient map. Let $A_S \subset G_S$ be a maximal torus. By [Mil17, Proposition 17.20], there exists a maximal torus A' of H with $\varphi(A') = A_S$. Since T is contained in the center of H, the multiplication $T \times A' \to H$ is a homomorphism, and the image TA' is a quotient of the torus $T \times A'$, hence a torus [Mil17, Definition 12.14 and the paragraph after it]. Since $A' \subset TA'$ and A' is a maximal torus of H, we have $TA' = A'$, and thus $T \subset A'$. Since A' is connected and $A' \subset \varphi^{-1}(A_S)$, we have $A' \subset A$. On the other hand, since $\varphi(A') = A_S$, $T \subset A' \cap \operatorname{Ker}\varphi$, $\varphi(\varphi^{-1}(A_S)) = A_S$, and $\dim\operatorname{Ker}\varphi = \dim T$ by (2), [Mil17, Remark 5.42] implies that

$$\dim A' \geq \dim A_S + \dim T = \dim\varphi^{-1}(A_S) = \dim A.$$

Since A is irreducible [Mil17, Corollary 1.35], $A' \subset A$ is closed, and $\dim A' \geq \dim A$, we have $A' = A$. Hence A is a maximal torus of H with $T \subset A$ and $\varphi(A) = A_S$.

Let A'' be a torus of G_R containing A. Since A'' is commutative and $T \subset A \subset A''$, the torus A'' is contained in the centralizer of T in G_R, and thus $A'' \subset H$ as A'' is connected. Since A is a maximal torus of H, we have $A'' = A$. Thus A is a maximal torus of G_R. Finally, the homomorphism $A/T \to A_S$ induced by φ is surjective, and the kernel $(A \cap \operatorname{Ker}\varphi)/T$ is finite since $(A \cap \operatorname{Ker}\varphi)^\circ \subset (\operatorname{Ker}\varphi)^\circ = T$. Thus $A/T \to A_S$ is an isogeny. □

5.4. **Weight symmetry and the minimizer of the reduction.** In this subsection, we prove Theorem 5.6. We use a maximal torus of G_S (Proposition 5.5) since Theorem 2.16(1) only provides a minimizing grading defined by a cocharacter of a possibly larger torus.

Theorem 5.6. *Notation and conditions as in Set-up 5.4. Let $A \subset G_R$ be a maximal torus with $T \subset A$, so that $A \subset H$, let $A_S := \varphi(A)$, and let T_S be the subgroup of $\operatorname{Aut}_{\mathrm{gr}}(S)$ generated by $\lambda_S(\mathbb{G}_m)$ and A_S. Assume that (X, ω) and (Y, ω_Y) are symplectic singularities, and that $\dim R_{d,\chi} = \dim R_{d,-\chi}$ for any $d \geq 0$ and $\chi \in X^*(A)$. Then the degree valuation v_{λ_S} of the grading of S is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$. Moreover, the following hold.*

(1) A_S is a maximal torus of G_S, and $\dim S_{d,\beta} = \dim S_{d,-\beta}$ for any $d \geq 0$ and $\beta \in X^*(A_S)$.

(2) T_S is an algebraic torus with

$$X_*(T_S) \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{R}\lambda_S \oplus (X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}),$$

the T_S-action on Y is good with vertex o_Y, and the linear map $c\lambda_S + \eta \to c\lambda_S - \eta$, where $c \in \mathbb{R}$ and $\eta \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$, preserves the Reeb cone of the T_S-action and the function $\xi \to \widehat{\mathrm{vol}}_{Y \ni o_Y}(\mathrm{wt}_\xi)$ on the Reeb cone.

Proof. The proof is long, so we divide it into several steps.

Step 1. In this step we prove that $\dim S_{d,\beta} = \dim S_{d,-\beta}$ for any $d \geq 0$ and $\beta \in X^*(A_S)$.

Since A is connected, commutative, and contains T, any element of A commutes with T, and $A \subset H$. Each R_d is a rational representation of A (Definition-Lemma 4.1(1)), and $R_{d,\chi}$ is the subspace of R_d on which A acts by $\chi \in X^*(A)$ (Definition 2.13). By Definition-Proposition 5.2(1) and (4), I is a homogeneous ideal with $g(I) = I$ for any $g \in A$. For $d \geq 0$ and $\chi \in X^*(A)$, let $(R/I)_{d,\chi}$ be the subspace of $R_d/(I \cap R_d)$ on which A acts by χ.

Let $k := \dim T$, and let $a_1, \ldots, a_k$ be a basis of $\mathfrak{t}$. By Definition-Lemma 5.1(1), $g(\mu(a_i)) = \mu(a_i)$ for any $g \in A$ and any i. By Proposition 5.3(2), the Koszul complex of $\mu(a_1), \ldots, \mu(a_k)$ is a free resolution of R/I. We grade the term $\bigwedge^j(R^{\oplus k})$ of the Koszul complex so that the standard basis elements $e_{i_1} \wedge \cdots \wedge e_{i_j}$ have degree jr, and we let A act on $\bigwedge^j(R^{\oplus k})$ through the coefficients. Since $\mu(a_1), \ldots, \mu(a_k) \in R_r$ are A-invariant, the differentials of the Koszul complex are homogeneous of degree 0 and A-equivariant. Taking the degree d part and then the subspace on which A acts by χ [Mil17, Theorem 12.12], we get an exact sequence

$$0 \to R_{d-kr,\chi}^{\oplus\binom{k}{k}} \to \cdots \to R_{d-r,\chi}^{\oplus\binom{k}{1}} \to R_{d,\chi} \to (R/I)_{d,\chi} \to 0,$$

where $R_{e,\chi} := 0$ for $e < 0$. Since R is finitely generated and $R_0 = \mathbb{C}$, we have $\dim R_e < +\infty$ for any e. Thus

$$\dim(R/I)_{d,\chi} = \sum_{j=0}^{k} (-1)^j \binom{k}{j} \dim R_{d-jr,\chi}.$$

By assumption, the right-hand side does not change if we replace χ with $-\chi$. Thus

$$\dim(R/I)_{d,\chi} = \dim(R/I)_{d,-\chi}$$

for any $d \geq 0$ and $\chi \in X^*(A)$.

The group A_S is a torus, as a quotient of A [Mil17, Definition 12.14 and the paragraph after it], and $A_S \subset \varphi(H) = G_S$ (Proposition 5.5(2)). By Proposition 5.5(1), each S_d is a rational representation of A_S (Definition-Lemma 4.1(1)), and $S_{d,\beta}$ is the subspace of S_d on which A_S acts by $\beta \in X^*(A_S)$ (Definition 2.13). Moreover, $S_d = (R_d/(I \cap R_d))^T$ (Definition-Proposition 5.2(2)), and for any $g \in A$, $\varphi(g)$ acts on S_d as g acts on $R_d/(I \cap R_d)$ (Set-up 5.4(7)). Let $\beta \in X^*(A_S)$, and let $\chi := \beta \circ \varphi|_A \in X^*(A)$. Since $\varphi(A) = A_S$, we have $S_{d,\beta} = S_d \cap (R/I)_{d,\chi}$. Since $T \subset \operatorname{Ker}\varphi$ (Proposition 5.5(2)), we have $\chi|_T = 1$, hence T acts trivially on $(R/I)_{d,\chi}$ and $(R/I)_{d,\chi} \subset S_d$. Thus $S_{d,\beta} = (R/I)_{d,\chi}$. Similarly, $S_{d,-\beta} = (R/I)_{d,-\chi}$, hence $\dim S_{d,\beta} = \dim S_{d,-\beta}$.

Step 2. In this step we prove that A_S is a maximal torus of G_S, that T_S is an algebraic torus with the decomposition in (2), that the T_S-action on Y is good with vertex o_Y, and that there exists $\xi \in X_*(T_S)$ in the Reeb cone of the T_S-action such that wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$.

By Proposition 5.5(1) and Definition-Lemma 4.1(3), there exists a maximal torus A'_S of G_S with $A_S \subset A'_S$. By Proposition 5.5(4), the identity component A' of $\varphi^{-1}(A'_S)$ is a maximal torus of G_R with $\varphi(A') = A'_S$. Since A is connected and $A \subset \varphi^{-1}(A'_S)$, we have $A \subset A'$, hence $A = A'$ by the maximality of A. Thus $A_S = \varphi(A') = A'_S$ is a maximal torus of G_S. Together with **Step 1**, we obtain (1).

For any $s \in \mathbb{G}_m$, $\lambda_{S,s}$ acts on S_d by the scalar s^d, hence commutes with any element of $\mathrm{Aut}_{\mathrm{gr}}(S)$. Thus T_S is the image of the homomorphism $\mathbb{G}_m \times A_S \to \mathrm{Aut}_{\mathrm{gr}}(S)$, $(s, g) \to \lambda_{S,s} g$, and T_S is an algebraic torus, as a quotient of $\mathbb{G}_m \times A_S$ [Mil17, Definition 12.14 and the paragraph after it]. Since $\dim Y > 0$, Lemma 4.2(1) and Proposition 5.5(1) imply that there exist integers $d_1, d_2 \geq 0$, $f \in S_{d_1}$, and $h \in S_{d_2}$ with $\{f, h\} \neq 0$. Then $\{f, h\} \in S_{d_1+d_2-r}$ (Definition-Proposition 5.2(3)). Let $s \in \mathbb{G}_m$ with $\lambda_{S,s} \in A_S$. Since $\lambda_{S,s} \in G_S$ preserves the Poisson bracket of S,

$$s^{d_1+d_2-r}\{f, h\} = \lambda_{S,s}^*\{f, h\} = \{\lambda_{S,s}^* f, \lambda_{S,s}^* h\} = s^{d_1+d_2}\{f, h\},$$

hence $s^r = 1$. If $(s, g) \in \mathbb{G}_m \times A_S$ satisfies $\lambda_{S,s} g = 1$, then $\lambda_{S,s} = g^{-1} \in A_S$, and hence $s^r = 1$ and $g = \lambda_{S,s}^{-1}$. Thus the kernel of $\mathbb{G}_m \times A_S \to T_S$ is finite, and [Mil17, Theorem 12.9] implies

$$X_*(T_S) \otimes_{\mathbb{Z}} \mathbb{R} = \mathbb{R}\lambda_S \oplus (X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}).$$

Hence $\dim T_S = \dim A_S + 1$.

Next, we show that the T_S-action on Y is good with vertex o_Y. Let $\mathfrak{m} := \bigoplus_{d>0} S_d$, which is the maximal ideal of S defining o_Y. For any closed point $y \in Y$, the morphism $\mathbb{G}_m \to Y$, $s \to \lambda_{S,s}(y)$, extends to a morphism $\mathbb{A}^1 \to Y$ sending 0 to o_Y, since $S_0 = \mathbb{C}$ and $f(\lambda_{S,s}(y)) = s^d f(y)$ for any $f \in S_d$. Thus o_Y is contained in the closure of any T_S-orbit. Since T_S preserves $\mathfrak{m}$, the point o_Y is fixed by T_S, hence o_Y is the only closed point contained in the closure of any T_S-orbit. Since $T_S \subset \mathrm{Aut}_{\mathrm{gr}}(S)$ acts faithfully on Y, the T_S-action on Y is good with vertex o_Y (Definition 2.4). Let $\mathfrak{t}^+_{S,\mathbb{R}} \subset X_*(T_S) \otimes_{\mathbb{Z}} \mathbb{R}$ be the Reeb cone of the T_S-action.

For $d \geq 0$ and $\beta \in X^*(A_S)$ with $S_{d,\beta} \neq 0$, the torus T_S acts on $S_{d,\beta}$ by the character $\gamma_{d,\beta} \in X^*(T_S)$ with $\gamma_{d,\beta}(\lambda_{S,s}g) = s^d\beta(g)$ for any $s \in \mathbb{G}_m$ and $g \in A_S$. Since $S = \bigoplus_{d,\beta} S_{d,\beta}$ [Mil17, Theorem 12.12] and $\gamma_{d,\beta}$ determines (d,β), the non-zero subspaces $S_{d,\beta}$ are the character spaces of the T_S-action on S. Moreover, $\langle \gamma_{d,\beta}, c'\lambda_S + \eta' \rangle = c'd + \langle \beta, \eta' \rangle$ for any $c' \in \mathbb{R}$ and $\eta' \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$, and $S_{0,\beta} = 0$ for any $\beta \neq 0$ since $S_0 = \mathbb{C}$. Thus, by Definition 2.4, for any $c' \in \mathbb{R}$ and $\eta' \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$, we have $c'\lambda_S + \eta' \in \mathfrak{t}^+_{S,\mathbb{R}}$ if and only if

$$c'd + \langle \beta, \eta' \rangle > 0 \text{ for any } d > 0 \text{ and } \beta \in X^*(A_S) \text{ with } S_{d,\beta} \neq 0,$$

and in this case,

$$\mathrm{wt}_{c'\lambda_S + \eta'}(f) = \min\{c'd + \langle \beta, \eta' \rangle \mid f_{d,\beta} \neq 0\}$$

for any $0 \neq f = \sum_{d,\beta} f_{d,\beta} \in S$ with $f_{d,\beta} \in S_{d,\beta}$.

For any $s \in \mathbb{G}_m$ and $g \in A_S$, we have $\lambda^*_{S,s}\omega_Y = s^r\omega_Y$ (Proposition 5.5(1)), and $g^*\omega_Y = \omega_Y$ (Lemma 2.12) since $g \in G_S$ preserves the Poisson bracket of S. Thus $t^*\omega_Y \in \mathbb{C}^*\omega_Y$ for any $t \in T_S$. By Theorem 2.16(1) applied to (Y, ω_Y) and T_S, there exist an algebraic torus T'_S acting faithfully on Y with $T_S \subset T'_S$ and $t^*\omega_Y \in \mathbb{C}^*\omega_Y$ for any $t \in T'_S$, and a cocharacter $\xi \in X_*(T'_S)$ in the Reeb cone of the T'_S-action, such that the grading of S defined by ξ is a minimizing grading of (Y, ω_Y).

We show that $T'_S = T_S$. Since T'_S is commutative and contains $\lambda_S(\mathbb{G}_m)$, we have $T'_S \subset \mathrm{Aut}_{\mathrm{gr}}(S)$, and $T'_S \cap \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ is a closed subgroup of T'_S (Definition-Lemma 4.1(1)). The identity component of $T'_S \cap \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ is smooth [Mil17, Theorem 3.23] and connected, hence a torus [Mil17, Definition 12.14 and the paragraph after it] which is contained in G_S and contains the connected group A_S, hence is equal to A_S by (1). For any $t \in T'_S$, there exists $s \in \mathbb{G}_m$ with $t^*\omega_Y = s^r\omega_Y = \lambda^*_{S,s}\omega_Y$, hence $\lambda^{-1}_{S,s}t \in \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)$ (Lemma 2.12). Thus $T'_S = \lambda_S(\mathbb{G}_m) \cdot (T'_S \cap \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y))$, and

$$\dim T'_S \leq 1 + \dim(T'_S \cap \mathrm{Aut}_{\mathrm{gr}}(S, \omega_Y)) = 1 + \dim A_S = \dim T_S.$$

Since T'_S is connected and T_S is a closed subgroup of T'_S, we get $T'_S = T_S$.

Thus $\xi \in X_*(T_S) \cap \mathfrak{t}^+_{S,\mathbb{R}}$. Let $S = \bigoplus_{e \geq 0} S^\xi_e$ be the grading of S defined by ξ (Definition 2.4). Since $\xi \in \mathfrak{t}^+_{S,\mathbb{R}}$, we have $\langle \gamma_{d,\beta}, \xi \rangle > 0$ for any $d > 0$ and $\beta \in X^*(A_S)$ with $S_{d,\beta} \neq 0$. Thus $S^\xi_0 = S_0 = \mathbb{C}$ (see also [LLX18, paragraph after Definition 3.1]) and $\bigoplus_{e>0} S^\xi_e = \mathfrak{m}$. Hence the vertex of the grading $S = \bigoplus_{e \geq 0} S^\xi_e$ is o_Y, and the degree valuation of the grading is wt_ξ (Definitions 2.5 and 2.4). Since the grading is a minimizing grading of (Y, ω_Y), wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$ (Definition 2.14).

Step 3. In this step we prove that the linear map σ of $X_*(T_S) \otimes_{\mathbb{Z}} \mathbb{R}$ with $\sigma(c\lambda_S + \eta) = c\lambda_S - \eta$ for any $c \in \mathbb{R}$ and $\eta \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$ preserves $\mathfrak{t}^+_{S,\mathbb{R}}$, and that $\mathrm{vol}_{Y \ni o_Y}(\mathrm{wt}_{\sigma(\xi')}) = \mathrm{vol}_{Y \ni o_Y}(\mathrm{wt}_{\xi'})$ for any $\xi' \in \mathfrak{t}^+_{S,\mathbb{R}}$.

By **Step 2**, σ is well defined. Let $\xi' = c'\lambda_S + \eta'$ with $c' \in \mathbb{R}$ and $\eta' \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$. By **Step 1**, $S_{d,\beta} \neq 0$ if and only if $S_{d,-\beta} \neq 0$. Thus, replacing β with $-\beta$ in the description of $\mathfrak{t}^+_{S,\mathbb{R}}$ in **Step 2**, we see that $\xi' \in \mathfrak{t}^+_{S,\mathbb{R}}$ if and only if $\sigma(\xi') = c'\lambda_S - \eta' \in \mathfrak{t}^+_{S,\mathbb{R}}$.

Now assume that $\xi' \in \mathfrak{t}^+_{S,\mathbb{R}}$, and let N be a positive integer. Let $\mathfrak{a}_N(\mathrm{wt}_{\xi'}) \subset \mathcal{O}_{Y,o_Y}$ be the ideal of the elements f with $\mathrm{wt}_{\xi'}(f) \geq N$, and let

$$J_N := \bigoplus_{c'd+\langle\beta,\eta'\rangle\geq N} S_{d,\beta}.$$

By **Step 2**, J_N is the ideal of the elements $f \in S$ with $\mathrm{wt}_{\xi'}(f) \geq N$. By Proposition 5.5(1), S is finitely generated. Since $S_0 = \mathbb{C}$ and $S_d = \bigoplus_\beta S_{d,\beta}$ for any d, there exist generators $h_1, \ldots, h_p$ of the $\mathbb{C}$-algebra S with $0 \neq h_i \in S_{d_i,\beta_i}$ and $d_i > 0$. Let $w > 0$ be the minimum of the numbers $c'd_i + \langle\beta_i, \eta'\rangle$ for $1 \leq i \leq p$. Any $S_{d,\beta}$ is spanned by the monomials $h_1^{l_1} \cdots h_p^{l_p}$ with $\sum_i l_i(d_i, \beta_i) = (d, \beta)$, and for such a monomial,

$$c'd + \langle\beta, \eta'\rangle = \sum_i l_i(c'd_i + \langle\beta_i, \eta'\rangle) \geq w \sum_i l_i.$$

Thus $S/J_N \cong \bigoplus_{c'd+\langle\beta,\eta'\rangle<N} S_{d,\beta}$ is spanned by the images of finitely many monomials, hence $\dim_{\mathbb{C}} S/J_N < +\infty$. Since $N > 0$, we have $J_N \subset \mathfrak{m}$, hence J_N is $\mathfrak{m}$-primary and

$$\mathcal{O}_{Y,o_Y}/J_N\mathcal{O}_{Y,o_Y} \cong S/J_N.$$

Since $\mathrm{wt}_{\xi'}(u) = 0$ for any $u \in S\backslash\mathfrak{m}$, we have $\mathfrak{a}_N(\mathrm{wt}_{\xi'}) = J_N\mathcal{O}_{Y,o_Y}$. Thus

$$\dim_{\mathbb{C}} \mathcal{O}_{Y,o_Y}/\mathfrak{a}_N(\mathrm{wt}_{\xi'}) = \sum_{c'd+\langle\beta,\eta'\rangle<N} \dim S_{d,\beta}.$$

Since $\sigma(\xi') = c'\lambda_S - \eta' \in \mathfrak{t}^+_{S,\mathbb{R}}$, replacing β with $-\beta$ and using **Step 1**, we get

$$\begin{aligned}\dim_{\mathbb{C}} \mathcal{O}_{Y,o_Y}/\mathfrak{a}_N(\mathrm{wt}_{\sigma(\xi')}) &= \sum_{c'd-\langle\beta,\eta'\rangle<N} \dim S_{d,\beta} = \sum_{c'd+\langle\beta,\eta'\rangle<N} \dim S_{d,-\beta} \\ &= \dim_{\mathbb{C}} \mathcal{O}_{Y,o_Y}/\mathfrak{a}_N(\mathrm{wt}_{\xi'}).\end{aligned}$$

By [LLX18, Definition 2.1], $\mathrm{vol}_{Y\ni o_Y}(\mathrm{wt}_{\xi'})$ is determined by the numbers $\dim_{\mathbb{C}} \mathcal{O}_{Y,o_Y}/\mathfrak{a}_N(\mathrm{wt}_{\xi'})$ for positive integers N. Thus $\mathrm{vol}_{Y\ni o_Y}(\mathrm{wt}_{\sigma(\xi')}) = \mathrm{vol}_{Y\ni o_Y}(\mathrm{wt}_{\xi'})$.

Step 4. In this step we prove that $a(\mathrm{wt}_{\sigma(\xi')}, Y) = a(\mathrm{wt}_{\xi'}, Y)$ for any $\xi' \in \mathfrak{t}^+_{S,\mathbb{R}}$, which finishes the proof of (2).

Since ω_Y is non-degenerate and $\dim Y > 0$, we have $\dim Y = 2m$ for a positive integer m. Let $\Omega := \omega_Y^m$. By Lemma 2.9, Ω is a nowhere vanishing algebraic $2m$-form on Y_{reg}, and $(Y, 0)$ is klt. Recall from **Step 2** that $\lambda_{S,s}^*\omega_Y = s^r\omega_Y$ and $g^*\omega_Y = \omega_Y$ for any $s \in \mathbb{G}_m$ and $g \in A_S$. By **Step 2**, any element (s, g) of the kernel of $\mathbb{G}_m \times A_S \to T_S$ satisfies $s^r = 1$. Thus the character $(s, g) \to s^{mr}$ of $\mathbb{G}_m \times A_S$ is trivial on the kernel of $\mathbb{G}_m \times A_S \to T_S$, and hence there exists $\alpha \in X^*(T_S)$ whose composite with $\mathbb{G}_m \times A_S \to T_S$ is $(s, g) \to s^{mr}$ [Mil17, Theorem 12.9]. Then $t^*\Omega = \alpha(t)\Omega$ for any $t \in T_S$. Hence $\langle\alpha, \lambda_S\rangle = mr$ and $\alpha|_{A_S} = 1$. Let $\xi' = c'\lambda_S + \eta' \in \mathfrak{t}^+_{S,\mathbb{R}}$ with $c' \in \mathbb{R}$ and $\eta' \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$. Since the T_S-action on Y is good (**Step 2**) and $\sigma(\xi') \in \mathfrak{t}^+_{S,\mathbb{R}}$ (**Step 3**), Lemma 2.7 implies

$$a(\mathrm{wt}_{\sigma(\xi')}, Y) = \langle\alpha, c'\lambda_S - \eta'\rangle = mrc' = \langle\alpha, c'\lambda_S + \eta'\rangle = a(\mathrm{wt}_{\xi'}, Y).$$

By **Step 3** and Definition 2.2, $\widehat{\mathrm{vol}}_{Y\ni o_Y}(\mathrm{wt}_{\sigma(\xi')}) = \widehat{\mathrm{vol}}_{Y\ni o_Y}(\mathrm{wt}_{\xi'})$. Together with **Steps 2 and 3**, we obtain (2).

Step 5. In this step we prove that v_{λ_S} is a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$.

Recall that $\xi \in X_*(T_S) \cap \mathfrak{t}^+_{S,\mathbb{R}}$ is the cocharacter in **Step 2** such that wt_ξ is a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$. By **Steps 3 and 4**, $\sigma(\xi) \in \mathfrak{t}^+_{S,\mathbb{R}}$ and $\widehat{\mathrm{vol}}_{Y\ni o_Y}(\mathrm{wt}_{\sigma(\xi)}) = \widehat{\mathrm{vol}}_{Y\ni o_Y}(\mathrm{wt}_\xi)$, hence $\mathrm{wt}_{\sigma(\xi)}$ is also a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$. Since $(Y, 0)$ is klt, Theorem 2.3 implies that $\mathrm{wt}_{\sigma(\xi)} = q\,\mathrm{wt}_\xi$ for some real number $q > 0$. Let $d \geq 0$ and $\beta \in X^*(A_S)$ with $S_{d,\beta} \neq 0$. By **Step 2**, $\mathrm{wt}_{\xi'}(f) = \langle\gamma_{d,\beta}, \xi'\rangle$

for any $\xi' \in \mathfrak{t}^+_{S,\mathbb{R}}$ and any $0 \neq f \in S_{d,\beta}$. Thus $\langle \gamma_{d,\beta}, \sigma(\xi) - q\xi \rangle = 0$. Since T_S acts faithfully on S, the characters $\gamma_{d,\beta}$ with $S_{d,\beta} \neq 0$ span $X^*(T_S) \otimes_{\mathbb{Z}} \mathbb{R}$ [Mil17, Theorem 12.9]. Thus $\sigma(\xi) = q\xi$.

Write $\xi = c\lambda_S + \eta$ with $c \in \mathbb{R}$ and $\eta \in X_*(A_S) \otimes_{\mathbb{Z}} \mathbb{R}$. By **Step 2**, $\mathfrak{t}^+_{S,\mathbb{R}}$ is an intersection of open half-spaces, hence convex. Thus $c\lambda_S = \frac{1}{2}(\xi + \sigma(\xi)) \in \mathfrak{t}^+_{S,\mathbb{R}}$. Since $\dim Y > 0$, there exist $d > 0$ and $\beta \in X^*(A_S)$ with $S_{d,\beta} \neq 0$, hence $cd > 0$ by **Step 2**. Thus $c > 0$. Since $\sigma(\xi) = c\lambda_S - \eta$ and $q\xi = qc\lambda_S + q\eta$, the decomposition in **Step 2** implies $c = qc$ and $-\eta = q\eta$. Thus $q = 1$ and $\eta = 0$, that is, $\xi = c\lambda_S$.

Then $\lambda_S = c^{-1}\xi \in \mathfrak{t}^+_{S,\mathbb{R}}$. Since λ_S acts on S_d with weight d, we have $\mathrm{wt}_{\lambda_S} = v_{\lambda_S}$ (Definitions 2.5 and 2.4), and $\mathrm{wt}_\xi = cv_{\lambda_S}$ by **Step 2**. Since $\widehat{\mathrm{vol}}_{Y \ni o_Y}$ is invariant under multiplying valuations by positive real numbers [Li15, Section 2.1, paragraph after Remark 2.2], v_{λ_S} is a minimizer of $\widehat{\mathrm{vol}}_{Y \ni o_Y}$. □

6. Proofs of the main theorems

In this section, we prove Theorems 1.6, 1.8, and 1.9, combining the results of Sections 2–5.

Proof of Theorem 1.6. By Definition 2.8, X is a normal affine variety with $\dim X > 0$. Since ω is non-degenerate on X_{reg}, we have $\dim X = 2n$ for an integer $n \geq 1$. Thus $R = \bigoplus_{d \geq 0} R_d$ and ω satisfy the conditions of Set-up 2.10.

By Theorem 2.16(2), we have $r \in \{1, 2\}$. Let λ be the $\mathbb{G}_m$-action defined by the grading $R = \bigoplus_{d \geq 0} R_d$. By Theorem 2.17, the log Fano cone $(X, \frac{2}{r}\lambda)$ is K-polystable for the torus $\lambda(\mathbb{G}_m)$. Moreover, the grading $R = \bigoplus_{d \geq 0} R_d$ is primitive by Definition 2.14, (X, ω) is a symplectic singularity by assumption, and $T \subset \mathrm{Aut}_{\mathrm{gr}}(R, \omega)$ is an algebraic torus. Thus the conditions of Theorem 3.1 hold, and hence there exists a ring automorphism $\rho : R \to R$ satisfying (1)–(3) such that $\rho(R_{d,\chi}) = R_{d,-\chi}$ for any $d \geq 0$ and $\chi \in X^*(T)$.

Let $d \geq 0$ and $\chi \in X^*(T)$. Since ρ is bijective, $\rho(R_{d,\chi}) = R_{d,-\chi}$, and $\rho(zf) = \bar{z}\rho(f)$ for any $z \in \mathbb{C}$ and $f \in R$ by (1), ρ maps any basis of the $\mathbb{C}$-vector space $R_{d,\chi}$ to a basis of the $\mathbb{C}$-vector space $R_{d,-\chi}$. Thus $\dim R_{d,\chi} = \dim R_{d,-\chi}$. □

Proof of Theorem 1.8. By Definition 2.8, X is a normal affine variety with $\dim X > 0$. Since ω is non-degenerate on X_{reg}, we have $\dim X = 2n$ for an integer $n \geq 1$. Thus $R = \bigoplus_{d \geq 0} R_d$ and ω satisfy the conditions of Set-up 2.10, and the grading $R = \bigoplus_{d \geq 0} R_d$ is primitive by Definition 2.14. Let $E = \mathrm{Proj}\, R$ and Δ_E be as in Definition-Proposition 2.15.

Since (X, ω) is a symplectic singularity, $(X, 0)$ is klt by Lemma 2.9. By Theorem 2.17, (E, Δ_E) is a K-polystable log Fano pair. Thus the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R)$ is reductive by Proposition 4.4. Moreover, by the definition of $\mathfrak{h}$, the isomorphism $R_r \to \mathrm{Lie}\, G_R$, $a \to \{a, -\}$, of Lemma 4.2(2) maps $\mathfrak{h}$ onto $\mathrm{Lie}\, T$. Thus the conditions of Proposition 4.7 hold for T and $\mathfrak{h}$.

(1) By Proposition 4.7(1), G_R is a connected reductive group. By Lemma 4.2(2), the map $\mathfrak{g} \to \mathrm{Lie}\, G_R$, $a \to \{a, -\}$, is an isomorphism of Lie algebras. Moreover, G_R is an almost direct product of a torus Z and a semisimple group G' [Mil17, Proposition 21.60(c)], that is, the multiplication map $Z \times G' \to G_R$ is an isogeny. Since an isogeny induces an isomorphism of Lie algebras [Mil17, Theorem 3.23, Remark 5.42, and 10.14], we have $\mathrm{Lie}\, G_R \cong \mathrm{Lie}\, Z \oplus \mathrm{Lie}\, G'$ as Lie algebras, where $\mathrm{Lie}\, Z$ is abelian and $\mathrm{Lie}\, G'$ is semisimple [Mil17, 21.56]. Thus $\mathfrak{g}$ is reductive.

(2) The isomorphism in (1) is the isomorphism of Lemma 4.2(2). Thus (2) follows from Proposition 4.7(2).

(3) Let C be a Lie algebra involution of $\mathfrak{g}$ with $C|_{\mathfrak{h}} = -\mathrm{id}_{\mathfrak{h}}$, and let $d \geq 0$. Since the isomorphism in (1) maps $\mathfrak{h}$ onto $\mathrm{Lie}\, T$, C corresponds to a Lie algebra involution of $\mathrm{Lie}\, G_R$ whose restriction to $\mathrm{Lie}\, T$ is $-\mathrm{id}_{\mathrm{Lie}\, T}$. Thus by (1), (2), and Lemma 4.8(1), there exists a unique algebraic group involution c of G_R whose differential is C under the isomorphism in (1), and $c(t) = t^{-1}$ for any $t \in T$.

Since R is generated by finitely many homogeneous elements of positive degree and $R_0 = \mathbb{C}$, we have $\dim R_d < +\infty$. By Definition-Lemma 4.1(1), R_d is a rational representation of G_R. For

any $\chi \in X^*(T)$, the subspace of R_d on which T acts by χ is $R_{d,\chi}$, and Theorem 1.6, applied to $T \subset \mathrm{Aut}_{\mathrm{gr}}(R,\omega)$, implies that $\dim R_{d,\chi} = \dim R_{d,-\chi}$. Thus by Lemma 4.8(2), the representation R_d of G_R is isomorphic to the representation of G_R on the vector space R_d given by $g \cdot_c f := c(g) \cdot f$. In other words, there exists a $\mathbb{C}$-linear bijection $\tau_d : R_d \to R_d$ such that

$$\tau_d(g \cdot f) = c(g) \cdot \tau_d(f)$$

for any $g \in G_R$ and $f \in R_d$.

By Definition-Lemma 4.1(2), for any $a \in \mathfrak{g}$, the derivative at the identity of the action of G_R on R_d in the direction of $\{a,-\} \in \mathrm{Lie}\, G_R$ is the map $f \to \{a,f\}$. Since the differential of c maps $\{a,-\}$ to $\{C(a),-\}$, taking the derivatives at the identity of G_R of both sides of $\tau_d(g \cdot f) = c(g) \cdot \tau_d(f)$ in the direction of $\{a,-\}$, we obtain

$$\tau_d(\{a,f\}) = \{C(a), \tau_d(f)\}$$

for any $a \in \mathfrak{g}$ and $f \in R_d$. □

Proof of Theorem 1.9. By Definition 2.8, X is a normal affine variety with $\dim X > 0$. Since ω is non-degenerate on X_{reg}, we have $\dim X = 2n$ for an integer $n \geq 1$. Let ℓ be the degree of ω, and let λ be the $\mathbb{G}_m$-action defined by the grading $R = \bigoplus_{d \geq 0} R_d$. Thus $R = \bigoplus_{d \geq 0} R_d$ and ω satisfy the conditions of Set-up 2.10, with ℓ in place of r. By Definition-Lemma 5.1, applied to the grading $R = \bigoplus_{d \geq 0} R_d$, we have $\mu(\mathfrak{t}) \subset R_\ell$. Let $I := R\mu(\mathfrak{t})$, so that $S = (R/I)^T$ (Definition-Proposition 5.2).

Step 1. In this step we reduce to the case where $T \subset \mathrm{Aut}_{\mathrm{gr}}(R,\omega)$.

Let $\pi : T \to \mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ be the group homomorphism defining the T-action, and let $\bar{T} := \pi(T)$. Since R_d is a rational representation of T for any $d \geq 0$, π is a homomorphism of algebraic groups under the embedding of Definition-Lemma 4.1(1). Thus $\bar{T}$ is a closed subgroup of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ [Mil17, Proposition 1.68], $\bar{T}$ is a torus as a quotient of T [Mil17, Definition 12.14 and the paragraph after it], and the differential $d\pi : \mathfrak{t} \to \mathrm{Lie}\, \bar{T}$ is surjective [Mil17, Theorem 3.23, Remark 5.42, and 10.14]. By Definition-Lemma 4.1(1), $\bar{T}$ acts on R by graded Poisson automorphisms. Let $\bar{\mu}$ be the moment map of the $\bar{T}$-action. Since $a \cdot f = d\pi(a) \cdot f$ for any $a \in \mathfrak{t}$ and $f \in R$, the uniqueness in Definition-Lemma 5.1 implies that $\mu = \bar{\mu} \circ d\pi$. Thus $\mu(\mathfrak{t}) = \bar{\mu}(\mathrm{Lie}\, \bar{T})$, and hence $I = R\bar{\mu}(\mathrm{Lie}\, \bar{T})$. Moreover, T acts on R/I through $\bar{T}$, so $(R/I)^T = (R/I)^{\bar{T}}$. Therefore, the Hamiltonian reduction of X by $\bar{T}$ is $Y = \mathrm{Spec}\, S$, and Definition-Proposition 5.2(2) and (3), applied to the $\bar{T}$-action, define the grading $S = \bigoplus_{d \geq 0} S_d$ and the Poisson bracket of S with $\{f + I, h + I\} = \{f,h\} + I$ for any $f,h \in R^T$. Possibly replacing T with $\bar{T}$, we may assume that $T \subset \mathrm{Aut}_{\mathrm{gr}}(R,\omega)$.

Step 2. In this step we construct a minimizing grading $R = \bigoplus_{e \geq 0} R'_e$ of (X,ω) such that Definition-Lemma 5.1 and Definition-Proposition 5.2, applied to the grading $R = \bigoplus_{e \geq 0} R'_e$, define the moment map μ of the T-action, the ideal $I = R\mu(\mathfrak{t})$, and the ring $S = (R/I)^T$.

Since $\dim X > 0$, there exists $d > 0$ with $R_d \neq 0$. Let $N := \gcd\{d > 0 \mid R_d \neq 0\}$, and let $R'_e := R_{Ne}$ for any $e \geq 0$. Since $R_d = 0$ unless $N \mid d$, we have $R = \bigoplus_{e \geq 0} R'_e$, the grading $R = \bigoplus_{e \geq 0} R'_e$ is primitive, and $R'_0 = \mathbb{C}$. Let λ' be the $\mathbb{G}_m$-action defined by the grading $R = \bigoplus_{e \geq 0} R'_e$. Then $\lambda_t = \lambda'_{t^N}$ for any $t \in \mathbb{G}_m$. Since $\bigoplus_{e > 0} R'_e = \bigoplus_{d > 0} R_d$, o is also the vertex of the grading $R = \bigoplus_{e \geq 0} R'_e$.

Let $\zeta := \exp(2\pi i/N)$. Since $R_d = 0$ unless $N \mid d$, the automorphism λ_ζ is the identity of X. Thus $\zeta^\ell \omega = \lambda_\zeta^* \omega = \omega$, and hence $N \mid \ell$ since $\omega \neq 0$. Let $r := \ell/N$. Then

$$(\lambda'_{t^N})^* \omega = \lambda_t^* \omega = t^\ell \omega = (t^N)^r \omega$$

for any $t \in \mathbb{G}_m$. Since any element of $\mathbb{G}_m$ is of the form t^N, ω is homogeneous of degree r with respect to the grading $R = \bigoplus_{e \geq 0} R'_e$. By Definition 2.5, we have $v_\lambda = N v_{\lambda'}$. Since the normalized volume is invariant under multiplying valuations by positive real numbers [Li15, Section 2.1, paragraph after Remark 2.2], $v_{\lambda'}$ is a minimizer of $\widehat{\mathrm{vol}}_{X \ni o}$. Thus $R = \bigoplus_{e \geq 0} R'_e$ is a

minimizing grading of (X,ω) (Definition 2.14), and $R=\bigoplus_{e\geq 0}R'_e$ and ω satisfy the conditions of Set-up 2.10, with λ' in place of λ.

Since the non-zero graded pieces of $R=\bigoplus_{e\geq 0}R'_e$ are those of $R=\bigoplus_{d\geq 0}R_d$, the linear algebraic groups $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ and G_R of Definition-Lemma 4.1 for the grading $R=\bigoplus_{e\geq 0}R'_e$ are equal to $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ and G_R for the grading $R=\bigoplus_{d\geq 0}R_d$. Since $\mu(\mathfrak{t})\subset R_\ell=R'_r$, the uniqueness in Definition-Lemma 5.1 implies that μ is also the moment map of the T-action for the grading $R=\bigoplus_{e\geq 0}R'_e$. Thus Definition-Proposition 5.2, applied to the grading $R=\bigoplus_{e\geq 0}R'_e$, defines the ideal $I=R\mu(\mathfrak{t})$, the ring $S=(R/I)^T$, and the Poisson bracket of S with $\{f+I,h+I\}=\{f,h\}+I$ for any $f,h\in R^T$. By Definition-Proposition 5.2(2), the grading of S for the grading $R=\bigoplus_{e\geq 0}R'_e$ is $S=\bigoplus_{e\geq 0}S'_e$, where $S'_e:=S_{Ne}$, and we have $S_d=0$ unless $N\mid d$. Thus o_Y is also the vertex of the grading $S=\bigoplus_{e\geq 0}S'_e$.

Step 3. In this step we prove (1).

Assume that $\dim Y>0$. By Definition-Lemma 4.1(3), there exists a maximal torus A of G_R with $T\subset A$. By **Step 2**, $R=\bigoplus_{e\geq 0}R'_e$ is a minimizing grading of (X,ω), and G_R is also the identity component of $\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ for the grading $R=\bigoplus_{e\geq 0}R'_e$. Thus Theorem 1.6, applied to the grading $R=\bigoplus_{e\geq 0}R'_e$ and A, implies that

$$\dim R'_{e,\chi}=\dim R'_{e,-\chi} \tag{6.1}$$

for any $e\geq 0$ and $\chi\in X^*(A)$, where $R'_{e,\chi}:=\{f\in R'_e\mid t\cdot f=\chi(t)f \text{ for any } t\in A\}$.

Let λ'_S be the $\mathbb{G}_m$-action defined by the grading $S=\bigoplus_{e\geq 0}S'_e$. We check the conditions of Set-up 5.4 for the grading $R=\bigoplus_{e\geq 0}R'_e$. By **Step 2**, $R=\bigoplus_{e\geq 0}R'_e$ and ω satisfy the conditions of Set-up 2.10, where r is the degree of ω. By **Steps 1 and 2**, $T\subset\mathrm{Aut}_{\mathrm{gr}}(R,\omega)$ is an algebraic torus, μ is the moment map of the T-action, and $S=\bigoplus_{e\geq 0}S'_e$ is the grading of Definition-Proposition 5.2(2), all for the grading $R=\bigoplus_{e\geq 0}R'_e$. Thus Set-up 5.4(1) and (2) hold. Since (Y,ω_Y) is a symplectic singularity, Y is a normal affine variety, and ω_Y is a closed non-degenerate algebraic 2-form on Y_{reg} (Definition 2.8). Thus S is a normal domain, and Set-up 5.4(3) and (4) hold, since $\dim Y>0$ and ω_Y induces the Poisson bracket of S. By **Step 2**, o_Y is the vertex of the grading $S=\bigoplus_{e\geq 0}S'_e$. Thus Set-up 5.4 holds for the grading $R=\bigoplus_{e\geq 0}R'_e$, with λ'_S in place of λ_S.

Moreover, (X,ω) and (Y,ω_Y) are symplectic singularities, and the equality (6.1) for any $e\geq 0$ and $\chi\in X^*(A)$ is the condition on A in Theorem 5.6 for the grading $R=\bigoplus_{e\geq 0}R'_e$. Thus by Theorem 5.6, $v_{\lambda'_S}$ is a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$.

Let λ_S be the $\mathbb{G}_m$-action defined by the grading $S=\bigoplus_{d\geq 0}S_d$. Since $S_d=0$ unless $N\mid d$, and $S'_e=S_{Ne}$ for any $e\geq 0$, we have $v_{\lambda_S}=Nv_{\lambda'_S}$ by Definition 2.5. Since the normalized volume is invariant under multiplying valuations by positive real numbers [Li15, Section 2.1, paragraph after Remark 2.2], v_{λ_S} is a minimizer of $\widehat{\mathrm{vol}}_{Y\ni o_Y}$. This proves (1).

Step 4. In this step we prove (2).

Assume that $\dim Y=0$. Since (Y,ω_Y) is a symplectic singularity, Y is a normal affine variety (Definition 2.8), and hence S is a domain. Thus S is a reduced finitely generated $\mathbb{C}$-algebra with $\dim S=0$, and hence S is a product of finitely many copies of $\mathbb{C}$ [Mil17, Proposition 11.1 and A.61]. Since a product of at least two copies of $\mathbb{C}$ has zero divisors, we have $S=\mathbb{C}$. □

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